Inequality, Markups, and the Phillips Curve

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Abstract

Micro evidence shows that poorer households are more price elastic than richer ones. We build this fact into a two-agent New Keynesian model with sticky prices and flexible wages that keeps the standard three-equation form. Each firm sells to rich and poor households and sets its desired markup from the price elasticity of its customer mix. Poor households live on wages and transfers, while rich households earn wages and also receive the profits, so inequality falls when output is above potential and wages rise relative to profits. Spending then shifts toward the price-sensitive poor, firms face more elastic demand, and their desired markup falls. This composition channel acts as a real rigidity, and the feedback from output to inequality behind it is stronger in more unequal economies. The slope of the Phillips curve factors in closed form into the conventional slope and the composition channel, so with equal price elasticities inequality leaves the slope unchanged. Calibrated to micro estimates of price elasticities by income, the composition channel makes the Phillips curve 4.3 percent flatter under Rotemberg pricing than in an otherwise identical model without the composition channel. Fiscal policy shapes this channel, because taxing profits and rebating them to poorer households lowers inequality and desired markups. Empirically, the response of measured markups to oil shocks is larger in countries with higher income inequality, consistent with the feedback from output to inequality that our model shares with standard two-agent models.

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1 Introduction

How does the income distribution shape the response of inflation and output to aggregate shocks? The answer matters for monetary policy: if the distribution affects how prices respond to output, it also affects the central bank's trade-off between inflation and output. The question became more pressing after 2021, when cost shocks raised inflation (Gagliardone and Gertler, 2023) and renewed the debate on markups, profits and cost pass-through (Conlon et al., 2023; Bilbiie and Känzig, 2024).

Most of the literature answers this question through aggregate demand alone. In heterogeneous-agent and two-agent New Keynesian models, households differ in their marginal propensity to consume (Kaplan et al., 2018; Auclert, 2019), but firms face the same price elasticity of demand from all households, so the distribution does not affect their desired markups. Micro evidence suggests that this assumption is restrictive: price sensitivity declines with income. Using Swiss household scanner data and the 2015 Swiss franc appreciation, Auer et al. (2024) estimate how the price elasticity falls with income, and in their baseline quantification it is 6.6 at an annual income of 20,000 Swiss francs and 3.0 at 120,000. The composition of demand then shapes firms' desired markups, so the distribution also matters on the supply side.

We show that this channel is a real rigidity, a force that makes desired prices respond less to output, and that it flattens the Phillips curve. Section 2 develops the model, a two-elasticity version of the Two-Agent New Keynesian (TANK) framework of Debortoli and Galí (2018, 2024), which we call TE-TANK. Fixing one elasticity per household type gives the Phillips curve in closed form. Its slope is the product of two terms: the slope without distributional feedback and the composition channel, present only when elasticities differ (Proposition 2). With a common elasticity the distribution does not enter the Phillips curve, as in Debortoli and Galí (2018), so in TE-TANK inequality moves the slope only through the elasticity gap. The composition channel flattens the curve. Section 2.6 shows that, for a given demand block, a flatter curve leaves more of a cost-push shock in inflation, because the output contraction that the central bank's rule produces then buys less disinflation.

The composition channel arises because desired markups move with the composition of demand. A positive demand shock compresses inequality, shifts demand toward the price-sensitive poor, and lowers desired markups, which damps inflation. A cost-push shock runs the mechanism in reverse: inequality rises, demand tilts toward the less elastic rich, and desired markups rise on top of the initial cost pressure. A positive technology shock also raises inequality, so the channel dampens the fall in inflation. The composition channel is a real rigidity in the sense of Ball and Romer (1990), a countercyclical desired markup generated by demand composition; it operates only

when prices are sticky.

We measure the channel against a markup-matched TANK model: a single-elasticity economy whose elasticity is chosen so that its steady-state markup equals TE-TANK's. Under a common price-adjustment cost the two economies then differ only in the composition channel. In the calibration of Section 3, with the high elasticity for hand-to-mouth (poor) households, the composition channel lowers the slope by 4.3 percent, under Rotemberg and under Calvo pricing alike.

The sign holds whenever the government does not rebate all profits as equal per-capita transfers, so that consumption is unequal in steady state. If elasticities also vary with consumption within type, it holds only when the poor's elasticity does not fall too steeply with consumption relative to the rich's. The net comparison with the markup-matched model depends on the pricing friction. A firm that raises its price loses mostly its elastic poor customers. Section 3.3 shows that this firm-level curvature, absent in a symmetric Rotemberg equilibrium, steepens TE-TANK's curve under Calvo pricing by more than the composition channel flattens it.

The paper makes two contributions. First, with fixed elasticities by household type, we derive in closed form a supply block in which the split of income between wages and profits moves desired markups. Second, we sign and quantify the composition channel and separate it from the parts of the comparison with common-elasticity TANK that depend on the pricing friction.

Under Rotemberg pricing, the flatter curve raises inflation pass-through after a cost-push shock by 3.3 percent against the markup-matched benchmark and deepens the output contraction by the same percentage. Section 3.4 shows that the two gaps are equal by construction: matching the markup also equates the demand blocks, so the output contraction is the same fixed multiple of inflation in both.

Fiscal redistribution works on the supply side, because it changes the elasticity of demand that firms face. Rebating profits to hand-to-mouth households lowers steady-state inequality and, only when elasticities differ, the desired markup (Proposition 1). Transfers to poorer households thus lower desired markups and weaken the real rigidity. Section 3.5 shows that, under Rotemberg pricing, moving from full profit rebating to none flattens the Phillips curve of TE-TANK and leaves the slopes of both common-elasticity benchmarks unchanged, and that it lowers cost-push pass-through in all three economies. Under Calvo pricing the same move steepens TE-TANK's curve instead.

The main empirical application uses oil shocks, because external evidence documents their distributional effects. With German administrative data and the oil-supply news shock of Känzig (2021), Broer et al. (2025) show that an oil-price increase lowers the earnings growth of low-income workers, while top incomes are largely unaffected.

Their evidence concerns earnings dispersion among workers, which has no counterpart in our model: both household types earn the same wage, and inequality moves only with the split of income between wages and profits. Their mechanism differs from ours, and our model does not derive the distributional incidence of oil shocks.

Descriptive evidence from US quarterly data and a cross-country panel is a consistency check on the model, not a test of the elasticity gap. Section 4 shows that countries with higher long-run income concentration have higher average markup levels, a descriptive correlation in the direction of the model's steady-state relation between inequality and the desired markup. Markups rise after the expansionary technology shock and the contractionary oil supply-news shock of Mori and Peersman (2026), and consumption inequality rises after the technology shock. Markups also rise after an expansionary monetary shock, contrary to the sticky-price core (Nekarda and Ramey, 2020) that TE-TANK shares with common-elasticity models. After the oil shock, the response of the inverse labor share, a markup proxy, is larger, or less negative, in more unequal countries, but imprecisely. This sign is consistent with the feedback from output to inequality that TE-TANK shares with common-elasticity TANK, which predicts it under both pricing conventions. Per unit of a common cost shock, the two models also predict the same sign for the cross-country difference in inflation responses. This difference is small and insignificant, and the test has little power against either model.

Two papers that also let elasticities differ across households are closest. Mongey and Waugh (2025) study how the composition of demand shifts markups when elasticities vary with wealth, with oligopolistic firms but without nominal rigidities, monetary policy, or supply shocks. Olivi et al. (2024) let baskets and elasticities vary across households in a multi-sector New Keynesian model with productivity shocks and study optimal monetary policy in a cost-of-living crisis. In their model the Phillips-curve slope depends on the expenditure distribution, and redistribution moves markups through an endogenous wedge. Their elasticities differ because non-CES preferences over varieties, common to all households, make them fall with spending in their calibration. Each household's contribution to the cyclical markup term in their slope combines a composition effect, from its elasticity relative to the average, with the response of its elasticity to its own spending, and the term raises markups in a boom when markups rise with wealth. We instead fix one CES elasticity per household type in a one-sector model. Desired markups then move only with the division of spending between the two types, which the split of income between wages and profits sets, and we sign the comparison with a markup-matched TANK model. A separate literature studies the flattening of the Phillips curve (Blanchard, 2016; Hooper et al., 2020; Hazell et al., 2022) and links it to population aging (Ambrocio, 2026) and market concentration (Andrés et al., 2026). We study how the slope shapes shock transmission, not its historical de-

cline.

The evidence that price elasticities differ by income comes from several sources. Households with more time for shopping search more and pay lower prices for the same goods (Aguiar and Hurst, 2007; Kaplan and Menzio, 2016), in line with Harrod (1936), who argues that demand becomes less elastic as income rises. Scanner-data estimates are less clear: some find only small differences across income groups (Argente and Lee, 2021; Faber and Fally, 2022), while Handbury (2021) finds that higher-income households are less price sensitive in most product modules once their stronger taste for quality is controlled for. More broadly, desired markups that move with the income and wealth distribution are documented and modeled elsewhere:[2] retail markups rise with household income and with local income dispersion (Sangani, 2024).[3]

The distribution also reaches prices through other margins. Non-homothetic preferences shift demand and earnings across sectors (Andreolli et al., 2024). In Kaplan and Menzio (2016), buyers who search less weaken competition, but the search margin operates under flexible prices and never reaches a nominal rigidity. In Cravino et al. (2020), the goods bought by the richest households have stickier prices than those bought by middle-income households, but realistic changes in inequality barely move the aggregate output response and change the price response only modestly. Variable markups also flatten the Phillips curve when higher entry costs reduce the number of firms (Fujiwara and Matsuyama, 2026). We place the cyclical composition of demand across household types, set by the split of income between wages and profits, inside the Phillips curve.

2 A Two-Agent New Keynesian Model with Heterogeneous Elasticities

The model extends the TANK framework of Debortoli and Galí (2018, 2024) with one new assumption on demand; it keeps their labor-market closure and departs from them otherwise only in allowing decreasing returns to labor. The new assumption is

[2] Enders (2020) also obtains countercyclical markups from consumption-weighted elasticities, but under flexible prices, with elasticities that move as monetary injections reach segmented asset markets; in our model the desired markup is constant under flexible prices and moves only when prices are sticky.

[3] Stroebel and Vavra (2019) find that local house-price increases raise retail prices through markups rather than costs, concentrated in the high-homeownership areas where the shock makes buyers less price-sensitive. Döpper et al. (2025) estimate an elasticity of markups with respect to consumer price sensitivity of -0.73 (standard error 0.025) within brand, category, market and retailer, net of common time effects. Faber and Fally (2022) show that rich and poor households buy from different parts of the firm-size distribution within the same product group. What remains open is the business-cycle frequency: this evidence is regional, cross-sectional, or secular, and none of it identifies the aggregate demand-composition channel. Near-uniform pricing across the stores of a chain (DellaVigna and Gentzkow, 2019) also weakens local tests of the mechanism without disabling the aggregate channel.

that hand-to-mouth households have a higher price elasticity of demand than households with asset market access, in line with the micro evidence that price sensitivity declines with income (Berry et al., 2004; Aguiar and Hurst, 2007; Auer et al., 2024). A fraction λ of households are hand-to-mouth (poor, denoted P); the remaining $1-\lambda$ have full asset market access (rich, denoted R); and $\varepsilon_P > \varepsilon_R$. The elasticity gap makes firms' desired markups depend on the composition of demand: when spending shifts toward rich households, firms face a less elastic customer base and raise desired markups.

The mechanism operates through the division of income between wages and profits. When the output gap is positive, markups and profit income fall relative to labor income, demand shifts toward the more price-sensitive poor households, and desired markups fall further. We call this loop from the output gap to the distribution and back into price setting the demand-composition feedback. Its strength depends on fiscal policy: δ determines how much profit income the government collects and τ how it is redistributed across the two types. Redistribution therefore has supply-side effects, because it changes the elasticity of demand firms face. With a common elasticity ε, by contrast, the desired markup is the constant $\varepsilon/(\varepsilon-1)$ whatever the income distribution, so common-elasticity TANK cannot connect inequality to markups.

2.1 Households

Households have unit mass, a fraction λ hand-to-mouth (poor) and $1-\lambda$ Ricardian (rich), and both types share the same preferences over consumption and leisure,

$$U_t^h = \left(\frac{(C_t^h)^{1-\sigma}}{1-\sigma} - \frac{(N_t^h)^{1+\varphi}}{1+\varphi} \right) Z_t, \tag{1}$$

where $h \in \{P, R\}$ indexes the household type, C_t^h is consumption, N_t^h is hours worked, σ is the inverse intertemporal elasticity of substitution, φ is the inverse Frisch elasticity, and Z_t is a preference shock with $\log Z_t = \rho_z \log Z_{t-1} + \varepsilon_t^z$. Each type aggregates consumption over varieties with its own CES elasticity of substitution ε_h, which implies the type-specific demand schedule

$$C_t^h(i) = \left(\frac{P_t(i)}{P_t^h} \right)^{-\varepsilon_h} C_t^h, \tag{2}$$

where P_t^h is the price index of type h. Appendix D.3 derives this demand function.[4] We set $\varepsilon_P = 6.6$ and $\varepsilon_R = 3.0$, the baseline quantification of Auer et al. (2024) that Section 3 discusses.

[4] Appendices A–D follow the references.

For rich households, utility maximization yields the familiar Euler equation and equity-pricing equation, which Appendix D.2 derives. Poor households cannot save or borrow and consume all labor income plus transfers each period. Employment is demand-determined and common across types, $N_t^P = N_t^R \equiv N_t$, and the real wage follows the schedule of Debortoli and Galí (2018, p. 15) in average consumption $C_t \equiv (1-\lambda)C_t^R + \lambda C_t^P$,

$$\frac{W_t}{P_t} = \mathcal{M}^w \, C_t^\sigma N_t^\varphi, \tag{3}$$

with a constant wage markup $\mathcal{M}^w$ large enough that both types are willing to supply the hours firms demand; it is a level parameter that enters no coefficient of the log-linear system (Appendix D.2).

The labor market is therefore not competitive. We read the schedule as standing in for a union that sets one wage for all workers as a markup over the marginal rate of substitution at average consumption, and firms choose employment at that wage. Both household types supply the hours firms demand as long as the wage is at least their own marginal rate of substitution, $(C_t^h)^\sigma N_t^\varphi$. This participation condition is tighter for rich households, whose consumption exceeds the average, and reads $\mathcal{M}^w \geq H_t^\sigma$, with $H_t \equiv C_t^R / C_t$ the rich household's consumption relative to the average. Because H_t exceeds one, the bound H_t^σ rises with the inverse intertemporal elasticity of substitution σ, so a larger σ requires a larger wage markup. As Debortoli and Galí (2024) do, we impose the participation condition at the steady state and for shocks that are not too large. We do not calibrate $\mathcal{M}^w$, because it scales only the levels of hours and output. At the baseline steady state the participation bound is 1.235, and if no profits are rebated to the poor it rises to 1.342.[5]

2.2 Firms, Demand Composition, and Markups

A continuum of monopolistically competitive firms produces differentiated goods using the technology

$$Y_t(i) = A_t N_t(i)^{1-\alpha}, \tag{4}$$

where A_t follows $\log A_t = \rho_a \log A_{t-1} + \varepsilon_t^a$ and $\alpha \in [0, 1)$ governs decreasing returns to labor. The calibration sets $\alpha = 1/3$, whereas Debortoli and Galí (2018) have a linear production function. Firms face Rotemberg quadratic price adjustment costs: changing the price from $P_{t-1}(i)$ to $P_t(i)$ costs $(\zeta/2)\left[P_t(i)/P_{t-1}(i) - 1\right]^2 Y_t$. These costs are paid in

[5] Appendix D.8.1 treats an alternative closure in which the wage equals the rich household's marginal rate of substitution, $W_t/P_t = (C_t^R)^\sigma N_t^\varphi$, and hand-to-mouth households are rationed at the common hours. It is the schedule (3) with the time-varying wage markup $\mathcal{M}_t^w = H_t^\sigma$. Real marginal cost then depends on the distribution, the slope of the Phillips curve gains a further factor below one, and the composition channel is unchanged.

the same composites of varieties that households consume.

Aggregate demand for variety i reflects the two household types:

$$Y_t(i) = \sum_h s_h \left(\frac{P_t(i)}{P_t^h} \right)^{-\varepsilon_h} Y_t^h, \tag{5}$$

where $s_h \in \{1-\lambda, \lambda\}$ is the population share of type h. We call $H_t = C_t^R / C_t$, the ratio of rich to average consumption defined above, the heterogeneity index; it also equals Y_t^R / Y_t, and a higher H_t means more consumption inequality. Rich households then account for the expenditure share $(1-\lambda)H_t$ and, by market clearing, $H_t^P = [1-(1-\lambda)H_t]/\lambda$. The effective (expenditure-weighted) elasticity faced by a firm in a symmetric allocation is then

$$\varepsilon_{\text{eff},t} = (1-\lambda)H_t\varepsilon_R + \lambda H_t^P \varepsilon_P = H_t(1-\lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P. \tag{6}$$

This object carries demand composition into price setting: a higher H_t shifts demand toward the less price-sensitive rich, so the elasticity firms face falls. The firm chooses $P_t(i)$ to maximize the present value of profits subject to (5). Appendix D.5 states the full problem. In a symmetric equilibrium all firms choose the same price, and because firms do not price discriminate across household types, both types face the same price level, $P_t^R = P_t^P = P_t$.

We treat ε_R and ε_P as fixed within type and calibrate their between-type gap using Auer et al. (2024). The fixed gap stands in for a demand system in which the elasticity is derived from preferences, such as Mongey and Waugh (2025), where elasticities vary endogenously with wealth; it keeps $\varepsilon_{\text{eff},t}$ moving only with the composition of demand H_t.[6]

Optimal markup Under flexible prices ($\zeta = 0$), the first-order condition yields the desired markup

$$\mathcal{M}_t = \frac{\varepsilon_{\text{eff},t}}{\varepsilon_{\text{eff},t} - 1} = \frac{H_t(1-\lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P}{H_t(1-\lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P - 1}. \tag{7}$$

When rich households consume more relative to the average (higher H_t) and are less price-sensitive ($\varepsilon_R < \varepsilon_P$), firms optimally charge higher markups. Appendix D.5.1 gives the derivation.

[6] If elasticities also vary with consumption within type, the channel scales by $1-\gamma$, where γ is the cross-type difference in elasticity-consumption gradients relative to the between-type gap. Against a markup-matched benchmark with the same average within-type gradient, the channel keeps its sign for $\gamma < 1$ and reverses for $\gamma > 1$. Appendix D.13 derives the scaling and reports the pass-through gap under assumed values of γ, which the cross-sectional evidence does not identify.

The desired markup (7) solves the first-order condition, and one maintained assumption on pricing makes it optimal: at every steady state considered, the symmetric price is each firm's global profit maximizer when all other firms charge it. With an elasticity gap this is not automatic, because a higher price loses mostly the elastic poor customers. Appendix D.1 gives a sufficient condition and verifies global optimality numerically at every calibration in the paper; Appendix D.9 shows how it fails once the elasticity gap is wide enough.

2.3 Fiscal and Monetary Policy

Fiscal policy The government collects a fraction δ of firm profits and redistributes the revenue across household types according to the transfer-tilt parameter $\tau \in [0,1]$, with $\tau = 0$ the most progressive scheme, equal per-capita transfers. This split of profits is the liquid–illiquid structure of Debortoli and Galí (2018): the fraction $1-\delta$ is paid as dividends on tradable shares, which only rich households hold, and the illiquid fraction δ is allocated by their transfer rule, which we read as fiscal policy. The government budget balances each period, and Appendix D.2 gives the details. Only one combination of the two fiscal parameters matters for allocations, inflation and the interest rate,

$$R \equiv \delta(1-\tau). \tag{8}$$

Eliminating transfers and the value of shares from the budget constraints leaves δ and τ in each household's disposable resources only through R: at a given R, a change in (δ, τ) shifts rich income between dividends and transfers but leaves the resources of both types unchanged. Fiscal pairs with the same R therefore generate the same steady state and the same first-order dynamics. The share price depends on δ separately, but it enters no equilibrium condition other than its own pricing equation (Appendix D.10.1).

The parameter R measures how much profit income is rebated according to population shares. Every steady-state object and every coefficient of the log-linear system, $\left(\bar{H}, \bar{\mathcal{M}}, \delta_2, \kappa_y, \kappa_h, \Gamma, \chi_y, \kappa\right)$, which Sections 2.4 and 2.5 define, is therefore a function of R alone given the non-fiscal parameters. At $R = 0$, poor households receive none; at $R = 1$, profits are fully collected and returned as equal lump-sum transfers, which reproduces the allocation of the representative-agent New Keynesian model (RANK). We call the set of steady states indexed by $R \in [0,1]$ the fiscal branch. The baseline calibration $(\delta, \tau) = (1, 0.7)$ gives $R = 0.30$, a free fiscal parameter that we do not match to an observed tax and transfer system, and sets the hand-to-mouth share to $\lambda = 0.40$, at the upper end of the available estimates (Appendix C.2).

Monetary policy The central bank follows a Taylor rule:

$$i_t = \rho + \phi_\pi \pi_t + \phi_y \hat{y}_t + v_t, \tag{9}$$

where ρ is the steady-state real rate, ϕ_π and ϕ_y are the policy response coefficients, and v_t is a monetary policy shock. We impose the Taylor principle, $\phi_\pi > 1$, a non-negative output response, $\phi_y \geq 0$, and a positive effective IS elasticity in (14) below, $\chi_y < 1$. A sufficient condition in primitives is $\lambda(1-R)[\varphi + \alpha + \sigma(1-\alpha)] \leq 1-\lambda$, which at the baseline calibration holds for $R \geq 0.25$; below that point $\chi_y < 1$ still holds, with a maximum of 0.725 at $R = 0$. For $R < 0.25$ admissibility therefore rests on a numerical check: along the fiscal branch, χ_y reaches one only for hand-to-mouth shares of $\lambda = 0.50$ and above. Under these three conditions the equilibrium is determinate. Given $\chi_y < 1$, the demand-composition feedback cannot overturn determinacy, because the slope stays strictly positive (Proposition 2). The feedback can, however, violate $\chi_y < 1$ itself where the sufficient condition fails: at $\lambda = 0.8$ and $R = 0$, $\chi_y = 2.12$ and the equilibrium is indeterminate under the calibrated rule. Appendix D.10.5 gives the argument. In the alternative closure of Appendix D.8.1, χ_y reaches one on the branch only from $\lambda = 0.66$, against $\lambda = 0.50$ under the wage schedule, so the schedule narrows the admissible region.

2.4 Steady State

The steady state has a recursive structure. Define the composite parameters

$$A = (1-\lambda)(\varepsilon_R - \varepsilon_P), \qquad B = \varepsilon_P, \qquad C = \frac{1-\lambda R}{1-\lambda}, \qquad D = \frac{(1-\alpha)\lambda(1-R)}{1-\lambda}. \tag{10}$$

The heterogeneity index and the flex-price markup are jointly determined by

$$\bar{H} = C - \frac{D}{\bar{\mathcal{M}}}, \tag{11}$$

$$\bar{\mathcal{M}} = \frac{A\bar{H} + B}{A\bar{H} + B - 1}. \tag{12}$$

The first equation follows from the rich household's budget constraint: C captures the concentration of profit-related income among rich households, and $D/\bar{\mathcal{M}}$ the equalizing effect of common labor income. The second is (7) in steady state. The elasticity gap enters only through A, and fiscal policy only through R, via C and D.

The fixed point is a feedback between redistribution, inequality, and market power. A decline in R shifts profit income toward rich households and raises $\bar{H}$. Because rich households are less price-sensitive, the effective demand elasticity falls and the desired

markup rises. The higher markup shifts income from wages toward profits and raises inequality further. Substituting (12) into (11) reduces the fixed point to a quadratic in $\bar{H}$, and output then follows recursively; Appendices D.4 and D.6 give the algebra and state the quadratic.

Proposition 1 (Existence, Uniqueness, and Fiscal Comparative Statics)**.**

(i) ***Existence and uniqueness.*** *Let $\lambda \in (0,1)$, $\alpha \in [0,1)$, and $\varepsilon_P \geq \varepsilon_R > 1$, as in Appendix D.1. For every fiscal pair $(\delta, \tau) \in [0,1]^2$, the steady-state system (11)–(12) has exactly one solution with strictly positive consumption for both household types. Under the maintained pricing assumption of Section 2.2, for which Appendix D.1 gives a sufficient condition, the symmetric price supporting this solution is each firm's global profit maximizer, so the solution is the unique symmetric zero-inflation steady state. If $R < 1$, this solution satisfies $\bar{H} \in \big(1, 1/(1-\lambda)\big)$ and $\bar{\mathcal{M}} > 1$; if $R = 1$, it satisfies $\bar{H} = 1$.*

(ii) ***Fiscal comparative statics.*** *Along this branch, $\bar{H}$ is strictly decreasing in R. If $\varepsilon_P > \varepsilon_R$, the steady-state markup is strictly increasing in $\bar{H}$ and therefore strictly decreasing in R. A higher transfer share to poor households thus lowers both steady-state inequality and the desired markup.*

Proof: Appendix D.10.2.

The model nests both benchmark economies. When $\varepsilon_R = \varepsilon_P = \varepsilon$, it reduces to common-elasticity TANK with fixed desired markup $\varepsilon/(\varepsilon - 1)$; that economy can retain consumption inequality when $R < 1$, but its markup does not respond to it. At $R = 1$, $\bar{H} = 1$. TE-TANK then has the same steady state as RANK with the population-weighted elasticity $\bar{\varepsilon} = (1-\lambda)\varepsilon_R + \lambda\varepsilon_P$, and under Rotemberg pricing the same first-order dynamics. Appendix D.10.2 proves this equivalence. In this Rotemberg economy, elasticity dispersion therefore has aggregate effects only when consumption is unequal; under Calvo pricing the firm-level curvature of Section 3.3 survives at $R = 1$.

2.5 Log-Linear Equilibrium

Lowercase letters denote logarithms, $\hat{x}_t$ denotes the percentage deviation of x_t from its steady-state value, $\hat{\imath}_t \equiv i_t - \rho$ is the deviation of the nominal rate from its zero-inflation steady state in percentage points, and $\tilde{y}_t \equiv \log(Y_t/Y_t^n)$ is the output gap relative to flexible-price output Y_t^n. The log-linear equilibrium consists of a distributional relation, a dynamic IS curve, a New Keynesian Phillips curve, and the Taylor rule (9). The distributional relation ties the heterogeneity index to the output gap period by period, so substituting it out returns the familiar three-equation system.

Distribution and aggregate demand Log-linearizing the rich household's budget constraint, normalized by aggregate output, gives

$$\hat{h}_t = -\chi_y \, \tilde{y}_t, \qquad \chi_y = \Gamma\left(\frac{\varphi+\alpha}{1-\alpha}+\sigma\right), \qquad \Gamma = \frac{(1-\alpha)\lambda(1-R)}{(1-\lambda)\,\bar{\mathcal{M}}\bar{H}}. \tag{13}$$

Appendix D.4 gives the derivation. When the output gap is positive, labor income rises relative to profit income and inequality falls. The composite Γ scales the demand-composition feedback and vanishes in the RANK limit $R = 1$; χ_y turns it into the loading of inequality on the output gap. At the baseline calibration, $\Gamma = 0.188$ and $\chi_y = 0.5637$: a one percent positive output gap lowers the heterogeneity index by 0.56 percent.

Since $\text{sign}(\chi_y) = \text{sign}(1-R)$, inequality moves opposite to the output gap for any disturbance whenever $R < 1$. Demand shocks move output and the gap in the same direction, so inequality falls in a demand-driven boom; a cost-push shock lowers both, so inequality rises as output falls. A positive technology shock raises potential output, and under the calibrated rule actual output rises by less, so the gap turns negative and inequality rises with output.

Dynamic IS curve The log-linearized Euler equation of rich households, combined with market clearing and the hand-to-mouth constraint, gives

$$\tilde{y}_t = \mathbb{E}_t[\tilde{y}_{t+1}] - \frac{1}{\sigma(1-\chi_y)}\left(\hat{\imath}_t - \mathbb{E}_t[\pi_{t+1}] - r_t^n\right), \tag{14}$$

where $r_t^n = (1-\rho_z)z_t + \sigma\mathbb{E}_t[\Delta\hat{y}_{t+1}^n]$ is the natural real rate. As in common-elasticity TANK (Campbell and Mankiw, 1989; Bilbiie, 2008), hand-to-mouth households create a New Keynesian cross (Bilbiie, 2025): a lower real rate raises Ricardian consumption and labor demand, hand-to-mouth households spend the extra wage income, and the IS semi-elasticity rises from $1/\sigma$ to $1/[\sigma(1-\chi_y)]$. In the RANK limit, $\chi_y = 0$ and (14) is the textbook IS curve. Appendix D.7 gives the derivation.

New Keynesian Phillips curve Appendix D.8 log-linearizes the firm's first-order condition around the zero-inflation steady state, which yields

$$\pi_t = \beta\,\mathbb{E}_t[\pi_{t+1}] + \kappa_h\,\hat{h}_t + \kappa_y\,\tilde{y}_t + u_t, \tag{15}$$

with

$$\kappa_y = \delta_2 G, \qquad G \equiv \frac{\varphi + \alpha}{1 - \alpha} + \sigma, \tag{16}$$

$$\kappa_h = \frac{(1-\lambda)\bar{H}}{\zeta}\big(1 - \overline{MC}\big)(\varepsilon_P - \varepsilon_R), \tag{17}$$

$$\delta_2 = \frac{1}{\zeta}[\bar{H}(1-\lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P - 1] = \frac{\varepsilon_{\text{eff}}(\bar{H}) - 1}{\zeta}, \tag{18}$$

where $\overline{MC} = 1/\bar{\mathcal{M}}$ is steady-state real marginal cost, δ_2 is the marginal-cost loading, the response of inflation to a one percent rise in real marginal cost, and u_t is an exogenous cost-push wedge, which Section 2.6 specifies. The coefficient κ_y is the output-gap loading with the demand-composition feedback switched off: the conventional marginal-cost block, evaluated at the TE-TANK steady state. Real marginal cost is the real wage over the marginal product of labor, and under the wage schedule of Section 2.1 the wage responds to average consumption and hours, so $\widehat{mc}_t = G\tilde{y}_t$ as in RANK: the distribution does not enter marginal cost, and with a common elasticity the Phillips curve is the RANK Phillips curve whatever the distribution (Debortoli and Galí, 2018).

The coefficient κ_h is the composition channel, present only when $\varepsilon_P \neq \varepsilon_R$. A boom tilts spending toward the more price-elastic poor, so $\varepsilon_{\text{eff},t}$ rises and the desired markup falls. The channel changes what firms charge, through who buys the goods, not what they pay. Since $1 - \overline{MC} = 1/\varepsilon_{\text{eff}}(\bar{H})$, κ_h is δ_2 times an elasticity:

$$\begin{aligned} \eta_\mu &\equiv \left.\frac{\partial \log \mathcal{M}_t}{\partial \log H_t}\right|_{\bar{H}} = \frac{(1-\lambda)\bar{H}(\varepsilon_P - \varepsilon_R)}{\varepsilon_{\text{eff}}(\bar{H})\big[\varepsilon_{\text{eff}}(\bar{H}) - 1\big]} \geq 0, \\ \kappa_h &= \delta_2 \eta_\mu = \frac{(1-\lambda)\bar{H}}{\zeta}\big(1 - \overline{MC}\big)(\varepsilon_P - \varepsilon_R), \end{aligned} \tag{19}$$

where η_μ is the steady-state elasticity of the desired markup with respect to the heterogeneity index, strictly positive if and only if $\varepsilon_P > \varepsilon_R$. By (13), a one percent positive output gap lowers the heterogeneity index by χ_y percent, and each one percent fall in the index lowers the desired markup by η_μ percent. The lower desired markup partly offsets the higher marginal cost in the boom, so inflation responds less to output. Under a common elasticity, $\eta_\mu = 0$ and $\kappa_h = 0$.

Reduced-form slope Substituting the distributional relation (13) gives the reduced-form NKPC with slope $\kappa \equiv \kappa_y - \kappa_h \chi_y$:

$$\pi_t = \beta\, \mathbb{E}_t[\pi_{t+1}] + \kappa\, \tilde{y}_t + u_t. \tag{20}$$

Substituting $\chi_y = \Gamma G$ and $\kappa_h = \delta_2 \eta_\mu$, the slope factors exactly:

$$\kappa = \kappa_y - \kappa_h \chi_y = \delta_2 G \cdot \left(1 - \Gamma \eta_\mu\right). \tag{21}$$

The first factor is the feedback-off slope κ_y and the second the composition channel, which equals one under a common elasticity. Setting $\hat{h}_t \equiv 0$ shuts the demand-composition feedback off and returns the slope to κ_y; so does a common elasticity, for any distribution of consumption, because with $\eta_\mu = 0$ the slope is the RANK slope $(\varepsilon - 1)G/\zeta$ at every R, the invariance result of Debortoli and Galí (2018). The composition channel is a real rigidity in the sense of Ball and Romer (1990): it makes the price a firm wants to set respond less to real activity, so a given nominal rigidity yields a flatter Phillips curve.

The composition channel is a countercyclical desired markup generated by demand composition: in a boom the labor share rises, inequality falls, and spending shifts toward the price-elastic poor, so the desired markup falls and desired prices respond less to real activity. Rotemberg and Woodford (1992, p. 1195) discuss this explanation of countercyclical markups and object that it would be a coincidence if every shock that raises demand also shifted it toward more elastic customers. In TE-TANK the shift is not a coincidence: by (13) inequality depends on the output gap alone, so every shock that puts output above potential raises the labor share and tilts spending toward the elastic poor, whatever its source. It is a supply-side object and differs from the New Keynesian cross $1/[\sigma(1 - \chi_y)]$ in (14).

Proposition 2 (Endogenous Flattening of the Phillips Curve). *Let $\sigma > 0$, $\varphi \geq 0$, $\alpha \in [0, 1)$, $\zeta > 0$, $\lambda \in (0, 1)$, $\varepsilon_P \geq \varepsilon_R > 1$, and $(\delta, \tau) \in [0, 1]^2$, so that $R \in [0, 1]$, and let the real wage follow the schedule (3). Then the reduced-form slope κ in (20) satisfies the following.*

(i) ***Factorization.*** *The slope factors as in (21), into the feedback-off slope $\kappa_y = \delta_2 G > 0$ and the composition channel $1 - \Gamma \eta_\mu$.*

(ii) ***Positive slope.*** *$0 < 1 - \Gamma \eta_\mu \leq 1$, hence*

$$\kappa \in (0, \kappa_y]. \tag{22}$$

The reduced-form slope is strictly positive everywhere on the fiscal square $(\delta, \tau) \in [0, 1]^2$.

(iii) ***Flattening.*** *$\kappa_h = \delta_2 \eta_\mu \geq 0$, with equality if and only if $\varepsilon_P = \varepsilon_R$, and*

$$\kappa < \kappa_y \iff \kappa_h \chi_y > 0 \iff R < 1 \text{ and } \varepsilon_P > \varepsilon_R. \tag{23}$$

The demand-composition feedback therefore flattens the Phillips curve relative to the same steady state with $\hat{h}_t \equiv 0$ exactly when consumption is unequal and the elasticities differ,

without ever reversing the sign of the slope. With a common elasticity, $\kappa = \kappa_y = (\varepsilon - 1)G/\zeta$ at every point of the fiscal square.

Proof: Appendix D.10.3.

The composition channel lies in $(0, 1]$, so it lowers the slope. This comparison holds the steady state fixed and removes only the demand-composition feedback; it is not a comparison with common-elasticity TANK. Appendix D.1 states which of the parameter restrictions bind.

2.6 Benchmark Comparisons and Cost-Push Transmission

Three benchmarks isolate different parts of the mechanism. The *feedback-off* benchmark is TE-TANK itself with $\hat{h}_t \equiv 0$: it keeps $(\bar{H}, \bar{\mathcal{M}}, \delta_2)$ and has slope κ_y. The *mean-matched* TANK replaces the two elasticities by their population mean $\bar{\varepsilon}$. Its steady-state markup is lower, so a comparison with it mixes dispersion with a markup-level effect. The *markup-matched* TANK(ε^*) uses the single elasticity whose markup $\varepsilon^*/(\varepsilon^* - 1)$ equals $\bar{\mathcal{M}}$. It then also shares $\bar{H}$, Γ, χ_y and, under Rotemberg pricing, δ_2, has $\eta_\mu = 0$, and differs from TE-TANK only in κ_h; its slope is the feedback-off slope κ_y, which is also the RANK(ε^*) slope. By (21),

$$\kappa^{\mathrm{TE}} = \kappa^{\mathrm{TANK}^*}\left(1 - \Gamma\eta_\mu\right), \tag{24}$$

so the composition channel $1 - \Gamma\eta_\mu$ is the entire difference in slopes that is due to elasticity dispersion alone, holding the markup level fixed. We call such a comparison dispersion-pure.

The two common-elasticity benchmarks give different magnitudes for the same mechanism. At the baseline calibration, cost-push inflation pass-through in TE-TANK is 3.3 percent higher than in markup-matched TANK($\varepsilon^* = 3.93$) and 17.9 percent higher than in mean-matched TANK($\bar{\varepsilon} = 4.44$). The larger number adds a markup-level effect. At a given mean elasticity, the rich account for more than their population share of spending, so dispersion lowers the effective elasticity and raises the steady-state markup; the higher markup flattens the curve through δ_2, and this step alone raises pass-through by 14.1 percent. This markup-level step depends on how the cost-push wedge is normalized, and Appendix C.3 shows that under an alternative normalization the mean-matched gap shrinks to 0.6 percent. The markup-matched gap is invariant to the normalization under Rotemberg pricing with a common adjustment cost ζ, because the two economies then share δ_2. Its sign follows from Proposition 2 and the pass-through solution (25) below: the two economies share the demand block, so only the slope differs. We therefore quote it as the headline magnitude of the Rotemberg

model; Section 3.3 shows that under Calvo pricing the two economies no longer share δ_2 and the net comparison changes sign.

IS amplification Against the mean-matched benchmark, dispersion also changes the IS amplification $1/(1-\chi_y)$ through the steady state. With strict dispersion, $\varepsilon_P > \varepsilon_R$, unequal consumption, $R < 1$, and $\chi_y^{\text{TANK}} < 1$, TE-TANK has a higher heterogeneity index and a higher markup than mean-matched TANK($\bar{\varepsilon}$), $\bar{H}^{\text{TE}} > \bar{H}^{\text{TANK}}$ and $\bar{\mathcal{M}}^{\text{TE}} > \bar{\mathcal{M}}^{\text{TANK}}$. At any $\bar{H} > 1$ its effective elasticity lies below $\bar{\varepsilon}$, so its markup is higher, and the higher markup raises $\bar{H}$ further through the fixed point (11)–(12). Because $\bar{\mathcal{M}}\bar{H}$ is the denominator of Γ in (13), TE-TANK then has the smaller loading, $\chi_y^{\text{TE}} < \chi_y^{\text{TANK}}$, and an IS curve that is more rate-sensitive than RANK's but less than the benchmark's, $1 < 1/(1-\chi_y^{\text{TE}}) < 1/(1-\chi_y^{\text{TANK}})$ (Appendix D.10.4). The ordering is therefore a markup-level effect. Markup-matched TANK(ε^*) shares $\bar{\mathcal{M}}\bar{H}$ and hence χ_y, so against it the elasticity gap acts only through the slope of the Phillips curve.

Cost-push transmission The cost-push wedge u_t in (15) follows an AR(1), $u_t = \rho_u u_{t-1} + \varepsilon_t^u$, with cost-push shock ε_t^u; it shifts marginal cost independently of the output gap.[7] We report pass-through in two units. Under the wedge normalization the shock is a unit cost-push wedge u_t. Under the structural normalization it is a unit structural shifter $\hat{u}_t$ of real marginal cost or of the log desired markup, which enters the Phillips curve as $u_t = \delta_2 \hat{u}_t$, with δ_2 the marginal-cost loading; a cost shock such as an oil-price increase would enter in this way. Inflation per unit of $\hat{u}_t$ is therefore δ_2 times inflation per unit of u_t, and the two normalizations give the same comparison between economies that share δ_2. For the cost-push results below, maintain $\beta \in (0,1)$, $\sigma > 0$, $\kappa > 0$, $\phi_\pi > 1$, $\phi_y \geq 0$, $\rho_u \in [0,1)$, and $0 \leq \chi_y < 1$, in both economies when two are compared. Then $\tilde{\sigma} \equiv \sigma(1-\chi_y) = \sigma(1-\Gamma G) > 0$, the reciprocal of the IS semi-elasticity in (14).

With the wedge as the only disturbance, $z_t = a_t = v_t = 0$, so that $r_t^n = 0$ and $\hat{y}_t = \tilde{y}_t$, the unique bounded solution of (20), (14) and (9) is proportional to the wedge, $\pi_t = \Psi_\pi u_t$ and $\tilde{y}_t = \Psi_y u_t$ with $\Psi_y = -\Theta\Psi_\pi$, where

$$\Psi_\pi = \frac{1}{(1-\beta\rho_u) + \kappa\Theta} > 0, \qquad \Theta = \frac{\phi_\pi - \rho_u}{\tilde{\sigma}(1-\rho_u) + \phi_y} > 0. \tag{25}$$

The coefficient Θ is the output contraction that the policy rule produces per unit of inflation, and $\kappa\Theta$ is the disinflation that this contraction buys through the Phillips curve. Impact inflation per unit of the wedge therefore falls with the slope κ and rises with

[7] The calibration sets $\rho_u = 0.9$, $\sigma_u = 0.05$; the paper's headline results are TE-TANK/TANK ratios of impact responses to the same unit shock, which are invariant to σ_u.

$\tilde{\sigma}$, that is, as the IS curve becomes less rate-sensitive. Two economies with the same non-elasticity parameters, policy rule and wedge process differ in pass-through only through the product $\kappa\Theta$: TE-TANK has higher pass-through than a common-elasticity benchmark if and only if $\kappa^{\text{TE}}\Theta^{\text{TE}} < \kappa^{\text{TANK}}\Theta^{\text{TANK}}$ (Appendix D.10.5).

A sufficient condition for greater pass-through is $\kappa^{\text{TE}} \leq \kappa^{\text{TANK}}$ and $\chi_y^{\text{TE}} \leq \chi_y^{\text{TANK}}$, with at least one inequality strict, because Θ rises with χ_y through $\tilde{\sigma} = \sigma(1-\chi_y)$. Under Rotemberg pricing, the condition holds structurally against the *markup-matched* benchmark whenever $R < 1$ and $\varepsilon_P > \varepsilon_R$: the two economies share χ_y, and $\kappa^{\text{TE}} = \kappa^{\text{TANK}^*}(1-\Gamma\eta_\mu) < \kappa^{\text{TANK}^*}$. Against the *mean-matched* benchmark it also holds structurally: TE-TANK has the smaller χ_y, and because the common-elasticity slope is $(\bar{\varepsilon}-1)G/\zeta$ at every R while the effective elasticity of TE-TANK lies below $\bar{\varepsilon}$ under the same two conditions, $\kappa^{\text{TE}} < \kappa^{\text{TANK}(\bar{\varepsilon})}$ (Appendix D.10.5). Since $\Psi_y = -\Theta\Psi_\pi$, a positive wedge also generates stagflation, and the output contraction is larger in TE-TANK whenever the pass-through amplification dominates any offsetting difference in Θ.

The economics is the demand-composition feedback in reverse. A cost-push recession opens a negative output gap and raises inequality, so demand shifts toward the less price-sensitive rich and firms raise desired markups on top of the initial cost pressure. A flatter Phillips curve means that a given output contraction buys less disinflation, so more of the wedge stays in prices. When inflation is driven by the output gap instead, as under demand and technology shocks, a flatter curve shrinks the inflation response.

A cross-sectional prediction Under Rotemberg pricing, a more unequal economy raises its markup by more after the same cost-push wedge, the sign that common-elasticity TANK also predicts. Let $\hat{\mu}_t$, the markup gap, be the log deviation of the realized markup from its steady state. The realized markup is price over the marginal cost of labor, the model's only input, so $\hat{\mu}_t$ is minus the log deviation of the labor share (Appendix D.4). We call it the labor-cost markup; it is the model counterpart of the inverse labor share used in Section 4.2. Under a structural shifter of real marginal cost, $\hat{\mu}_t$ is not the markup over total marginal cost, which falls on impact because prices absorb only part of the shifter (Appendix D.8). Across steady states, indexed by inequality $\bar{H}$, which rises as the profit rebate R falls along the fiscal branch, the impact markup response $\partial\hat{\mu}_t/\partial u_t$ is strictly increasing in $\bar{H}$ under an elasticity condition that the calibrated economy satisfies and for any output coefficient in the Taylor rule. Appendix D.11 states the condition, (D.40), and proves this result as Proposition D.1(iii); Section 4.2 compares its sign with the data. In common-elasticity TANK the sign holds per unit of the cost-push wedge and per unit of a structural shifter alike, so it is the prediction the two models share; Appendix B.7 reports its calibrated gradient. TE-

TANK's own gradient per unit of a structural shifter, the unit in which an oil shock enters, depends on the pricing convention. Under Rotemberg pricing it is positive, 9.8 percent from $R = 1$ to $R = 0$, below the benchmark's, because δ_2 falls with inequality and shrinks the wedge that a given shifter creates. Under Calvo pricing it is larger, 21.7 percent. Per unit of the wedge it is 39.8 percent under Rotemberg and 6.8 percent under Calvo pricing.

Divine coincidence Absent the wedge ($u_t = 0$), setting $\pi_t = \tilde{y}_t = 0$ satisfies (20) at all times, so the model keeps the divine coincidence: zero inflation also closes the output gap. Demand and technology shocks leave the flexible-price H_t, and hence the flexible-price desired markup, at its steady-state value, so heterogeneous elasticities create no additional cost-push wedge; they change only the pass-through of u_t, through κ. The steady-state markup level is the distortion; its cyclical response is the rigidity. With sticky prices the rigidity multiplies the slope through the composition channel $1 - \Gamma\eta_\mu$.[8]

3 Quantitative Analysis

3.1 Calibration

Table 1 reports the quarterly calibration, and Appendices C.1 and C.2 give the sources. Its upper panel lists the parameters common to the three economies and its lower panel the model-specific ones. The price adjustment cost ζ is common to the three economies and is set so that the markup-matched benchmark has the same marginal-cost loading δ_2 under Rotemberg pricing as under Calvo pricing with the Calvo parameter $\theta = 0.753$. All quantitative results use this Rotemberg baseline unless stated otherwise; Sections 3.3 and 3.5 report how the main results change under Calvo pricing with a common reset probability. The TANK and RANK columns use the markup-matched $\varepsilon^* = 3.93$, which reproduces the TE-TANK markup $\bar{\mathcal{M}} = 1.34$.

The gradient of the elasticity in income disciplines the sign and order of magnitude of the mechanism, and the elasticity level moves its magnitude. In the TE-TANK column, the demand elasticities follow the rich–poor gradient that Auer et al. (2024) estimate from Swiss household scanner data; their level is a normalization. The gap of 3.6 points is their slope in log income evaluated between annual incomes of 20,000 and 120,000 Swiss francs. Assigning these two incomes to the hand-to-mouth and the

[8]The divine coincidence is a flexible-wage property. Under nominal wage rigidity it breaks for disturbances that move the flexible-wage marginal rate of substitution, which in this model's shock set is only the technology shock.

Ricardian households is an assumption, because the model's types differ by asset access, and their steady-state consumption ratio of 1.905 is much smaller than the ratio of the two incomes. The model's rich-to-poor consumption ratio is below the ratio in the Consumer Expenditure Survey (CEX), which we did not target. Because the survey ranks households by consumption rather than by asset access, this gap does not show that the model understates the channel. Table C.4 in Appendix C.2 shows that, at the estimated elasticity gap, the elasticity level moves the dispersion-pure pass-through gap from 1.7 to 6.1 percent over the published range of aggregate markups. At the baseline elasticities, the gap has a supremum of 6.72 percent over all admissible (λ, R), which no admissible point attains.

TABLE 1: Baseline Calibration

Description	Symbol	TE-TANK	TANK	RANK
Common parameters				
Discount factor	β		0.995	
Inverse intertemporal elasticity of substitution	σ		1.00	
Inverse Frisch elasticity	φ		1.00	
Decreasing returns to labor	α		1/3	
Price adjustment cost	ζ		105.63	
Model-specific parameters				
Share of hand-to-mouth	λ	0.40	0.40	0.00
Share of profits collected	δ	1.00	1.00	1.00
Transfer tilt toward rich	τ	0.70	0.70	0.00
Rich elasticity	ε_R	3.00	3.93	3.93
Poor elasticity	ε_P	6.60	3.93	3.93

Notes: Common parameters are shared across all three economies. TANK and RANK use the markup-matched elasticity $\varepsilon^* = 3.93$, at which $\varepsilon/(\varepsilon - 1)$ equals TE-TANK's steady-state markup $\bar{\mathcal{M}} = 1.34$. RANK is the limit $\lambda = 0$. With the common elasticity of the TANK column, the parameterization with $\lambda = 0.40$ and $R = 1$ is the same economy: at $R = 1$ consumption is equal, $\bar{H} = 1$ by Proposition 1, so the loading χ_y in (13) is zero, and Section 2.3 states that every fiscal pair with $R = 1$ gives the same allocations and dynamics. Section 2.4 shows that TE-TANK at $R = 1$ is instead RANK with the population-weighted elasticity $\bar{\varepsilon} = 4.44$. The wage markup $\mathcal{M}^w$ of (3) is not calibrated: it scales only the levels of hours and output, and participation requires $\mathcal{M}^w \geq 1.235$ at the baseline and $\mathcal{M}^w \geq 1.342$ along the whole fiscal branch. Appendix C.1 reports the full parameterization, including the Taylor rule and shock persistence.

3.2 The Flattening of the Phillips-Curve Slope

Table 2 shows that elasticity dispersion flattens the Phillips curve by 4.34 percent at the baseline calibration, holding the steady state fixed. Its rows apply the two factors of (21) in turn, and its columns give the economies whose slope each row equals, the slope, and its change relative to κ_y. Row 1 is the *feedback-off* slope $\kappa_y = 0.0833$, which is also the slope of RANK(ε^*) and of *markup-matched* TANK(ε^*), because with a common elasticity the distribution does not enter the slope. In row 2, the composition channel $1 - \Gamma\eta_\mu$ lowers the slope to 0.0797 in TE-TANK.

TABLE 2: The Phillips-Curve Slope, Factor by Factor

	Economy	κ	vs. κ_y
Feedback off, $\kappa_y = \delta_2 G$	RANK(ε^*), TANK(ε^*)	0.0833	—
+ composition channel	TE-TANK	0.0797	−4.34%

Notes: The two factors of Proposition 2(i), applied in turn at the baseline calibration of Table 1. Row 1 is the feedback-off slope $\kappa_y = \delta_2 G$, which is also the slope of RANK(ε^*) and of markup-matched TANK(ε^*). Row 2 multiplies by the composition channel $1 - \Gamma\eta_\mu$, which requires $\varepsilon_P \neq \varepsilon_R$. The final column is the reduction relative to κ_y.

The composition channel is the paper's headline object. It vanishes under a common elasticity. Under Rotemberg pricing the markup-matched benchmark shares TE-TANK's demand block and hence Θ, so by (25) the two economies differ in pass-through only through the slope, and the composition channel turns a 4.3 percent flatter Phillips curve into 3.3 percent higher cost-push inflation pass-through. Table C.5 in Appendix C.3 shows that TE-TANK also has a flatter slope than mean-matched TANK($\bar{\varepsilon}$). Its higher steady-state markup and inequality also give it a smaller χ_y (Appendix D.10.4). Both inequalities of the sufficient condition that Section 2.6 derives therefore hold, and the pass-through gap is 17.9 percent, which also contains the markup-level effect.

3.3 Rotemberg versus Calvo Pricing

The composition channel flattens the Phillips curve by the same 4.3 percent under Rotemberg and under Calvo pricing, but the net comparison with the markup-matched benchmark changes sign between the two conventions. Table 3 reports TE-TANK relative to markup-matched TANK(ε^*) at the baseline calibration under both, and a third

column adds Kimball demand curvature within each household type under Calvo pricing. Its upper panel gives the slope and the two pass-through measures, its lower panel the two factors whose product is the slope ratio. The Rotemberg column holds the adjustment cost ζ fixed across the two economies and reproduces the numbers of Sections 2.6 and 3.2, per unit of the cost-push wedge and per unit of a structural shifter alike. The Calvo column holds the reset probability $1-\theta$ fixed instead. There the slope is 3.2 percent higher in TE-TANK, pass-through per unit of the wedge 2.3 percent lower, and pass-through per unit of a structural shifter 5.4 percent higher.

TABLE 3: TE-TANK Relative to Markup-Matched TANK Under Rotemberg and Calvo Pricing

	Rotemberg (common ζ)	Calvo (common θ)	Calvo, Kimball within type ($\varsigma = 1.93$)
Phillips-curve slope κ	-4.3	$+3.2$	$+1.7$
Pass-through per unit wedge, Ψ_π	$+3.3$	-2.3	-1.2
Pass-through per unit real-marginal-cost shifter, $\delta_2\Psi_\pi$	$+3.3$	$+5.4$	$+5.1$
Factors of the slope ratio			
Composition channel, $1-\Gamma\eta_\mu$	-4.3	-4.3	-4.3
Firm-level curvature, Ξ^*/Ξ_{TE}	0.0	$+7.8$	$+6.3$

Notes: Each entry is TE-TANK relative to markup-matched TANK($\varepsilon^* = 3.93$) at the baseline calibration of Table 1, in percent, $100\,(\text{ratio}-1)$. The Rotemberg column holds the price adjustment cost ζ fixed across the two economies, so $\delta_2 = (\varepsilon_{\text{eff}}-1)/\zeta$ is shared; the two Calvo columns hold the Calvo parameter $\theta = 0.753$ fixed, so $\delta_2 = \lambda_\theta/\Xi$ with $\lambda_\theta = (1-\theta)(1-\beta\theta)/\theta$ and the curvature term Ξ of equation (26). The third column adds Kimball demand within each type, with superelasticity $\varsigma = 1.93$ (Beck and Lein, 2020) in both economies, so Ξ gains $\varsigma/(\varepsilon_{\text{eff}}-1)$. In the upper panel, κ is the reduced-form slope, Ψ_π impact inflation per unit of the cost-push wedge u_t, and $\delta_2\Psi_\pi$ impact inflation per unit of a structural shifter $\hat{u}_t = u_t/\delta_2$, the two normalizations that Section 2.6 defines. The lower panel reports the two factors of equation (27), each in percent; their ratios multiply to the slope ratio in the first row. Under Rotemberg pricing the curvature factor equals one by construction.

Under CES demand, the adjustment cost of Rotemberg (1982) and the reset probability of Calvo (1983) give first-order Phillips curves of the same form (Roberts, 1995), and one adjustment cost reproduces each reset probability (Born and Pfeifer, 2020, p. 1018). Ascari and Rossi (2012) show that trend inflation breaks this equivalence. In TE-TANK it breaks for a different reason: a firm's demand mixes two CES demands with different elasticities and is not CES itself. Mrázová and Neary (2017, footnote 18) note that market demand aggregated across two groups with different CES preferences is superconvex, so its elasticity falls as the price rises. This curvature is known, and Olivi et al. (2024) show that the superelasticity of a firm's demand, including the change in the mix of its customers as it moves its own price, enters the Calvo slope. Any nonzero superelasticity therefore breaks the equivalence, because it enters the Calvo slope and not the Rotemberg slope. Our contribution is narrower: the between-type term is negative, and under Calvo pricing it reverses the comparison

with markup-matched TANK.

The difference is a firm-level curvature that enters the slope under Calvo pricing and not under Rotemberg pricing. A firm that raises its own price loses mostly its elastic poor customers, so the demand that remains is less elastic: at the baseline, a one percent increase in the own relative price lowers the elasticity of residual demand by 0.63 percent. The superelasticity of the firm's demand, the percent change in its elasticity per percent change in its relative price, is therefore negative. This is the opposite of the curvature in Kimball (1995), in which the elasticity rises with the own relative price. A rising elasticity makes a firm reluctant to move its price away from the others, a strategic complementarity that flattens the Phillips curve; a falling elasticity makes it willing to move further, which weakens the strategic complementarity and steepens the curve. Under Calvo pricing a resetting firm's price differs from the average price, so the curvature of its own demand enters the reset condition. The marginal-cost loading becomes $\delta_2 = \lambda_\theta / \Xi$, with $\lambda_\theta = (1-\theta)(1-\beta\theta)/\theta$ and

$$\Xi = \frac{1-\alpha+\alpha\varepsilon_{\text{eff}}}{1-\alpha} - \frac{\text{Var}_\omega(\varepsilon)}{\varepsilon_{\text{eff}}(\varepsilon_{\text{eff}}-1)}, \qquad \text{Var}_\omega(\varepsilon) = \omega_R\omega_P(\varepsilon_P-\varepsilon_R)^2, \tag{26}$$

where the first term is the usual curvature of a monopolist with decreasing returns, $\text{Var}_\omega(\varepsilon)$ is the expenditure-weighted variance of the elasticity across the firm's customers, and ω_h are the steady-state expenditure shares. The same variance works against the concavity of profit in Appendix D.5.2. Under Rotemberg pricing all firms charge the same price in every period, so the curvature enters only that second-order condition, and $\delta_2 = (\varepsilon_{\text{eff}}-1)/\zeta$ carries no curvature term. Appendix D.8 gives the Calvo derivation.

The slope ratio against the markup-matched benchmark then factors exactly into the curvature and the composition channel,

$$\frac{\kappa^{\text{TE}}}{\kappa^{\text{TANK}^*}} = \frac{\Xi^*}{\Xi_{\text{TE}}}\left(1-\Gamma\eta_\mu\right), \qquad \Xi^* = \frac{1-\alpha+\alpha\varepsilon^*}{1-\alpha}, \tag{27}$$

where Ξ^* is the curvature of the benchmark, whose customers share one elasticity, and the composition channel $1-\Gamma\eta_\mu$ is the factor of (21). The lower panel of Table 3 reports the two factors. The composition channel is the same -4.3 percent in both columns, because it works through the aggregate state H_t, which neither pricing friction touches. The curvature factor is one under Rotemberg pricing and 1.078 under Calvo pricing, a 7.8 percent steeper curve, because the variance term lowers Ξ_{TE} to 2.751 against $\Xi^* = 2.967$. The product is the $+3.2$ percent in the first row. The same factor scales δ_2, which is why the two pass-through measures separate under Calvo pricing: per unit of the wedge, the steeper curve lets 2.3 percent less through, but a given structural shifter

creates a larger wedge, $u_t = \delta_2 \hat{u}_t$, and inflation per unit of the shifter is 5.4 percent higher. Under Calvo pricing TE-TANK's higher structural pass-through comes from the larger loading δ_2, not from a flatter curve.

The two conventions also differ in what an equal nominal rigidity means. Under Rotemberg pricing it is an equal adjustment cost ζ, and the paper's ζ reproduces the Calvo slope of the common-elasticity benchmark at ε^*, as Appendix C.1 describes. Under Calvo pricing it is an equal reset probability, and the adjustment cost that would give TE-TANK its own Calvo slope is $\zeta = 97.9$, below the calibrated value: the same reset probability corresponds to a steeper curve, and so to a lower adjustment cost. Because the two economies share the effective elasticity, the Rotemberg column holds the marginal-cost loading δ_2 fixed, the Calvo column the frequency of price changes.

The curvature enters under any friction in which a resetting firm's price differs from the average price, so symmetric Rotemberg pricing is the special case. Individual prices change infrequently and by amounts that are large relative to inflation (Dhyne et al., 2006), which neither friction matches literally, and the Rotemberg cost has no direct counterpart in the data (Born and Pfeifer, 2020). Survey data on Swiss manufacturing firms show state-dependent pricing (Lein, 2010). In menu-cost models the friction also selects which firms adjust (Golosov and Lucas Jr., 2007). Auclert et al. (2024) show that to first order these models nonetheless have a Phillips curve that is well approximated by the Calvo form, with the slope of a Calvo model whose frequency exceeds the observed one, so selection steepens the curve without changing its shape. In their framework a real rigidity that makes every firm's marginal cost depend on the aggregate price index, through intermediate inputs, scales the menu-cost Phillips curve by the same factor as the Calvo slope (their Proposition 5, pp. 174–175). The composition channel also shifts every firm's desired price through an aggregate state, H_t, so by the same logic we expect it to flatten a menu-cost Phillips curve by the factor $1 - \Gamma\eta_\mu$, although we do not solve such a model. The firm-level curvature also changes the width of the inaction band for given menu costs (Auclert et al., 2024, footnote 44), so whether it reverses the net comparison under menu costs depends on how such a model is calibrated and lies outside our model.

Two conclusions follow. The composition channel is robust to the pricing convention and to Kimball curvature within each type: under both conventions poorer households' higher elasticity makes the desired markup rise with inequality, the countercyclical desired markup is a real rigidity, and it flattens the Phillips curve by 4.3 percent at the baseline. The net comparison with the markup-matched benchmark is not: whether TE-TANK's curve is flatter or steeper than the benchmark's, and whether a cost-push wedge raises inflation by more or by less, depends on whether the firm-level curvature enters price setting. We keep Rotemberg pricing as the baseline, the conven-

tion of the heterogeneous-household models closest to TE-TANK (Debortoli and Galí, 2018; Kaplan et al., 2018; Bilbiie, 2025), because it gives the slope in closed form with the composition channel isolated.

The curvature factor is a property of the two-CES mixture with a fixed elasticity within each type. Within each type demand is CES, so the only reason a firm's elasticity moves with its own price is the change in the mix of its customers, and the sign of the curvature comes from the between-type variance alone. Beck and Lein (2020) estimate household-level superelasticities in scanner data and find them mostly positive, with a median of 1.93. This is Kimball curvature within each type, of the opposite sign to the between-type term. If each type has superelasticity ς, Ξ gains $\varsigma/(\varepsilon_{\text{eff}} - 1)$ in TE-TANK and in a benchmark with the same within-type curvature, while the variance term is unchanged. Appendix D.9 compares the firm-level superelasticity with other demand systems and with estimates.

The third column of Table 3 sets $\varsigma = 1.93$ in both economies and shows that this curvature shrinks the Calvo reversal without removing it. The composition channel is unchanged at -4.3 percent, and the curvature factor falls to $+6.3$ percent, so TE-TANK's slope is 1.7 percent steeper than the benchmark's. Pass-through is 1.2 percent lower per unit of the wedge and 5.1 percent higher per unit of a structural shifter. The net Calvo comparison changes sign only above a within-type superelasticity of 5.86, outside the bulk of the estimates in Beck and Lein (2020). The between-type variance is the mechanism, and the within-type term only scales it down.

Micro data can settle the sign of the net Calvo comparison, because they can measure how strongly a firm's demand elasticity moves with its own price. Aggregate data do not identify this demand curvature: Smets and Wouters (2007) report that the curvature of their Kimball aggregator is not identified and fix it. In the model, the between-type term lowers the superelasticity of a store's pooled demand below that of its customers' own demand for the same products, by 0.63 at the baseline. At a given average elasticity, a store whose customers differ more in price sensitivity, which the model ties to income through type, therefore has higher own-cost pass-through. Herreño et al. (2026) provide a framework that maps firm-level cost pass-through into the slope of the Phillips curve.

3.4 Impulse Response Functions

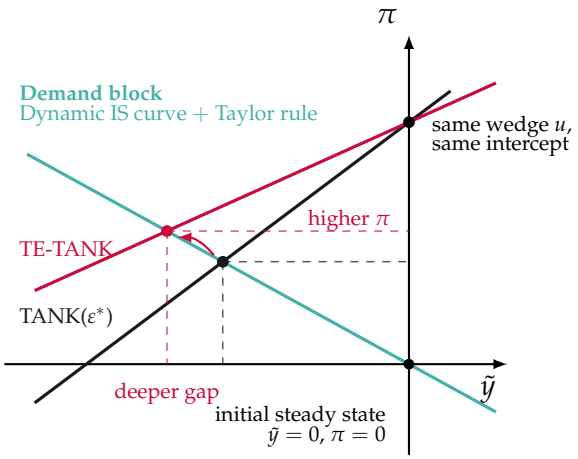
We solve the log-linearized model by first-order perturbation in Dynare. Figure D.2a in Appendix D.12 shows that a positive technology shock raises inequality and desired markups, so the elasticity gap shrinks the fall in inflation; Figure D.2b shows that after an expansionary monetary policy shock inequality falls and the markup gap $\hat{\mu}_t$ with it,

so the gap damps inflation instead.

Cost-push shock Under Rotemberg pricing, heterogeneous elasticities raise inflation and deepen the output loss after a cost-push shock. On impact, the wedge raises annualized TE-TANK inflation by 0.52 percentage points and lowers output by 0.46 percent. Relative to TANK(ε^*), Ψ_π rises from 2.494 to 2.577 and $|\Psi_y|$ from 8.875 to 9.170, the same percentage increase. The two gaps coincide because the two economies share $\bar{H}$, $\bar{\mathcal{M}}$, and χ_y, so the entire first-order difference runs through the slope of the Phillips curve.

Figure 1 shows why the flatter curve leaves more of the wedge in inflation. The horizontal axis is the output gap and the vertical axis inflation, with the steady state at the origin. The teal line is the common demand block. The same wedge shifts the red TE-TANK and the black TANK(ε^*) Phillips curves to a common intercept, and the flatter red curve meets the demand line at higher inflation and a deeper output gap.

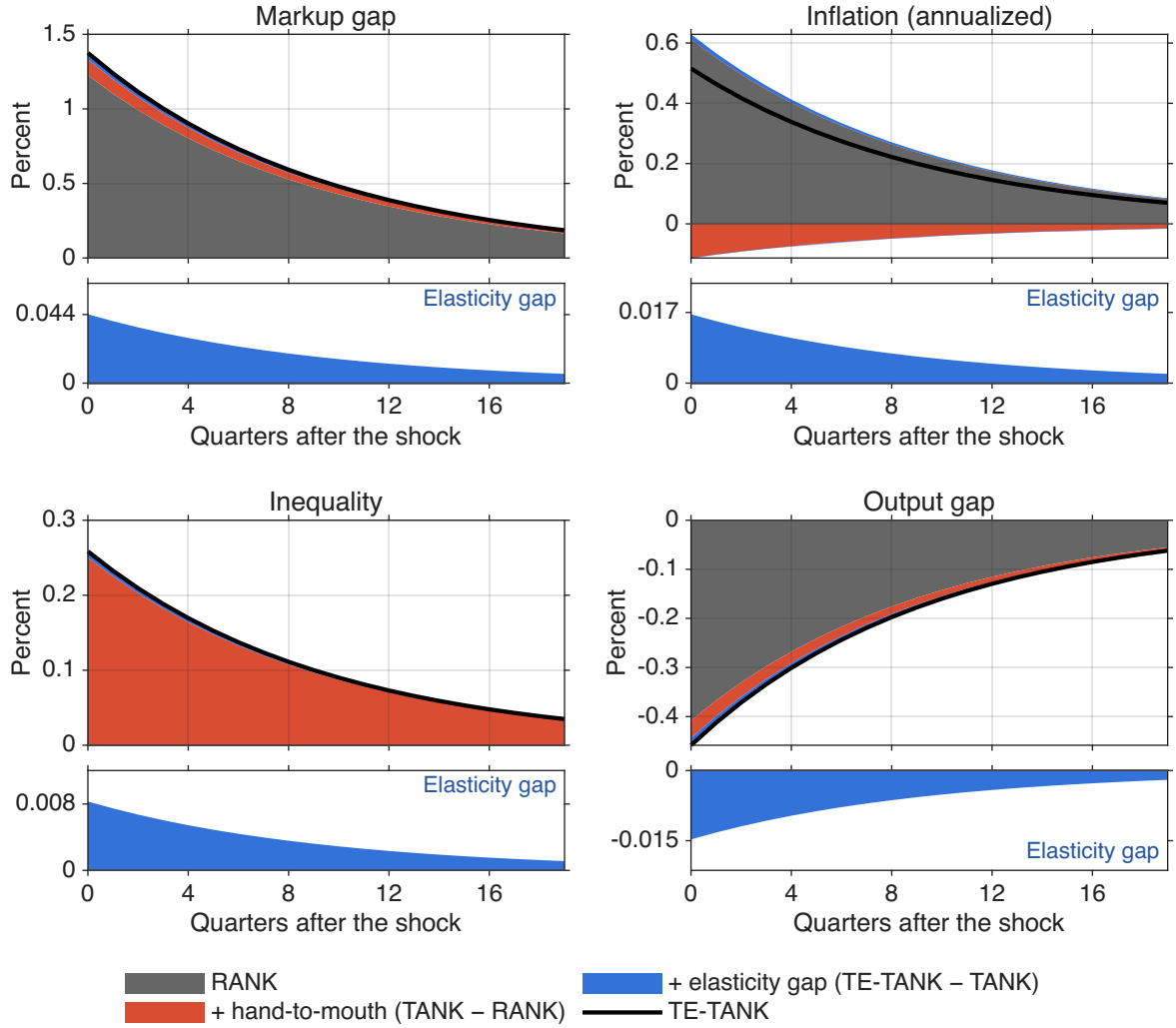
FIGURE 1: Why a Flatter Phillips Curve Raises the Inflationary Cost of a Cost-Push Shock



Notes: Stylized, with the slope difference exaggerated for legibility. The origin is the initial steady state; each economy slides back down the demand locus to the origin as the AR(1) wedge decays. The downward-sloping locus is the demand block, (14) and (9) solved forward, with slope $-1/\Theta$; it is common to the two economies because matching the markup gives them the same χ_y. The upward-sloping curves are (20) solved forward, $\pi = [\kappa/(1-\beta\rho_u)]\tilde{y} + u/(1-\beta\rho_u)$, so only κ differs. Calibrated, $\kappa^{\text{TE}} = 0.0797$ against $\kappa^{\text{TANK}^*} = 0.0833$ delivers $\Psi_\pi = 2.577$ against 2.494. RANK has $\tilde{\sigma} = 1$ and so does not share the locus.

Figure 2 shows that the elasticity gap raises the markup gap and inflation and deepens the fall in the output gap. The top panels plot the markup gap and annualized inflation, the bottom panels inequality $\hat{h}_t$ and the output gap, each in percent against quarters after the shock. The black line is TE-TANK, built from three stacked bands: gray is the RANK response, red adds hand-to-mouth aggregation (TANK minus RANK), and blue adds the elasticity gap (TE-TANK minus TANK). The elasticity-gap band is thin at the scale of the main panels, so the strip under each panel shows it on its own scale; on impact it is the 3.3 percent higher inflation pass-through that Section 3.2 reports, or

FIGURE 2: Cost-Push Shock: Sequential Channel Decomposition at the Markup-Matched Benchmark



Notes: Impulse responses to a one-standard-deviation cost-push shock ε_t^u to the AR(1) cost-push wedge u_t in equation (15) ($\rho_u = 0.9$, $\sigma_u = 0.05$). The **solid black line** is TE-TANK ($\varepsilon_R = 3.0$, $\varepsilon_P = 6.6$). The markup gap $\hat{\mu}_t$ is the labor-cost markup, the model's inverse labor share; the wedge u_t enters the Phillips curve and not marginal cost, so $\hat{\mu}_t$ does not subtract a cost behind the wedge. Three additive bands build it along the path RANK → TANK → TE-TANK: **gray** is the RANK response, **red** the contribution of hand-to-mouth aggregation (TANK minus RANK), and **blue** the contribution of the elasticity gap (TE-TANK minus TANK). Under each panel, a strip shows the elasticity-gap band on its own scale. The accounting is sequential, not a partition into independent channels. Both common-elasticity corners use the markup-matched $\varepsilon^* = 3.93$, so the blue band is the dispersion-pure contribution. Inflation is annualized; other variables are quarterly percentage deviations from steady state.

0.017 percentage points of annualized inflation. The band is also a small part of the impact rise in the markup gap, most of which is the sticky-price response shared with RANK and common-elasticity TANK. It reflects the composition channel in reverse: inequality rises with the negative output gap, demand tilts toward the less price-elastic rich, and firms add inflation on top of the wedge. The labor-cost markup, the model's

inverse labor share, and profits, which in the model are output less labor cost, thus rise during a cost-driven inflation with no change in conduct. If the wedge stands for a shifter of real marginal cost such as an energy price, the markup over total cost falls, because prices absorb only part of the shifter on impact (Appendix D.8). Bilbiie and Känzig (2024) show that in the New Keynesian model with sticky prices only, rising profits and inflation together arise after markup shocks, with a recession, rather than after the demand expansions that the post-2021 "greedflation" narrative invokes; demand shocks raise profits and inflation together only when wages are sticky enough relative to prices. Here the pattern follows from static profit maximization under a shift in customer composition.

RANK has the highest pass-through, $\Psi_\pi = 3.061$. Hand-to-mouth aggregation leaves the slope at its RANK value and makes the IS curve more rate-sensitive, $\tilde{\sigma} = 0.44$ against 1 in RANK, so the same Taylor rule buys more disinflation, and the red band in Figure 2 is negative for inflation.

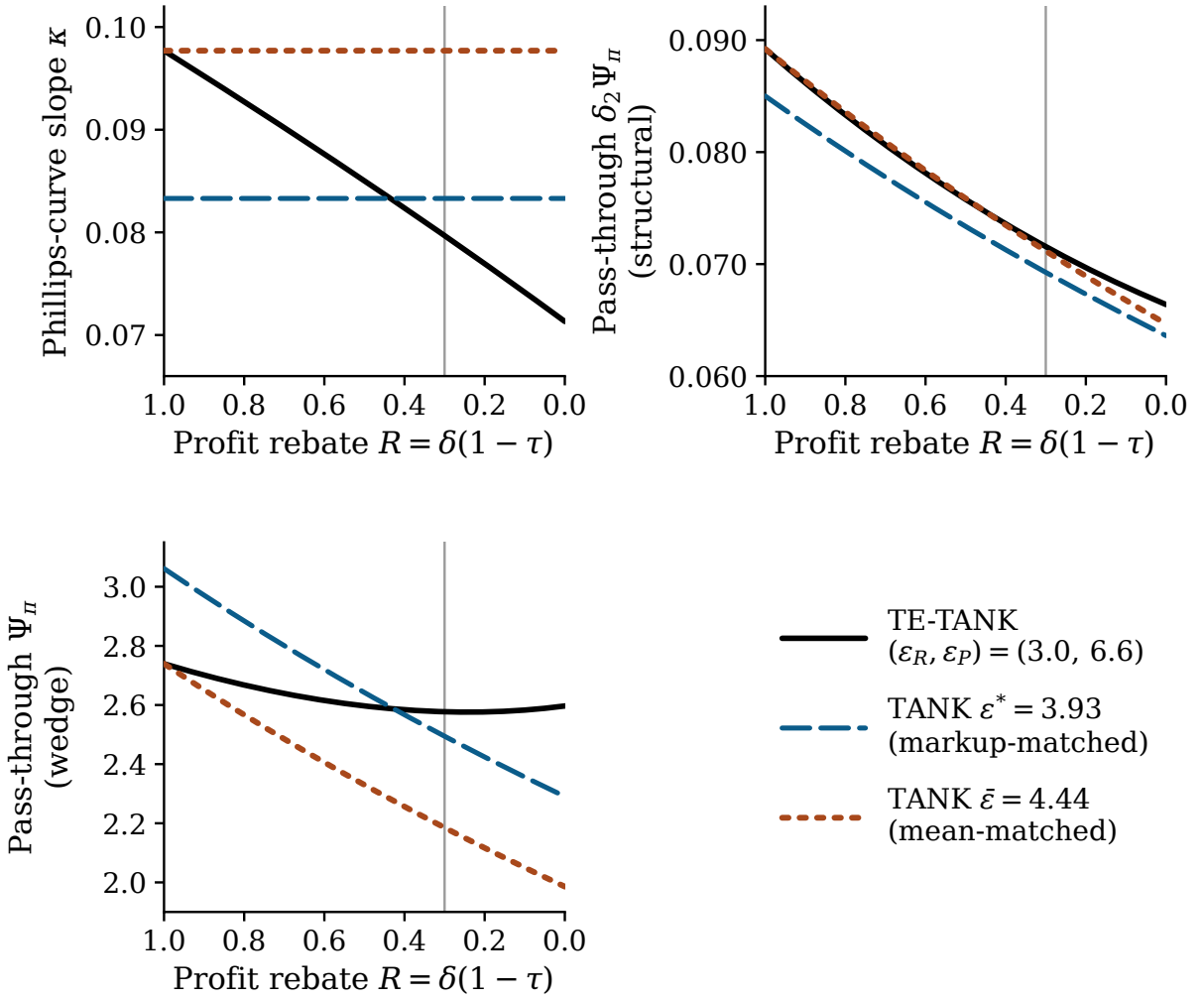
3.5 Fiscal Policy as a Supply-Side Instrument

Profit redistribution moves the Phillips curve in this environment, not only the demand block. We move along the fiscal branch from full rebating ($R = 1$, no steady-state inequality) to none ($R = 0$), holding all other primitives and the policy rule at baseline. A lower rebate raises inequality and, only when elasticities differ, the desired markup; it strengthens the demand-composition feedback, which with a common elasticity acts only through the IS curve. Figure 3 shows that along this branch, under Rotemberg pricing, the curve flattens only in TE-TANK, and that pass-through falls in all three economies. Rebating thus steepens the curve only when elasticities differ: with a common elasticity the slope is the RANK slope at every R, so the whole steepening comes from the elasticity gap. In Mongey and Waugh (2025), by contrast, a one-time transfer raises the aggregate markup, because it lowers the marginal value of wealth and so the price elasticity of poor and middle-income households. Their transfer is lump sum and deficit financed in a small open economy, not a redistribution of profits between types. With one fixed elasticity per type, the rebate here moves only the composition of spending between the two types, and Appendix D.13 shows when that within-household margin would reverse the sign.

In each panel the horizontal axis is R, falling from left to right, and the vertical rule marks the baseline. The top-left panel plots the slope κ. The top-right panel plots impact pass-through per unit of a structural shifter, $\delta_2 \Psi_\pi$, with $\delta_2 = (\varepsilon_{\text{eff}} - 1)/\zeta$ under Rotemberg pricing. The bottom-left panel plots Ψ_π, impact inflation per unit of the cost-push wedge u_t itself. The solid black line is TE-TANK, the dashed blue line

markup-matched TANK, and the dotted orange line *mean-matched* TANK. The slope is constant in the two benchmarks and falls in TE-TANK. Structural pass-through falls along the whole branch in all three economies, not only between its endpoints. Wedge pass-through falls in all three as well, but in TE-TANK by much less and not monotonically: it reaches its minimum at $R = 0.24$ and rises again toward $R = 0$.

FIGURE 3: The Phillips-Curve Slope and Cost-Push Pass-Through Along the Fiscal Branch



Notes: The reduced-form Phillips-curve slope κ (top left), impact cost-push inflation pass-through per unit of a structural shifter, $\delta_2\Psi_\pi$ (top right), and per unit of the cost-push wedge, Ψ_π (bottom left), along the admissible fiscal branch, with the profit rebate $R = \delta(1-\tau)$ falling from full rebating ($R = 1$, no steady-state inequality) to none ($R = 0$). Non-fiscal primitives and the policy rule are held at the baseline calibration. Section 2.6 defines the two normalizations: the structural shifter $\hat{u}_t = u_t/\delta_2$, with $\delta_2 = (\varepsilon_{\text{eff}} - 1)/\zeta$, shifts real marginal cost or the log desired markup. In the two benchmarks δ_2 is constant along the branch, so the two normalizations give the same percentage change. Table 4 reports the change in each line over the whole branch. The vertical rule marks the baseline calibration, $R = 0.30$ ($\delta = 1$ and $\tau = 0.7$). TE-TANK and *mean-matched* TANK coincide at $R = 1$, where the effective elasticity equals the population mean.

Table 4 reports the change between the two endpoints, under Rotemberg pricing in

Panel A and under Calvo pricing, with CES or Kimball demand within each type, in Panels B and C. In Panel A, the first column shows that the slope falls by 27.0 percent in TE-TANK and does not move in *markup-matched* TANK($\varepsilon^* = 3.93$) or *mean-matched* TANK($\bar{\varepsilon} = 4.44$). The last column shows that structural pass-through $\delta_2 \Psi_\pi$ falls by 25.6 percent in TE-TANK and by 25.2 and 27.5 percent in the two benchmarks. Read from $R = 0$ toward $R = 1$, more rebating lowers the desired markup in TE-TANK, weakens the demand-composition feedback in all three, steepens the curve in TE-TANK, and lets more of a given cost shock reach inflation in all three economies. The wedge column shows that per unit of the cost-push wedge u_t, pass-through Ψ_π also falls in all three, by 5.2 percent in TE-TANK and by 25.2 and 27.5 percent in the benchmarks. The output contraction $|\Psi_y| = \Theta \Psi_\pi$ rises in all three economies, because the output loss per unit of inflation Θ rises with inequality, and by most in TE-TANK, in which Ψ_π falls least. In the benchmarks δ_2 is constant along the branch, so their two columns coincide. Proposition D.1 and the benchmark paragraph in Appendix D.11 prove all Panel A signs except those of the TE-TANK pass-through entries and of its markup response per unit of a structural shifter, which are numerical.

The two normalizations differ in TE-TANK because the marginal-cost loading δ_2 enters twice. Two forces damp inflation after a cost shock, and Ψ_π depends on them only through their product: the slope κ, the disinflation per unit of lost output, and Θ, the output contraction the policy rule produces per unit of inflation. A lower rebate raises inequality, which makes demand more rate-sensitive, so Θ rises and Ψ_π falls. In the common-elasticity benchmarks the slope is constant, so this demand-side force is the whole change. In TE-TANK the effective elasticity also falls as spending tilts toward the price-insensitive rich, which lowers both δ_2 and the composition channel $1 - \Gamma \eta_\mu$ and so flattens the curve. This supply-side force works against the demand side: in the log split of $\kappa\Theta$ along the branch, $\Delta \log \delta_2 = -0.242$ and $\Delta \log(1 - \Gamma \eta_\mu) = -0.073$ against $+0.389$ from the demand side, so the supply side is about 0.8 times the demand side. Wedge pass-through Ψ_π therefore falls by much less than in the benchmarks, $\Delta \log \Psi_\pi = -0.053$, and not monotonically. The same fall in δ_2, however, scales down the wedge that a given structural shifter creates, $u_t = \delta_2 \hat{u}_t$, so $\Delta \log(\delta_2 \Psi_\pi) = -0.296$ and structural pass-through falls about as much as in the benchmarks. Holding the wedge fixed while δ_2 falls amounts to raising the primitive cost shock along the branch, so the size of the wedge change is a property of the normalization. The structural normalization is the one under which a cost shock enters, so the structural result is the primary one.

The TE-TANK entries depend on the pricing convention; with CES demand within each type, the markup-matched benchmark's entries do not. The mean-matched benchmark's slope stays constant and its pass-through changes little, a fall of 26.0

TABLE 4: Fiscal Redistribution and Cost-Push Transmission Along the Rebate Branch

Economy	% change along the branch, $R{=}1 \to R{=}0$					
	κ	$\partial\hat{\mu}/\partial u$ (wedge)	$\partial\hat{\mu}/\partial\hat{u}$ (structural)	$\lvert\Psi_y\rvert$	Ψ_π (wedge)	$\delta_2\Psi_\pi$ (structural)
Panel A. Rotemberg pricing (common ζ)						
TE-TANK $(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$	-27.0	$+39.8$	$+9.8$	$+39.8$	-5.2	-25.6
TANK$(\varepsilon^* = 3.93)$, markup-matched	$+0.0$	$+11.8$	$+11.8$	$+11.8$	-25.2	-25.2
TANK$(\bar{\varepsilon} = 4.44)$, mean-matched	$+0.0$	$+11.0$	$+11.0$	$+11.0$	-27.5	-27.5
Panel B. Calvo pricing (common θ)						
TE-TANK $(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$	$+6.0$	$+6.8$	$+21.7$	$+6.8$	-27.6	-17.5
TANK$(\varepsilon^* = 3.93)$, markup-matched	$+0.0$	$+11.8$	$+11.8$	$+11.8$	-25.2	-25.2
TANK$(\bar{\varepsilon} = 4.44)$, mean-matched	$+0.0$	$+13.3$	$+13.3$	$+13.3$	-26.0	-26.0
Panel C. Calvo pricing, Kimball demand within type ($\varsigma = 1.93$)						
TE-TANK $(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$	-1.0	$+14.1$	$+21.4$	$+14.1$	-22.7	-17.7
TANK$(\varepsilon^* = 3.93)$, markup-matched	$+0.0$	$+13.7$	$+13.7$	$+13.7$	-23.9	-23.9

Notes: Percentage change in each object between the endpoints of the admissible fiscal branch, from full profit rebating ($R = \delta(1-\tau) = 1$) to none ($R = 0$), holding all non-fiscal primitives and the policy rule at baseline. κ is the reduced-form Phillips-curve slope, $\partial\hat{\mu}/\partial u$ the impact markup response per unit of the cost-push wedge (Proposition D.1(iii)), $\partial\hat{\mu}/\partial\hat{u}$ the same response per unit of a structural shifter, in both cases the response of the labor-cost markup, the model's inverse labor share, and $|\Psi_y|$ the impact output contraction per unit of the wedge. Panel A holds the Rotemberg adjustment cost $\tilde{\zeta}$ fixed along the branch, so $\delta_2 = (\varepsilon_{\text{eff}} - 1)/\zeta$; Panels B and C hold the Calvo parameter fixed at $\theta = 0.753$, so $\delta_2 = \lambda_\theta/\Xi$ (Section 3.3). Demand is CES within each type in Panels A and B; in Panel C each type has Kimball demand with superelasticity $\varsigma = 1.93$, in TE-TANK and in the markup-matched benchmark. Ψ_π is impact inflation per unit of the wedge u and $\delta_2\Psi_\pi$ per unit of the structural shifter $\hat{u} = u/\delta_2$, the two normalizations that Section 2.6 defines. On a 2001-point grid in R, the benchmark slopes are constant, and every other entry is monotone along the branch except two in TE-TANK: wedge pass-through Ψ_π in Panel A, which is trough-shaped with its minimum at $R = 0.24$, and the slope in Panel C.

percent instead of 27.5, but its markup responses rise by 13.3 percent instead of 11.0. Under Calvo pricing with a common reset probability, the loading $\delta_2 = \lambda_\theta/\Xi$ of Section 3.3 rises by 14.0 percent along the branch instead of falling, because the curvature term Ξ falls with inequality, and every TE-TANK entry changes with it. Ξ falls because the effective elasticity falls and with it the decreasing-returns part of Ξ; the variance term is nearly constant along the branch. Panel B of Table 4 shows that the TE-TANK slope then rises, by 6.0 percent, while the benchmark slopes stay constant, and that structural pass-through falls in all three economies, by 17.5 percent in TE-TANK against 25.2 and 26.0 percent in the benchmarks. Wedge pass-through also falls in all three, and the markup responses and the output contraction rise in all three; only the TE-TANK slope changes sign between Panels A and B.

Kimball demand within each type moves every TE-TANK entry part of the way back toward Panel A. With the within-type superelasticity of Beck and Lein (2020), $\varsigma = 1.93$, Ξ gains the term $\varsigma/(\varepsilon_{\text{eff}} - 1)$, which rises as the effective elasticity falls, so δ_2 rises by only 6.4 percent. Panel C of Table 4 shows that TE-TANK's slope then

falls by 1.0 percent, close to the benchmark's constant slope, a near tie whose sign is not robust. Both pass-through measures fall and the markup responses rise in both economies. What survives the change of convention is the steady state, which involves neither ζ nor θ: under both, a lower rebate raises inequality and the desired markup, and it lowers pass-through in all three economies. Whether a lower rebate flattens or steepens TE-TANK's curve is a property of the pricing convention.

Under Rotemberg pricing, the markup response to a cost-push wedge rises with inequality in all three economies, so its sign does not separate TE-TANK from the benchmarks. The second column of Table 4 shows that the impact markup response per unit of the cost-push wedge rises by 39.8 percent in TE-TANK against 11.8 percent in the *markup-matched* benchmark: the same sign and a larger rise in TE-TANK. The third column shows that per unit of a structural shifter the TE-TANK response rises by 9.8 percent, less than the 11.8 percent of the benchmark, because the fall in δ_2 shrinks the wedge that the shifter creates. Pass-through falls in all three economies under both normalizations, so no sign along the branch separates TE-TANK from the benchmarks; the elasticity gap shows in magnitudes, a falling slope and a much smaller fall in wedge pass-through. The markup signs hold for any output coefficient in the Taylor rule, and in the benchmarks so does the pass-through sign; Appendix D.11 derives the benchmark signs.

4 Empirical Patterns

The patterns below are descriptive consistency checks, not tests of the elasticity gap. Most of the model's aggregate implications, including the markup responses to supply shocks, follow from the sticky-price core shared with RANK and common-elasticity TANK. An oil shock enters the model as a structural shifter of real marginal cost, not as a fixed Phillips-curve wedge. Per unit of such a shifter, TE-TANK and *markup-matched* TANK both predict a larger markup response and a smaller cumulative inflation response in more unequal countries, under either pricing convention. The cross-country evidence therefore speaks to the feedback from output to inequality that the two models share, not to the elasticity gap. Appendix B states the local-projection specification, identified shocks, and markup proxies, and its Table B.1 collects the full sign record.

4.1 Markup Levels and Inequality

Figure 4 shows that economies with higher long-run income concentration have higher average markup levels. Each point is one country. The horizontal axis is the country-mean post-tax top-10% income share from the World Inequality Database (WID), and

the vertical axis is the markup proxy, the inverse of the labor share from the Penn World Table (PWT) (Feenstra et al., 2015); the line is the OLS fit with its 95% band. With the income share measured as a fraction, the slope of the fitted line is 2.009 (heteroskedasticity-robust SE 0.450, $N = 138$); with the labor share from ILOSTAT, the statistics database of the International Labour Organization (ILO) (International Labour Organization, 2026), it is 2.245 (SE 0.347, $N = 185$). The inverse labor share has a long right tail, and without the 19 countries with a mean above 2.5, among them oil and gas exporters and offshore financial centers, the slope of a regression of the log markup on the income share falls from 0.829 (SE 0.172) to 0.351 (SE 0.110): the sign survives, but the magnitude falls to 0.42 of its full-sample value. The production-approach markup, the only measure not based on the labor share, gives a positive slope that rests on the 7 Latin American countries in its sample: without them it falls from 0.475 (SE 0.160) to -0.014 (SE 0.209). Table A.1 in Appendix A reports the log specification for all three markup sources and the within-country estimates.

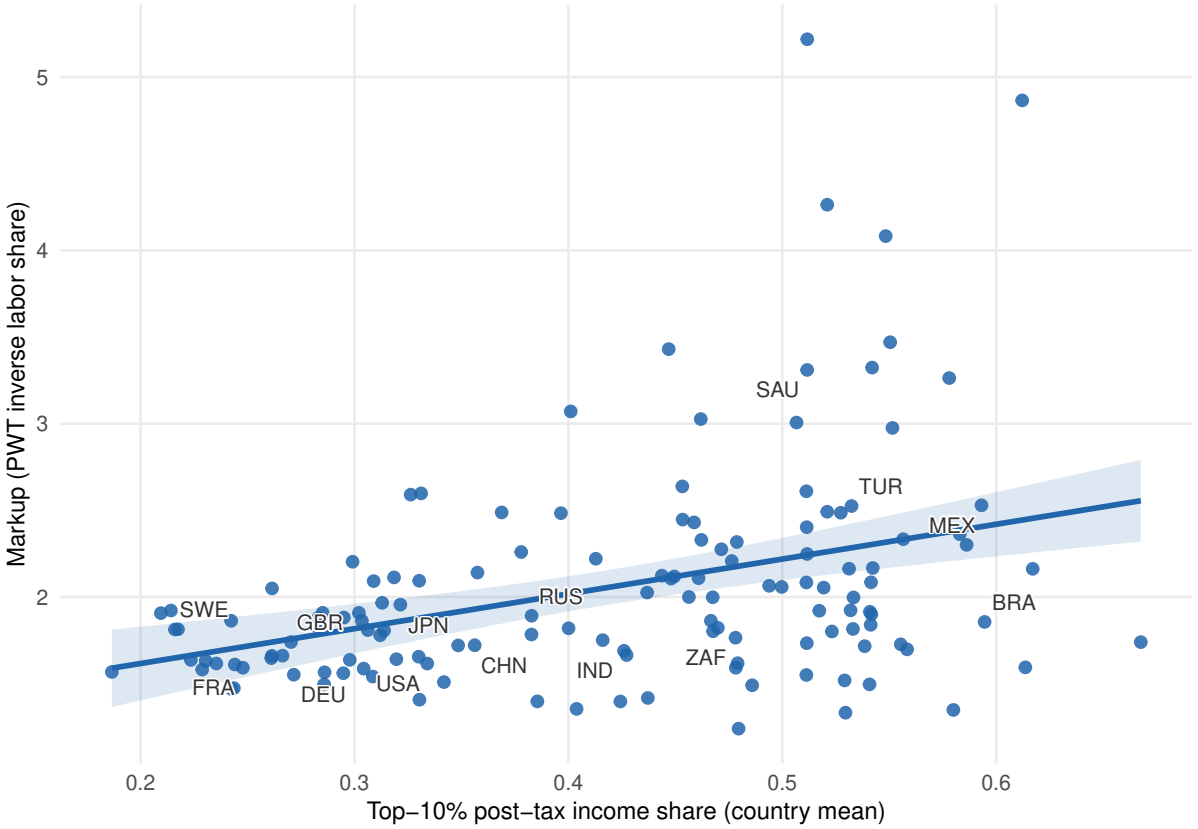
The relationship is descriptive and does not identify the model's steady-state link from inequality to the desired markup. Cross-country differences in market structure, regulation, technology, sectoral composition, and measurement may drive both variables, and causality can also run from markups to inequality, as in Boar and Midrigan (2025), where higher markups raise the income of concentrated firm owners. The within-country estimates are weaker and in one case wrong-signed, so the evidence speaks to long-run levels, not to changes within countries.

4.2 Shock-Conditional Evidence

US markups rise after the technology and oil supply shocks but also after an expansionary monetary policy shock, contrary to the sticky-price core that TE-TANK shares with common-elasticity models, and consumption inequality rises after the technology shock; Appendix B.3 and Table B.1 report these signs. Across countries, we estimate country-specific local projections for 35 economies, average them with the Mean Group estimator of Pesaran and Smith (1995), and split countries at the median WID post-tax top-10% income share. The shock is the common oil supply-news shock of Mori and Peersman (2026), an empirical analogue of a contractionary supply disturbance, not a structural implementation of u_t. Mori and Peersman (2026) keep the OPEC-announcement instrument of Känzig (2021) but add financial variables to the information set of his model, because without them his shock is predictable.

Figure 5 shows that the markup response to the oil shock, measured by the inverse labor share, is larger, or less negative, in more unequal economies. In both panels, the horizontal axis is the horizon in quarters and the vertical axis the cumulative change in

FIGURE 4: Markup Levels and Inequality Levels Across Countries

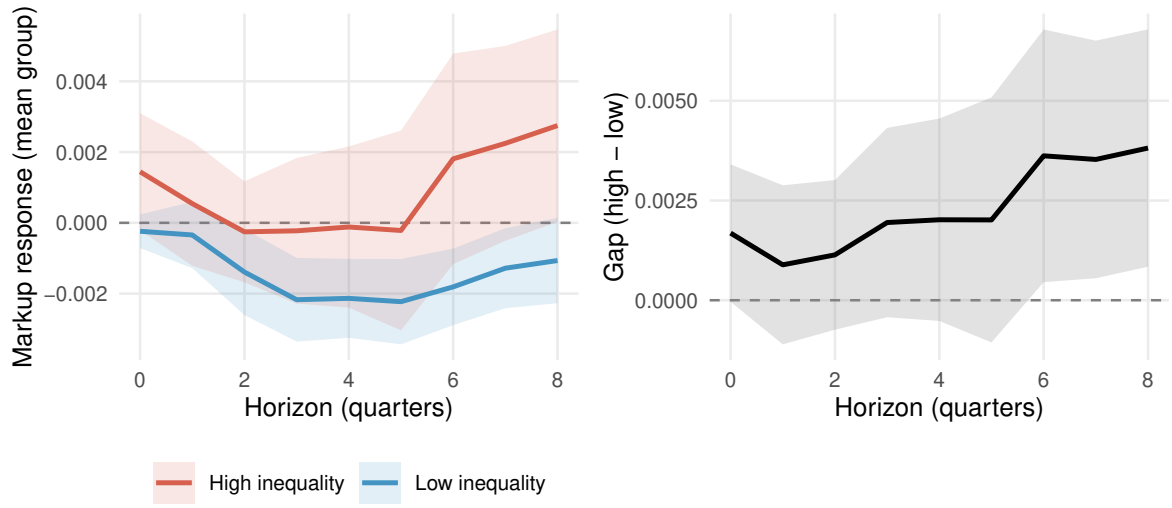


Notes: Each point is one country. The vertical axis is the markup level measured as the inverse labor share, the geometric time mean $\exp\left(\overline{\log(1/\mathrm{LS})}\right)$ of the labor share LS; the horizontal axis is the time-mean post-tax top-10% income share from the World Inequality Database (WID). Both means are taken over each country's own overlap window with WID. The labor share is from the Penn World Table, adjusted for the labor income of the self-employed, for 138 countries (5480 country-years, 1951 to 2019). The line is the OLS fit with a 95% confidence band based on conventional standard errors. Labels mark selected large economies.
Sources: Penn World Table 10.01 and WID.world.

the log inverse labor share after a one-standard-deviation shock. The left panel shows the high-inequality group in red and the low-inequality group in blue, the right panel their difference, and the shaded areas are 90% intervals. The high-minus-low gap is positive at every horizon, $+0.0020$ at $h = 4$ and $+0.0038$ at $h = 8$. The low-inequality response is negative at every horizon, while the high-inequality response is close to zero from the second to the fifth quarter and positive after that. Up to five quarters the gap is therefore a smaller fall in the more unequal countries, not a larger rise. A moving-block bootstrap that resamples the shared quarterly axis gives $p = 0.347$ and $p = 0.030$, so the common-shock-robust evidence is insignificant at four quarters and significant at the five percent level at eight. Känzig's original shock gives a similar gap, as Table B.5 in Appendix B.8 shows.

The sign and significance of the gap depend on how inequality is measured and

FIGURE 5: Cross-Country Markup Response by Inequality Group (Mean Group, Oil Shock)



Notes: Mean Group markup responses to a +1 SD Mori and Peersman (2026) oil supply-news shock, high- versus low-inequality countries (35 OECD and other European economies, split at the median WID post-tax top-10% share, without the 5 net oil exporters that Appendix B.4 lists). Right panel: the high-minus-low gap. Shaded bands are 90% intervals from the cross-country dispersion of the country coefficients; Appendix B.6 reports the common-shock-robust bootstrap. Inference on the gap uses the standard error of the difference, so the overlap of the left panel's bands is not informative about it.

on which oil exporters are excluded. The pre-tax top income share and the top wealth share give smaller positive gaps that are not significant, the disposable-income Gini gives gaps close to zero, and the market-income Gini reverses the sign (Appendix Table B.5). At eight quarters the gap is significant at the 5 percent level only under the baseline exclusion rule, and at the 10 percent level under the four alternative sets of excluded oil exporters (Appendix B.4). This significance is pointwise: a simultaneous band over horizons, from a country bootstrap, includes zero at four and at eight quarters (Appendix B.8). The median split also largely separates Western European economies, most of them in the low-inequality group, from the post-socialist economies of Central and Eastern Europe, most of them in the high-inequality group (Appendix Table B.3), so the gap can reflect other differences between these regions. Most countries enter the sample in 1995 or later, so the estimate rests mainly on the oil episodes after 1995.

The positive estimates are consistent with the feedback from output to inequality that TE-TANK shares with common-elasticity TANK, but they cannot confirm the elasticity-gap channel. Common-elasticity TANK predicts a positive gap per unit of a common cost shifter under both pricing conventions, and Appendix D.11 derives this sign. Per unit of the same shifter TE-TANK predicts a positive gradient too. Under Rotemberg pricing it is below the benchmark's: at the observed inequality spread, the

high-minus-low markup response is 0.098 times the baseline response, against 0.118 in the benchmark, because the fall in δ_2 with inequality partly offsets the stronger feedback; under Calvo pricing it is larger, 0.209. The estimated gap at eight quarters is also an order of magnitude larger than the largest model gap under either normalization and pricing convention, 0.000306 per unit of the Phillips-curve wedge. Power against that largest gap is only 5.3 percent at a nominal size of five percent, and lower against the smaller gaps per unit of a structural shifter, so a failure to reject could not refute the channel either. Appendices B.4–B.8 give the details.

The cross-country inflation response does not separate the two models either. Both predict a smaller cumulative price-level response in more unequal economies, under either normalization and pricing convention. On the same shock and split, the high-minus-low cumulative price-level gap is negative at both horizons, as both models predict, is not significant, and could be detected only if it were much larger than either model implies. Appendix B.5 reports this test.

5 Conclusion

Heterogeneous price elasticities place the consumption distribution inside the Phillips curve. In a two-agent New Keynesian model calibrated to Swiss micro evidence (Auer et al., 2024), desired markups depend on the composition of demand, so they rise whenever a shock shifts spending toward the less price-sensitive rich. The reduced-form slope factors in closed form into the slope without distributional feedback and the composition channel, a real rigidity that lowers the response of desired prices to real activity and flattens the curve when prices are sticky. The channel is a countercyclical desired markup, latent under flexible prices, and with a common elasticity the distribution does not enter the Phillips curve.

The composition channel has a definite sign. Relative to markup-matched TANK, which shares TE-TANK's steady state, the composition factor in the slope is -4.3 percent under Rotemberg and Calvo pricing alike, and with fixed elasticities its sign requires only that consumption be unequal in steady state. Under Rotemberg pricing this factor is the whole difference between the two economies, so a cost-push shock raises inflation by 3.3 percent more than in the benchmark. Section 3.3 shows that under Calvo pricing the net comparison reverses: a firm-level curvature, absent in a symmetric Rotemberg equilibrium, steepens TE-TANK's curve by more than the composition channel flattens it.

Fiscal redistribution operates on the supply side: rebating profits to hand-to-mouth households lowers the desired markup and weakens the composition channel. Under Rotemberg pricing it therefore steepens the Phillips curve of TE-TANK and leaves the

common-elasticity slope unchanged. The model is too stylized to design transfers, but it suggests that how profit income is shared affects desired markups and the slope of the Phillips curve. The evidence is a consistency check, not a test of the elasticity gap. More unequal countries have higher markups, a descriptive correlation. After an oil supply shock their price level responds less, as both models predict, but the difference is small and insignificant. Their markup proxy responds by more, or falls by less, imprecisely and depending on the inequality measure, consistent with the feedback from output to inequality that TE-TANK shares with common-elasticity TANK.

The main limit is that the model has no cost channel or nominal wage rigidity, and the sign of the feedback rests on a labor share that rises in booms. The evidence after demand shocks does not support such a labor share, since measured US markups rise after a monetary expansion (Nekarda and Ramey, 2020), and sticky wages could make it fall in booms.

TE-TANK relates to a New Keynesian model with non-homothetic demand as TANK relates to HANK: TANK fixes which households are constrained, a status that varies over time in HANK (Debortoli and Galí, 2018), and TE-TANK fixes each type's price elasticity. To first order, the spending-weighted elasticity that firms face moves through a between-household composition term and a within-household term, as each household's elasticity moves with its own consumption. TE-TANK keeps only the first, which yields the closed-form slope. The within-household term pushes markups the other way, because a household becomes less price sensitive as its own spending or wealth rises. In Olivi et al. (2024), whose cyclical markup term contains both terms, it raises markups in a boom when markups rise with wealth; in Mongey and Waugh (2025), together with the sorting of households across firms, it raises the aggregate markup after a transfer. Added to TE-TANK at assumed curvatures, it lowers the extra inflation after a cost-push shock to 1.8 percent at the curvature implied by the income range of the micro estimates and to -0.65 percent at the curvature fitted to the model's own consumption spread (Appendix D.13). Whether the composition of demand flattens or steepens the Phillips curve therefore depends on how a household's price sensitivity moves with its own consumption over the cycle, which the cross-sectional evidence used to calibrate the elasticity gap does not identify.

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Declaration of competing interest

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Data availability

All data used in this paper are publicly available from the sources documented in Appendices B and C. The code and constructed data that reproduce all results, tables and figures are available from the authors and will be deposited with the journal on acceptance.

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Appendix

Inequality, Markups, and the Phillips Curve

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A Markup and Inequality: Additional Evidence

Table A.1 shows that the between-country slope of the markup level on inequality, which Section 4.1 documents, is positive for all three markup measures and on both the post-tax and the pre-tax income share. The production-approach slope, however, rests on Latin America, and the within-country estimates are weak. The table's rows are three log markup measures: the Penn World Table and ILO inverse labor shares and the production-approach markup of De Loecker and Eeckhout (2018). The three sources differ in coverage, with the most country-years in the Penn World Table, and their intersection is too thin for a common-sample estimate. We regress each measure on the WID top-10% income share. The five columns are the between-country estimator on the post-tax share and on the pre-tax share, the between-country estimator on the post-tax share without Latin America, the within-country estimator with fixed effects, and the long difference. Without the 7 Latin American countries, the production-approach slope falls from 0.475 (SE 0.160) to −0.014 (SE 0.209), while the two inverse labor-share slopes become larger.

TABLE A.1: The Between-Country Markup-Inequality Slope: Proxy and Estimator Robustness

	Between	Between, pre-tax share	Between, excl. Latin America	Within FE	Long-diff.
Inverse LS: PWT `labsh`	0.829***	0.935***	1.018***	0.307**	0.346*
N_{cty}=138	(0.172)	(0.197)	(0.242)	(0.149)	(0.183)
Inverse LS: ILO labour income share	0.999***	0.964***	1.288***	0.101	−0.348
N_{cty}=185	(0.141)	(0.181)	(0.148)	(0.173)	(0.236)
De Loecker–Eeckhout production approach	0.475***	0.564***	−0.014	0.236	0.442
N_{cty}=41	(0.160)	(0.189)	(0.209)	(0.314)	(0.670)

Notes: Dependent variable: log markup; regressor: WID post-tax top-10% national income share; heteroskedasticity-robust standard errors in parentheses, clustered by country in the within column. Between: OLS on country means; within FE: country and year fixed effects; long-diff.: change from each country's first to last year. N_{cty} is the number of countries, which is the sample size of the between and long-difference columns. The second column repeats the between-country regression with the WID pre-tax share (`sptinc`) in place of the post-tax share (`sdiinc`); the pre-tax share covers more country-years, and for PWT the pre-tax slope on the post-tax country-years alone is 0.921 (0.198), so the gap to the first column is the tax base and not the coverage. The slopes of the markup level on the share are positive and significant at the 1% level on both bases for all three measures. The third column repeats the between-country regression without the Latin American and Caribbean countries (World Bank region), on 112 countries for PWT, 155 for ILO and 34 for De Loecker–Eeckhout. The within-country fixed-effects column is estimated on 5,451 country-years for PWT, 3,876 for ILO and 1,411 for De Loecker–Eeckhout. In the PWT sample, the within estimator drops 29 US country-years (1951–1979) that form singleton year cells. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The last two columns show that the association is weaker within countries than across them. In the Penn World Table sample, the within-country fixed-effects slope is 0.307 (SE 0.149) and the long-difference slope is 0.346 (SE 0.183). The ILO long dif-

ference has the wrong sign, and the other within-country estimates are positive but imprecise.

Post-tax is the paper's baseline because the model's inequality object is post-transfer. The heterogeneity index $H_t = C_t^R / C_t$ is consumption per rich household relative to population-average consumption, so the rich households' aggregate consumption share is $(1-\lambda)H_t$, which they finance out of income after taxes and transfers. The WID post-tax national income share (`sdiinc`) is the closest available counterpart to post-transfer resources, and hence to $\bar{H}$. The second column of Table A.1 replaces it with the pre-tax share (`sptinc`), the series most of this literature uses, and the between-country slope stays positive and significant at the 1% level for all three measures. The pre-tax share covers more country-years in the Penn World Table sample, but the table notes show that the gap between the first two columns comes from the tax base, not from the coverage.

B Local Projections

Table B.1 shows that the estimated signs match the shared TANK core after the technology shock and for the markup after the oil shock, contradict it for the markup after the demand shock, and leave every TE-TANK increment untested or inconclusive. After the technology shock, markups and consumption inequality both rise, as the sticky-price mechanism and the cyclicality relation of Section 2.5 predict; the point estimate of the markup rise is larger before 1975, although a pooled test does not reject equal paths on the baseline measure, while the inequality response is estimated on 1981Q4–2008Q4. After the demand shock, markups rise against the sticky-price prediction. After the oil shock, markups rise, also over a cost base that adds nonlabor to labor cost, and consumption inequality is an uninformative null after both the oil and the demand shock. The cross-country markup gap after the oil shock is, like the other estimates, a descriptive consistency check rather than a test of the TE increment, as Section 4 states. Per unit of a structural cost shifter, the unit in which an oil shock enters, its positive sign is the prediction of common-elasticity TANK and of TE-TANK alike, whose own gradient is smaller than the benchmark's under Rotemberg pricing and larger under Calvo pricing (Section B.7). The gap is positive under every country omission and oil-exporter exclusion, but its significance depends on the horizon, the sorter and the markup proxy: at $h = 8$ it is significant at the 5 percent level under all three inference methods, at $h = 4$ under none, and the market Gini reverses its sign. The inflation pass-through test has power of about 5.1 percent at $h = 8$ against a size of five percent, so it cannot discriminate between TE-TANK and markup-matched TANK even per unit of the Phillips-curve wedge, the unit in which they differ most under

Rotemberg pricing.

TABLE B.1: Model Predictions versus Empirical Estimates: Signs by Shock

	Model prediction		Empirics		Aligned?	
	Markup	Inequality	Markup	Inequality	Markup	Inequality
TFP (expansionary)						
TANK	↑	↑	↑[a]	↑[d]	consistent	consistent[d]
TE-TANK	⇑	⇑	+, imprecise[b]	—	tested–inconcl.[b]	not sep. tested
Demand (expansionary)[g]						
TANK	↓	↓	↑[c]	0[d]	inconsistent	tested–inconcl.[d]
TE-TANK	≈ 0	≈ 0	—	—[i]	not identified[h]	not sep. tested[i]
Cost-push (oil)						
TANK	↑	↑	↑[e]	0[d]	consistent[e]	tested–inconcl.[d]
TE-TANK	⇑	⇑	+, fragile[f]	—	tested–inconcl.[f]	not sep. tested

Notes: Model predictions are for Rotemberg pricing and for each economy's level response at the baseline steady state. There TE-TANK and markup-matched TANK share δ_2, so the increments are the same per unit of the Phillips-curve wedge and per unit of a structural cost shifter. The table does not show the cross-country gradient of the oil-shock markup response, which depends on the normalization and the pricing convention (Section B.7). ↑ / ↓ gives the response sign. TE-TANK rows give the increment relative to homogeneous-elasticity TANK: ⇑ denotes amplification and ≈ 0 a negligible increment; "—" means no dedicated estimate. "Consistent" and "inconsistent" require an informative estimate with the predicted or the opposite sign; "tested–inconcl." denotes an imprecise or design-sensitive estimate, "not sep. tested" no estimate of the TE increment, and "not identified" a design that cannot isolate it. [a] Point estimates larger before 1975 (Figure B.2); on the baseline measure a pooled test does not reject equal paths. [b] The cross-country split on the US technology shock, Panel A of Table B.5, is positive but imprecise. [c] Against the sticky-price prediction. [d] CEX P90/P10 consumption ratio; joint p-values 0.018 (TFP), 0.284 (oil) and 0.950 (demand). [e] A markup over unit labor plus nonlabor cost also rises; that cost base contains no energy, so the energy wedge is not ruled out. [f] The country-bootstrap sup-t band of the gap includes zero at both horizons, narrowly at $h = 8$; the time bootstrap gives $p = 0.347$ at $h = 4$ and 0.030 at $h = 8$, and the result depends on the sorter and the markup proxy. [g] The markup and inequality outcomes use different monetary shocks. [h] The design cannot separate κ_h from κ_y. [i] A level null cannot identify the TE increment.

B.1 Specification

For each horizon $h = 0, 1, \ldots, 12$ we estimate the local projection of Jordà (2005),

$$\mu_{t+h} - \mu_{t-1} = a_h + \beta_h s_t + \sum_{l=1}^{L+1} \gamma_{l,h} s_{t-l} + \sum_{l=1}^{L+1} d_{l,h} \Delta\mu_{t-l} + e_{t+h}, \tag{B.1}$$

where μ_t is the log markup level and s_t an identified shock, so that β_h is the cumulative markup response at horizon h, measured from $t-1$ and normalized to a one-standard-deviation shock. The lag length L is chosen by AIC (at most eight lags) from an auxiliary VAR in the shock and $\Delta\mu_t$, is held fixed across horizons, and is common to the markup proxies of an exhibit. The controls include one lag more than L. With this lag augmentation, ordinary least squares with Eicker–Huber–White standard errors gives valid inference without a HAC correction (Montiel Olea and Plagborg-Møller, 2021).

We use the HC3 formula, which is more conservative in small samples than HC1; only the smooth local projection uses HC1.

B.2 Markup Measures

The unconditional cyclicality of the markup depends on how it is measured (Nekarda and Ramey, 2020), so Table B.2 lists five measures that span the main methodological choices. Measure (i), the inverse labor share under Cobb–Douglas technology, is the Nekarda and Ramey (2020) baseline. The cross-country panel uses its economy-wide national-accounts counterpart, $\log(\text{GDP}/\text{compensation of employees})$, which Section B.4 defines. Measure (ii) adds overhead labor following Galí and Gertler (1999), and measure (v) is the price-over-unit-cost markup of Enders (2020).

TABLE B.2: Markup measures

Markup measure	Definition	Source / parameter
Inverse labor share (CD)	$\mu^{CD} = -\ln s_L$	BLS labor share s_L
CD + overhead labor	$\mu^{CD,OH} = -\ln s'_L,\ s'_L = \nu\, WS/Y$	ν = production-worker share of private employment; from 1964Q1
CES, labor margin	$\mu_L^{CES} = \mu^{CD} + \ln\left[\alpha_L\,(Y/ZL)^{1/\sigma-1}\right]$	BLS, $\sigma = 0.5$
CES, capital margin	$\mu_K^{CES} = \mu^{CD} + \ln\left[1-\alpha_K\,(Y/uK)^{1/\sigma-1}\right]$	BLS + BEA capital, const. u
Price / unit cost (Enders)	$\mu = \ln(P/UC)$	BLS NFC Prod. & Costs

Notes: The five markup measures used in the local-projection analysis. Measures (i)–(iv) are constructed from quarterly US national-accounts data; measure (v) is the price-over-unit-cost ratio for non-financial corporations. Measure (ii) divides business-sector wages and salaries WS, not compensation, by nominal output Y, multiplies this ratio by the variable-labor share ν of Nekarda and Ramey (2020), which counts only production and nonsupervisory workers as variable labor, and omits the constant Cobb–Douglas output elasticity, which cancels in log changes.

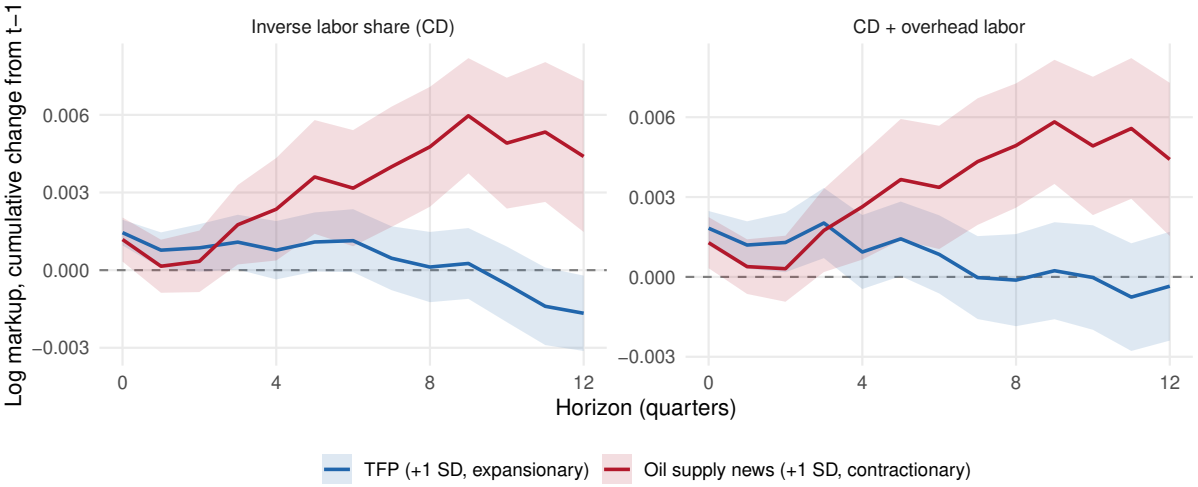
B.3 Unconditional Responses to Identified Shocks

The unconditional signs are mixed and test only the core that TANK and TE-TANK share. Markups rise after the technology and oil supply shocks, but they also rise, on all four proxies, after an expansionary Jarociński and Karadi (2020) monetary policy shock, whereas the sticky-price class predicts a decline.[9] Consumption inequality rises after the technology shock and is an uninformative null after the oil and demand shocks.

[9]Matching this sign requires machinery that the model does not contain: a cost channel, under which a lower policy rate lowers financing costs and marginal cost, or nominal wage rigidity (Erceg et al., 2000): Bilbiie and Känzig (2024) show that profits rise with inflation after a demand shock only when wages are sticky enough relative to prices. The composition of measured marginal cost also decides the sign of the estimate itself (Bils et al., 2018). Adding capital investment does not resolve the issue: in Bilbiie and Känzig (2024) it restores demand amplification, but procyclical profits still keep the inflation response muted.

Markups after both supply shocks Figure B.1 shows that markups rise after both the technology and the oil shock, on two proxies of Nekarda and Ramey (2020) estimated from equation (B.1): the Cobb–Douglas inverse labor share in the left panel and its overhead-labor variant in the right panel. The technology-shock response is largest on impact and fades to about zero within two years. The oil-shock response is small and imprecise on impact and builds over about two years. The technology-shock response tests the sticky-price core that TE-TANK shares with common-elasticity models: marginal cost falls and, with sticky prices, the ratio of price to marginal cost rises. The response replicates Nekarda and Ramey (2020) and is consistent with New Keynesian pricing in general, not only with our extension. Two qualifications apply. First, the point estimates of the US markup response to the technology shock are larger before 1975 than after, as Figure B.2 shows, although a pooled test does not reject equal paths on the baseline measure, so the technology-shock evidence may rest partly on the early sample. Second, the inverse labor share measures the labor margin, while energy costs enter marginal cost directly, so part of its rise after the oil shock could be the mechanical wedge between the two.

FIGURE B.1: Markup Responses to Supply Shocks: Unconditional Local Projections



Notes: Cumulative log-markup responses from equation (B.1) on US quarterly data. Left panel: measure (i) of Table B.2; right panel: measure (ii). Samples are the merged shock–proxy windows, and N counts their quarters: 1949Q1–2017Q4 ($N = 276$) for the technology shock, from 1964Q1 ($N = 216$) in the right panel; 1975Q1–2017Q4 ($N = 172$) for the oil shock, whose series begins in 1975. All US samples end in 2017Q4, where the Nekarda and Ramey (2020) markup series end.Blue: identified TFP shock, +1 SD expansionary (SVAR innovation). Red: Mori and Peersman (2026) oil supply news, +1 SD contractionary (sign of the authors' series). Shaded bands are ±1 standard error (approximately 68% pointwise; EHW/HC3).

The technology shock is identified in a first stage, as in Nekarda and Ramey (2020). A quarterly five-variable Cholesky SVAR with four lags and a quadratic trend orders utilization-adjusted TFP (Fernald, 2012) in log levels first, followed by log real GDP per

capita, the log GDP deflator, the three-month Treasury-bill rate and the log markup; the reduced-form residual of the TFP equation is the shock s_t in equation (B.1). Its correlation with the baseline shock is 0.995 when the markup is dropped from the SVAR, and at least 0.840 under a linear trend, TFP ordered last, first differences, a long-run restriction (Galí, 1999) or a post-1984 sample. The local projections and their bootstrap treat the identified shock as a given regressor, so first-stage estimation uncertainty is not propagated into the bands. The cross-country oil design is unaffected, since Mori and Peersman (2026) post their oil supply-news series.

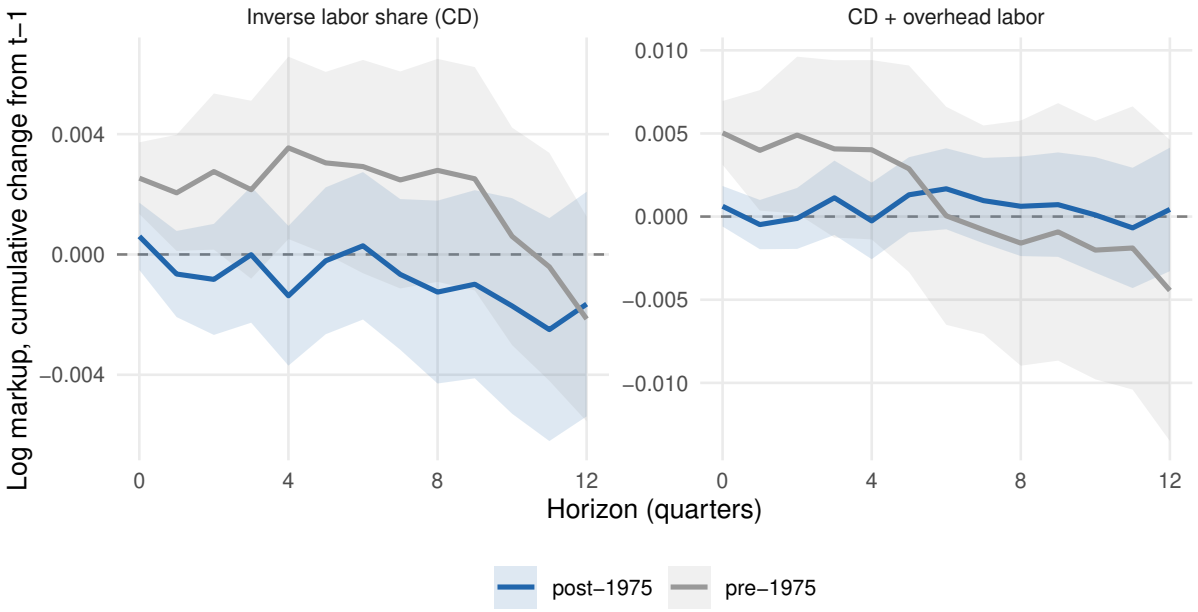
For the unconditional markup response to the technology shock, a joint bootstrap test across all horizons ($h = 0, \ldots, 12$) rejects the null that the entire path is zero ($p = 0.041$), and the sup-t simultaneous confidence band (Montiel Olea and Plagborg-Møller, 2019) excludes zero at the impact horizon. The impact response drives the rejection: the path is positive but imprecise thereafter, and the mild negative excursion beyond two years is indistinguishable from zero at every horizon, pointwise and jointly.

The impact rise is robust across measures. Four of the five measures are positive and significant at impact, the CES capital-margin measure being the exception, and the response holds on the Enders price-over-unit-cost measure, which does not reduce to the labor-share identity. Adding current and lagged hours growth as controls raises the impact response from 0.0014 (SE 0.0005) to 0.0022 (SE 0.0005) and leaves the response at $h = 4$ almost unchanged. Hours themselves respond to the shock, so conditioning on them changes the estimand and does not rule out labor hoarding. Smooth (Barnichon and Brownlees, 2019) and bias-corrected (Herbst and Johannsen, 2024) local projections leave the response intact, with an estimated finite-sample bias of at most 2.1 percent of the standard error. The result matches Nekarda and Ramey (2020): the nature of the shock, not its sign, governs markup cyclicality.

Technology shock before and after 1975 Figure B.2 shows that, in the first year after the technology shock, the markup response is larger before 1975 than after on both measures of Figure B.1. Each panel re-estimates equation (B.1) separately on the two windows, with the full-sample shock in the same units in both. On the baseline Cobb–Douglas measure the impact response is 0.0025 (SE 0.0007) before 1975 and 0.0006 (SE 0.0007) after, but a pooled local projection, fully interacted with a post-1975 indicator, does not reject equal paths in a sup-t bootstrap test over $h = 0, \ldots, 8$ ($p = 0.157$). For the overhead-labor measure, whose pre-1975 window starts in 1964Q1, the impact difference stays significant after a Bonferroni correction over the nine horizons ($p = 0.013$). The two CES measures of Table B.2 also respond more before 1975: the post-minus-pre difference is -0.0034 (SE 0.0017) on impact for the labor-margin measure

and -0.0049 (SE 0.0021) at $h = 4$ for the capital-margin measure, with joint p-values of 0.260 and 0.054.

FIGURE B.2: Markup Responses to the Technology Shock before and after 1975



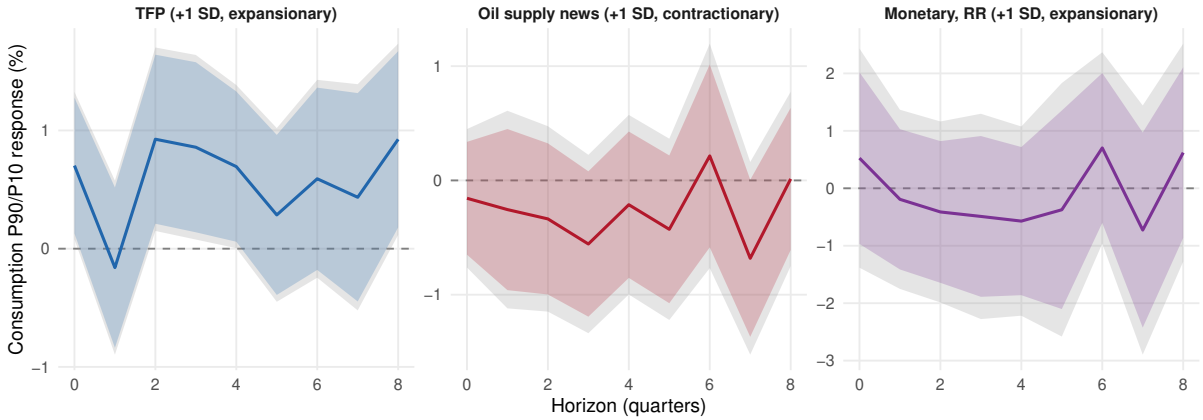
Notes: Cumulative log-markup responses from equation (B.1) to a +1 SD technology shock, estimated separately on two windows. Left panel: measure (i) of Table B.2; right panel: measure (ii). Gray: 1949Q1–1974Q4 ($N = 104$), from 1964Q1 ($N = 44$) in the right panel. Blue: 1975Q1–2017Q4 ($N = 172$). N is the number of quarters in each merged shock–proxy window. The shock is the full-sample SVAR shock, not re-estimated by window, and is scaled by its full-sample standard deviation, so the two lines are comparable in size. Shaded bands are 90% pointwise intervals (EHW/HC3), wider than the ± 1 standard-error bands of Figure B.1. Vertical axes differ across panels.

Oil shock: a broader cost base The Enders (2020) markup, the price deflator of nonfinancial corporations over their total unit cost (in the BLS definition, unit labor plus unit nonlabor cost, the latter including capital consumption, taxes on production, and net interest), also rises after the oil shock, so the rise survives a broader cost base. That cost base contains no energy, however, so the wedge between the labor margin and true marginal cost remains open.

Consumption inequality Figure B.3 shows that consumption inequality rises after the technology shock and responds imprecisely after the oil and monetary shocks. The outcome is the Coibion et al. (2017) P90/P10 ratio from the Consumer Expenditure Survey, and the monetary shock for this outcome is that of Romer and Romer (2004). After the technology shock the ratio rises and the path is jointly significant, so inequality is procyclical conditional on technology, as Section 2.5 predicts. After the oil and monetary shocks the responses are imprecise in both directions. These nulls are uninfor-

mative rather than evidence against the model: Broer et al. (2025) detect distributional effects of the Känzig (2021) oil shock, which uses the same instrument as the baseline shock, in large monthly administrative earnings data, so a null in a short quarterly survey ratio is expected.

FIGURE B.3: Consumption Inequality after Three Identified Shocks



Notes: Cumulative percentage responses of the Coibion et al. (2017) consumption P90/P10 ratio (BLS CEX, quarterly), estimated by lag-augmented local projections with EHW/HC3 standard errors and normalized to a +1 SD shock. Shocks are labeled in the panel headers; the oil shock is that of Mori and Peersman (2026) and the monetary shock that of Romer and Romer (2004). Colored bands: pointwise 90% intervals; gray bands: 90% sup-t simultaneous bands. Samples and N refer to the $h = 0$ regression: TFP 1981Q4–2008Q4 ($N = 109$), oil 1981Q3–2008Q4 ($N = 110$), monetary 1981Q4–2008Q4 ($N = 109$). Joint p-values (sup-t, $h = 0$–8): TFP 0.018, oil 0.284, monetary 0.950.

B.4 Cross-Country Panel: Design Detail and Robustness

The high-minus-low markup gap after the common Mori and Peersman (2026) oil shock, which Section 4.2 reports, stays positive under every oil-exporter exclusion set, and the pooled interaction with the inequality sorter stays positive, and significant at the 10 percent level, when seven country confounders enter jointly. Mori and Peersman (2026) keep the instrument of Känzig (2021), the change in oil futures prices around OPEC announcements, and add the S&P 500, the VXO volatility index and the US one-year interest rate to his oil-market VAR; without these financial variables, his shock and the instrument are predictable. We use the monthly shock series updated through December 2025 that Lorenzo Mori posts on his website (`https://sites.google.com/view/lorenzomori/data`), downloaded on 22 October 2026. The sample contains 35 OECD and other European economies through 2019Q4 and excludes 5 net oil exporters by the rule set out below. In the model a cost-push shock also raises inequality, through the split of income between wages and profits, and Broer et al. (2025) find that the oil shock lowers the earnings growth of low-income workers. The cross-country panel observes neither response, so the exercise

checks the distributional response and distribution-dependent pricing jointly rather than pass-through alone. In the closest model, Chan et al. (2024) bring an energy shock into a two-agent New Keynesian setting, in which the energy price compresses the labor share through energy–labor complementarity in production, not through differences in the elasticity of demand across types.

Using the same instrument, Broer et al. (2025) find textbook supply-shock responses of German inflation and activity. A fully anticipated monetary non-response changes the shock's aggregate and distributional effects little, whereas a non-response that repeatedly surprises markets raises activity, inflation and the employment prospects of low-income workers. The small effect of an anticipated non-response partly mitigates the concern that cross-country differences in policy reactions, rather than in demand composition, drive the heterogeneity documented below.

Mean Group design For each country i we estimate a separate local projection on the common global oil shock,

$$\begin{aligned} \mu_{i,t+h} - \mu_{i,t-1} &= a_{i,h} + \beta_{i,h}\, s_t + \sum_{l=1}^{L+1} \rho_{i,l,h}\, \Delta\mu_{i,t-l} + \sum_{l=1}^{L+1} \omega_{i,l,h}\, s_{t-l} + e_{i,t+h}, \\ \hat{\beta}_h^{\mathrm{MG}} &= \frac{1}{N} \sum_i \hat{\beta}_{i,h}. \end{aligned} \tag{B.2}$$

The Mean Group estimator of Pesaran and Smith (1995) averages the country coefficients, with standard errors from the cross-country dispersion; unlike a pooled estimator, it is consistent under slope heterogeneity. The lag length L is common and chosen by a panel AIC rule. The shock s_t is aggregated to the quarter and standardized with the sign of the authors' series, so $+1$ SD is contractionary. It is common to all countries and is not re-estimated, so identification comes entirely from how the response $\hat{\beta}_{i,h}$ differs across countries. The outcome is the inverse labor share $\mu_{i,t} = \log(\text{GDP}/\text{compensation of employees})$, and the WID top-10% share (Alvaredo et al., 2017) is an availability sorter rather than the model-matched consumption-dispersion object. Table B.3 lists the 35 economies with their markup coverage and group, and the excluded oil exporters in italics. Ending the sample in 2019Q4 keeps the finding from resting on the post-2021 inflation surge that Section 1 uses as motivation.

Oil-exporter exclusion The baseline excludes net oil exporters, because for them the oil supply shock is a positive terms-of-trade shock and oil rents lower the labor share,

TABLE B.3: Cross-Country Sample: Coverage and Inequality Split

ISO3	First	Last	N	H_i^{post} (%)	Group	ISO3	First	Last	N	H_i^{post} (%)	Group
DNK	1995Q1	2019Q4	100	18.6	*excl. (oil)*	POL	1995Q1	2019Q4	100	29.9	High
NOR	1995Q1	2019Q4	100	21.7	*excl. (oil)*	HRV	1995Q1	2019Q4	100	30.1	High
SWE	1993Q1	2019Q4	108	22.2	Low	PRT	1995Q1	2019Q4	100	31.2	High
FIN	1990Q1	2019Q4	120	22.6	Low	GRC	1995Q1	2019Q4	100	31.7	High
NLD	1995Q1	2019Q4	100	23.4	Low	LUX	1995Q1	2019Q4	100	31.9	High
SVK	1995Q1	2019Q4	100	23.5	Low	LVA	1995Q1	2019Q4	100	32.7	High
CZE	1995Q1	2019Q4	100	23.8	Low	BIH	2000Q1	2019Q4	80	32.9	High
CHE	1990Q1	2019Q4	120	24.0	Low	BGR	1995Q1	2019Q4	100	33.6	High
BEL	1995Q1	2019Q4	100	24.3	Low	MNE	2006Q1	2019Q4	56	33.7	High
FRA	1980Q1	2019Q4	160	24.8	Low	SRB	1995Q1	2019Q4	100	34.9	High
ITA	1995Q1	2019Q4	100	25.2	Low	LTU	1995Q1	2019Q4	100	35.0	High
SVN	1995Q1	2019Q4	100	25.8	Low	JPN	1994Q1	2019Q4	104	36.1	High
AUT	1995Q1	2019Q4	100	26.0	Low	EST	1995Q1	2019Q4	100	36.3	High
IRL	1995Q1	2019Q4	100	26.2	Low	ROU	1995Q1	2019Q4	100	36.5	High
HUN	1995Q1	2019Q4	100	26.2	Low	USA	1947Q1	2019Q4	292	36.6	High
AUS	1959Q3	2019Q4	242	27.1	Low	ISR	1995Q1	2019Q4	100	43.2	High
DEU	1991Q1	2019Q4	116	28.1	Low	TUR	1995Q1	2019Q4	100	51.8	High
GBR	1955Q1	2019Q4	260	28.2	*excl. (oil)*	*MEX*	2008Q1	2019Q4	48	59.5	*excl. (oil)*
CYP	1995Q1	2019Q4	100	28.9	Low						
ESP	1995Q1	2019Q4	100	29.2	Low						
CAN	1981Q1	2019Q4	156	29.5	*excl. (oil)*						
MLT	1995Q1	2019Q4	100	29.7	Low						

Notes: All 40 countries for which the first stage delivers a coefficient, sorted by the WID post-tax top-ten-percent income share H_i^{post}, the mean of the annual share over 1990–2019 with at least 5 annual observations. "First" and "Last" are the first and last quarters with a markup observation at or before 2019Q4, and N counts those quarters. A country enters the panel with at least 40 usable quarters of markup growth and shock. Group assigns each country to the low or high half at the median H_i^{post} of the baseline sample, giving 17 high and 18 low. The 5 net oil exporters are set in italics and marked "excl. (oil)".

which raises the markup proxy mechanically. Following the classification of Peersman and Van Robays (2012) by average net oil trade, a country counts as a net oil exporter if its annual net exports of petroleum and other liquids, production minus consumption in the EIA International Energy Statistics, are positive on average over its own window, which runs from the later of its first markup quarter and 1975 to 2019. The rule excludes Norway, Mexico, Canada, Denmark, and the United Kingdom. Only Norway has positive net exports in every year, and the United Kingdom is a borderline case. The main alternative set is a hand list of Norway, Mexico, and Canada.

The gap stays positive under every exclusion set, with the median recomputed in each case, but its size varies. At $h = 8$ it is $+0.0038$ (SE 0.0018) under the rule, the largest value, $+0.0030$ (SE 0.0018) when the United Kingdom is kept, $+0.0030$ (SE 0.0018) under the hand list, $+0.0029$ (SE 0.0017) when only Norway is excluded, and $+0.0030$ (SE 0.0016) when no country is excluded. The baseline gap is significant at the 5 percent level, and the four alternatives only at the 10 percent level. At $h = 4$ no

exclusion set gives a significant gap, and without any exclusion the gap is the smallest, at +0.0017 (SE 0.0016).

A pooled regression on the continuous sorter complements the Mean Group gap and allows checks for confounders:

$$\mu_{i,t+h} - \mu_{i,t-1} = a_i + \beta_h s_t + \gamma_h \left(s_t \times \tilde{H}_i\right) + \sum_k \theta_{k,h} \left(s_t \times X_{k,i}\right) + \text{lags} + u_{i,t+h}, \qquad \text{(B.3)}$$

where $\tilde{H}_i$ is the demeaned country-mean post-tax top-10% income share and $X_{k,i}$ contains time-invariant country controls. Time fixed effects are omitted because they would absorb the common shock. The pooled estimator imposes a common slope, so Mean Group is the primary estimator, while the pooled specification supplies Driscoll–Kraay inference (Driscoll and Kraay, 1998) and confounder checks.

Panel E of Table B.5 shows that the pooled interaction $\hat{\gamma}_h$ stays positive and significant at the 10 percent level under Driscoll–Kraay standard errors when seven time-invariant country traits enter jointly as shock interactions: imports, capital-account openness (Chinn and Ito, 2006), energy intensity, the capital share, employment protection, inflation targeting, and net energy imports. The panel reports the baseline and the joint specification, and its N column counts country-quarters at $h = 4$ and $h = 8$, not countries. The coefficient is a pooled slope on the continuous sorter, so it is a different estimand from the Mean Group gap, for which Section B.6 reports the shared-time bootstrap.

B.5 Inflation Pass-Through by Steady-State Inequality

On the same oil shock and median split, the high-minus-low gap in the cumulative price-level response is -0.0027 percentage points at $h = 4$ and -0.0235 at $h = 8$, with 90 percent bootstrap intervals from -0.2362 to 0.2574 and from -0.4036 to 0.3657. The point estimates are negative, the sign that both economies predict under either normalization and either pricing convention, but they are not distinguishable from zero, and the design has almost no power against the difference between the two models' predictions. Per unit of a structural cost shifter, the unit in which an oil shock enters (Section 2.6), the numerical complement and the homogeneous-elasticity benchmark of Appendix D.11 both predict a fall in pass-through along the branch, by 25.6 and 25.2 percent. Per unit of the Phillips-curve wedge both also predict a fall, by 5.2 and 25.2 percent, so the two economies differ in size, not in sign. That unit holds the wedge fixed while δ_2 falls, which amounts to a larger primitive cost shock in the more unequal economy (Section 3.5). These are the Rotemberg values. Under Calvo pricing TE-TANK's structural pass-through also falls along the branch, by 17.5 percent, against

25.2 percent in the benchmark (Section 3.5).

Outcome, and the model object it maps to The outcome is the cumulative price-level response, $\sum_{s=t}^{t+h} \pi_{i,s} = 100\,(\log P_{i,t+h} - \log P_{i,t-1})$ in percentage points, not an inflation rate. The projection (B.2) is re-estimated per country with this outcome on the same shock, aggregated by the same Mean Group estimator, and split at the same median; only the control-lag length is re-selected. The model object is instead the impact inflation response to a unit cost-push wedge, $\Psi_\pi = 1/[(1-\beta\rho_u) + \kappa\Theta]$ in (25), which falls with the Phillips-curve slope κ and with the output contraction Θ that the policy rule produces per unit of inflation. With a single AR(1) wedge and the linear policy block, equilibrium inflation inherits the wedge's persistence exactly, $\pi_{t+s} = \Psi_\pi \rho_u^s u_t$, so the cumulative response through horizon h is

$$\sum_{s=0}^{h} \pi_{t+s} = \Psi_\pi \frac{1-\rho_u^{h+1}}{1-\rho_u}\, u_t. \tag{B.4}$$

The multiplier depends only on ρ_u and h, both common across economies and across positions on the fiscal branch, so the comparative static of the cumulative response in steady-state inequality carries the sign of $\mathrm{d}\Psi_\pi/\mathrm{d}\bar{H}$ at every horizon. Per unit of a structural cost shifter the cumulative response is $\delta_2\Psi_\pi$ times the same multiplier, so under Rotemberg pricing TE-TANK and markup-matched TANK both imply a negative high-minus-low gap. Per unit of the Phillips-curve wedge the benchmark implies a negative gap as well, and so does TE-TANK over most of the branch, on which its Ψ_π falls (Appendix D.11).

Sample and lag length The inflation panel uses the same 35 countries and the same median split as the markup design, so its markup counterpart is the baseline gap of Section 4.2. The price index is the all-items consumer price index, monthly and not seasonally adjusted, taken at the last month of each quarter, with one source per country and no splicing: the OECD for most countries, the Eurostat HICP for Cyprus, Malta, and Romania, the IMF for Bosnia and Herzegovina, Montenegro, and Serbia, and the quarterly OECD index for Australia. A country's sample is its markup window intersected with the availability of its index. The rule of the markup design, the median across countries of each country's AIC choice with a ceiling of eight lags, gives $L = 6$, against $L = 2$ in the markup design. The choice is not sharp, because 14 of the 35 countries choose the ceiling, so we report the gap across a grid of lag lengths.

Inference and robustness At the headline lag length, the gap has $t = -0.03$ at $h = 4$ and $t = -0.14$ at $h = 8$ under the cross-country dispersion standard errors. The

moving-block bootstrap of Section B.6, with the markup design's block length of 10 quarters and $B = 4999$ draws, gives $p = 0.927$ and $p = 0.887$; blocks of eight and twelve quarters also give large p-values. Across the lag grid, the gap is negative at both horizons, standard errors lie between 0.053 and 0.212, and no $|t|$ exceeds 1.64. At $h = 4$ the gap is close to zero, so dropping single countries can turn it positive. Dropping Turkey, which moves the gap more than dropping any other country, gives $+0.0571$ (SE 0.0896); Turkey is kept in the baseline. Because the size of the point estimate depends on the lag length and on single countries, the conclusion on pass-through rests on the power calculation, not on the sign of any one cell.

Power With the convention of Section B.7, the minimum detectable effect at eighty percent power and five percent size is 0.4352 percentage points at $h = 4$ and 0.6772 at $h = 8$. The largest model-implied difference between TE-TANK and markup-matched TANK, per unit of the Phillips-curve wedge, is smaller by a factor of 12 at $h = 4$ and 41 at $h = 8$, so the power is 5.6 and 5.1 percent against a nominal size of five percent. No inflation-specific model grid exists to map the observed inequality spread into the fiscal branch, as $A(g_{xc})$ does for the markup gap, so the model-implied gap is the difference between the two economies' full-branch percentage changes in Ψ_π, as a fraction, times the unconditional Mean Group price-level response $\hat{\theta}_h$. We take the contrast per unit of the Phillips-curve wedge, the 5.2 percent fall in TE-TANK against the 25.2 percent fall in the benchmark. This contrast is the largest difference the fiscal block can generate under either normalization, and hence an upper bound on what any observed spread could imply. Per unit of a structural cost shifter, the unit in which an oil shock enters, the two economies fall by almost the same amount, -25.6 against -25.2 percent, so the contrast is a small fraction of the wedge contrast and of the opposite sign, and the test has even less power.

The policy-rule caveat The signs of the comparative statics do not depend on the output coefficient of the Taylor rule. Equation (D.41) signs homogeneous-elasticity pass-through for every $\phi_y \geq 0$, and Proposition D.1(iii) signs the markup comparative static under the elasticity condition (D.40) alone, and the homogeneous benchmark's markup comparative static needs no condition at all (part (b) of the benchmark paragraph in Appendix D.11). The size of both responses depends on the rule, through the output contraction per unit of inflation $\Theta = (\phi_\pi - \rho_u)/[\tilde{\sigma}(1 - \rho_u) + \phi_y]$ in (25), which rises with the inflation coefficient ϕ_π and falls with the output coefficient ϕ_y. Systematic rules differ across the countries in this panel, and the design holds none of them fixed, so cross-country dispersion in the inflation response to a common oil shock can reflect differences in reaction functions rather than in demand composition. The

sign mapping of equation (B.4) also assumes that the common oil shock reaches both groups as a domestic cost shifter of comparable size and persistence, which the design does not test.

B.6 Common-Shock Inference: Two Estimands, Three Standard Errors

The three inference methods apply to two estimands. Country-dispersion standard errors and the moving-block bootstrap use the Mean Group gap $\hat{\Delta}_h = \bar{\beta}_{\mathrm{hi},h} - \bar{\beta}_{\mathrm{lo},h}$, which averages heterogeneous country slopes before comparing groups and equals $+0.0020$ at $h = 4$ and $+0.0038$ at $h = 8$. Driscoll–Kraay standard errors use the pooled implied gap $\hat{\gamma}_h(\bar{H}_{\mathrm{hi}} - \bar{H}_{\mathrm{lo}})$ from equation (B.3), which fits one slope to the continuous sorter.

Country-dispersion standard errors treat the countries as independent, although all receive the same shock. The moving-block bootstrap (Künsch, 1989) resamples the common 1975Q1–2019Q4 time axis and re-estimates every country on the same quarters; with $B = 4999$ and blocks of 10 quarters, its percentile p-values are 0.347 at $h = 4$ and 0.030 at $h = 8$. Blocks of eight and twelve quarters give similar values, below five percent at $h = 8$. Driscoll–Kraay standard errors treat periods as the independent units; the pooled gap is close to the Mean Group gap at $h = 4$ and smaller at $h = 8$, and it is significant at the 10 percent level at $h = 4$ and at the 5 percent level at $h = 8$. All three methods reject a zero gap at the 5 percent level at $h = 8$ and none does at $h = 4$. At $h = 4$ the bootstrap gives the largest p-value, because it credits the sample with roughly the number of oil episodes rather than the number of countries or quarters.

B.7 Power and Model-Implied Magnitudes

The design cannot detect even the largest gap that the calibrated model gives: at $h = 8$ the power of the test against that gap is 5.3 percent, against a nominal size of five percent. The minimum detectable effect is $\mathrm{MDE}_h = (z_{0.975} + z_{0.80})\,\hat{\sigma}_h = 2.80\,\hat{\sigma}_h$, the smallest gap that a two-sided test at five percent size detects with eighty percent power, where $\hat{\sigma}_h$ is the standard deviation across the draws of the moving-block bootstrap of Section B.6. The model-implied gap is $\Delta_h^{\mathrm{model}} = A(g_{xc})\,\hat{\theta}_h$, where $\hat{\theta}_h$ is the estimated unconditional Mean Group response. The observed sorter gap g_{xc}, the difference between the mean H_i^{post} of the high and the low group divided by the mean over all baseline countries, equals 31.7 percent. The amplification $A(g)$ is the model's high-minus-low cost-push markup response, divided by the response of the baseline economy, for two steady states whose inequality differs by g. The headline value is computed per unit of the Phillips-curve wedge under Rotemberg pricing, the normalization that

gives the largest gradient, so the power calculation below is an upper bound. The model response is that of the labor-cost markup, the model counterpart of the inverse labor share used as the proxy. Scaling by the estimated response removes proportional measurement error in the labor-share proxy, which affects the gap and the average alike.

Per unit of a structural cost shifter the same object is 0.098 in TE-TANK under Rotemberg pricing, 0.209 under Calvo pricing, and 0.118 in the markup-matched benchmark under either; a positive gradient is the prediction the two models share and the one the sign of the estimated gap is compared with. TE-TANK's own gradient per unit of a structural shifter is positive but below the benchmark's under Rotemberg pricing, because the fall in δ_2 shrinks the wedge that a given shifter creates, and above it under Calvo pricing. All three values lie below the wedge value, so power against each is lower than the value reported below. The observed gap lies beyond the model grid, four gaps on the calibrated fiscal branch up to 16.8 percent, just below the ceiling of the fiscal block. Hence $A(g_{xc}) = 0.388$ is a linear extrapolation through the origin from the largest grid point; a quadratic fit through the origin gives $A = 0.391$, and the grid maximum gives 0.206. The mapping also assumes a unit elasticity between the top-10% income share and the model's inequality index, although income and consumption dispersion are not interchangeable, as Appendix C.2 discusses.

Table B.4 shows that the predicted gap lies far below the detectable one at both horizons. At $h = 8$ the minimum detectable effect is 0.00562, while the largest model gap, per unit of the Phillips-curve wedge, is 0.000306, smaller by a factor of 18.4; without extrapolation the factor is 34.6, and per unit of a structural cost shifter, where the model gap is smaller, the factor is larger still. Eighty percent power would require an effective sample about 338 times longer than the 45-year span, with precision scaling in the inverse square root of the sample length. Because the bootstrap credits the common shock with only about 18 independent time blocks, adding countries gains less, by an amount we have not evaluated. At $h = 4$ the mapped gap is negative only because the estimated unconditional response is negative there: A is positive, and per unit of the wedge the model predicts a positive gap at every horizon, because the impact response signed in Proposition D.1 decays geometrically without crossing zero.

TABLE B.4: Detectable Effects and Model-Implied Gaps: Cross-Country Design

	$\hat{\Delta}_h$	Bootstrap s.d.	MDE_h	Δ_h^{model}	$\Delta_h^{\text{model}}/\text{MDE}_h$	Power
h=4	0.0020	0.0023	0.00654	−0.000448	−0.069	0.054
h=8	0.0038	0.0020	0.00562	0.000306	0.054	0.053

Notes: $\hat{\Delta}_h$ is the Mean Group high-minus-low gap in the log inverse labor share after the oil supply-news shock; the bootstrap standard deviation is from the moving-block time bootstrap with $B = 4999$ and a block length of 10 quarters. $\text{MDE}_h = 2.80\,\hat{\sigma}_h$ is the smallest gap detectable at a two-sided five percent size with eighty percent power. $\Delta_h^{\text{model}} = A(g_{xc})\,\hat{\theta}_h$ is calibrated TE-TANK's predicted gap at the observed cross-country inequality spread, per unit of the Phillips-curve wedge under Rotemberg pricing, the normalization that gives the largest gradient. Per unit of a structural cost shifter the predicted gap is smaller, in TE-TANK under either convention and in the benchmark (Section B.7), so the power reported here is an upper bound. Power is that of the two-sided five percent test evaluated at Δ_h^{model}; it is bounded below by the size of the test, so values near 0.05 mean the test carries essentially no information about the model's prediction.

B.8 Cross-Country Robustness

Table B.5 shows that the sign of the gap survives changes of the shock, the markup proxy and the sample, but not of the inequality sorter, and that its magnitude and significance vary across all five panels. Panels A to E vary, in turn, the identifying shock, the markup proxy, the inequality sorter, the sample, and the confounder controls of the pooled interaction. In Panel A, the oil supply-news shock of Känzig (2021) gives a similar gap, +0.0026 (SE 0.0010) at $h = 4$ and +0.0038 (SE 0.0015) at $h = 8$, with time-bootstrap p-values of 0.226 and 0.057. The monthly-average proxy of Kilian (2024), built from twelve-month futures around OPEC announcements, preserves the positive sign but removes significance; Kilian argues that this proxy does not isolate oil supply news. In Panel B, the profit-side GOS proxy gives +0.0026 (SE 0.0014) and +0.0036 (SE 0.0018), while the annual production-approach proxy gives a positive gap at four years that is significant only at the 10 percent level. In Panel C, top-income and top-wealth sorters give positive but insignificant gaps, the market Gini reverses the sign, significantly at the 10 percent level at $h = 4$, and the disposable Gini gives gaps close to zero. In Panel D, every single-country omission preserves the positive gap, and dropping the low-labor-share countries leaves it significant at the 5 percent level at $h = 8$ only. The 90 percent sup-t simultaneous band of the baseline gap comes from a country bootstrap with 2000 draws that, like the dispersion standard errors, treats countries as independent. It runs from −0.0013 to +0.0053 at $h = 4$ and from −0.0001 to +0.0077 at $h = 8$, so it includes zero at both horizons, narrowly at $h = 8$. Panel E reports the pooled confounder check of Section B.4. Both factor-share-based outcomes

move with wage incidence, employment composition, and energy costs while desired markups are unchanged; the GOS proxy checks this without ruling it out.

TABLE B.5: Cross-Country Robustness of the High-Minus-Low Gap

	Gap (high−low)		N
	$h = 4$	$h = 8$	
Panel A: identifying shock and instrument			
Mori and Peersman (2026), headline (≤2019Q4)	0.0020 (0.0015)	0.0038** (0.0018)	35
Känzig (2021) oil supply-news shock	0.0026** (0.0010)	0.0038*** (0.0015)	35
Kilian (2024) m12, matched window (1989Q2–2017Q2)	0.0012 (0.0015)	0.0016 (0.0016)	35
US TFP SVAR shock (secondary)	0.0012 (0.0019)	0.0022 (0.0029)	39
Panel B: markup proxy			
GOS profit-share proxy (μ_{gos})	0.0026* (0.0014)	0.0036** (0.0018)	34
Baseline proxy, restricted to μ_{gos} country-quarters	0.0021 (0.0016)	0.0034* (0.0019)	34
De Loecker–Eeckhout annual production approach (annual frequency)	0.0222* (0.0131) at $h = 4$ years		17
Panel C: inequality sorter			
WID post-tax top-10% income share (baseline)	0.0020 (0.0015)	0.0038** (0.0018)	35
WID pre-tax top-10% income share	0.0015 (0.0016)	0.0029 (0.0019)	35
WID top-10% wealth share	0.0013 (0.0015)	0.0028 (0.0018)	35
Capital share	0.0011 (0.0014)	0.0010 (0.0018)	34
SWIID market Gini	−0.0027* (0.0015)	−0.0029 (0.0018)	35
SWIID disposable Gini	−0.0002 (0.0016)	0.0003 (0.0019)	35
Panel D: sample composition			
Leave-one-out range (min–max across 35 single-country drops)	[+0.0015, +0.0029]	[+0.0025, +0.0040]	35
Balanced subsample (≥80 usable quarters)	0.0034*** (0.0013)	0.0051*** (0.0017)	33
Excl. low labor-share countries (mean labor share <0.35)	0.0022 (0.0015)	0.0038** (0.0019)	32
Panel E: pooled interaction under confounder controls			
Pooled interaction $\hat{\gamma}_h$: baseline, no confounder controls	0.0206* (0.0112)	0.0287** (0.0117)	3446 / 3306
Pooled interaction $\hat{\gamma}_h$: all seven confounders jointly	0.0163* (0.0095)	0.0225* (0.0124)	2846 / 2734

Notes: High-minus-low Mean Group gap in the cumulative markup response at quarterly horizons, with country-dispersion standard errors in parentheses; Panel E instead reports the pooled interaction $\hat{\gamma}_h$ of equation (B.3) with Driscoll–Kraay standard errors. The baseline sorter is the WID post-tax top-10% income share and the baseline sample has 35 countries. The SWIID rows use country means of the SWIID Gini point estimates (Solt, 2020) over 1990–2019. The De Loecker–Eeckhout row uses the annual production-approach markups of De Loecker and Eeckhout (2018), and the Kilian row the monthly average IV(12) proxy of Kilian (2024, Figure 7a). The baseline shock is the oil supply-news shock of Mori and Peersman (2026); the Känzig row replaces it, as a robustness check, with the oil supply-news shock of Känzig (2021), identified with the same instrument. The GOS profit-share proxy is $\mu_{gos} = \log(\text{GDP}/(\text{GDP} - \text{GOS}))$, where GOS is gross operating surplus plus gross mixed income from Eurostat and OECD quarterly national accounts; Japan has no quarterly GOS series. The US TFP row does not apply the oil-exporter exclusion, which concerns the oil shock only. The leave-one-out row reports ranges and carries no stars. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

C Data, Calibration, and Compact Benchmarks

This appendix reports the full model calibration, the discipline of the distributional block, and the homogeneous-elasticity benchmarks. The replication code and data document series-construction detail beyond what the tables below need, such as raw source files and vintage handling. They are available from the authors and will be deposited with the journal on acceptance.

C.1 Calibration

Table C.1 reports the parameters common to all three economies. It adds a determinacy-consistent Taylor rule (Debortoli and Galí, 2018, 2024) and the persistence of the four shocks to the parameters of Table 1. The adjustment cost ζ reproduces the marginal-cost slope of a Calvo economy with decreasing returns at a price-stickiness parameter of 0.753, an average price duration of 4.0 quarters.[10] We impose full dividend taxation, $\delta = 1$, in all three economies. The wage markup $\mathcal{M}^w$ of the wage schedule scales only the levels of hours and output and enters no coefficient of the log-linear system, so we do not calibrate it (Appendix D.2).

TABLE C.1: Common Model Parameters

Description	Symbol	Value
Discount factor	β	0.995
Inverse intertemporal elasticity of substitution	σ	1.0
Inverse Frisch elasticity	φ	1.0
Decreasing returns to labor	α	1/3
Price adjustment cost	ζ	105.63
Taylor rule on inflation	ϕ_π	1.5
Taylor rule on output	ϕ_y	0.125
AR(1) preference shock	ρ_z	0.5
AR(1) technology shock	ρ_a	0.9
AR(1) monetary shock	ρ_ν	0.5
Cost-push persistence	ρ_u	0.9

[10] The mapping equates the Rotemberg slope $(\varepsilon^* - 1)/\zeta$ with the Calvo slope $\left[(1-\theta)(1-\beta\theta)/\theta\right] \cdot \left[(1-\alpha)/(1-\alpha+\alpha\varepsilon^*)\right]$, whose second factor corrects for decreasing returns; at $\varepsilon^* = 3.93$ and the ζ of Table C.1 it holds at $\theta = 0.753$.

Demand elasticities We set $\varepsilon_P = 6.6$ for poor and $\varepsilon_R = 3.0$ for rich households, the baseline of Auer et al. (2024) at annual incomes of 20,000 and 120,000 Swiss francs, so $\varepsilon_P/\varepsilon_R = 2.2$. Their estimates pin down the gradient of the elasticity in income, not its level.[11] Assigning the two income points to the model's hand-to-mouth and Ricardian households is an assumption: the types differ by asset access, and their steady-state consumption ratio of 1.905 is much smaller than the ratio of the two incomes. Their first approach identifies expenditure switching between imported and domestic goods, so applying it to domestic varieties assumes that the rich–poor gradient carries over from one margin to the other. The assumption is weaker for their second approach, which estimates substitution between any pair of products within a sector, on a sample that includes domestic goods, recovers essentially the same gradient, and changes little when that sample is restricted to imports. Three caveats remain: the data cover groceries, the identifying price variation is an import-price shock in both approaches, and the nested demand system applies one elasticity to all products in a sector. Panel A of Table C.4 reports how the markup and the dispersion-pure gap move with the elasticity level at the calibrated ratio, a path that includes elasticity gaps below 3.6 points.

Steady states Table C.2 shows that the three economies share the markup $\bar{\mathcal{M}} = 1.34$, because the homogeneous economies use $\varepsilon^* = 3.93$, and that TANK and RANK also share the Phillips-curve slope κ. TANK and TE-TANK also share $\bar{H} = 1.2347$ and differ only in the inequality loading κ_h, which lowers the TE-TANK slope.

C.2 Disciplining the Distributional Calibration

Table C.3 shows that every input of the distributional block is set externally, that the steady-state markup is an implied outcome, and that the CEX consumption ratio is held out as a validation moment rather than targeted. The hand-to-mouth share $\lambda = 0.40$ is at the upper end of evidence that places it near one-third (Kaplan et al., 2014). Kaplan et al. (2014, p. 80) also find that two-thirds of hand-to-mouth households are wealthy, holding sizable illiquid wealth. The label "poor" therefore refers to the lack of liquid wealth, and assigning the low-income elasticity of Auer et al. (2024)

[11] Auer et al. (2024) estimate the slope of the elasticity in log income at 2.19 (SE 0.55) and 1.93 (SE 0.87) in their two approaches and calibrate it to 2, a gap of 3.6 points between the two incomes. Neither baseline design identifies the level; they set the top-income elasticity to 3. Under stronger assumptions they obtain 4.92 and 1.76 at 20,000 francs, so the level is much less tightly determined than the gradient. Their sensitivity exercise varies the top-income elasticity from 1.5 to 5 at fixed differences, which implies a ratio between 1.72 and 3.39.

TABLE C.2: Steady-State Values

Variable	Symbol	TE-TANK	TANK	RANK
Heterogeneity Index	H	1.2347	1.2347	1.0000
Markup	M	1.3409	1.3409	1.3409
Output	Y	0.7384	0.7384	0.7384
Marginal Cost	MC	0.7458	0.7458	0.7458
NKPC Slope wrt Output	κ_y	0.0833	0.0833	0.0833
NKPC Slope wrt Inequality	κ_h	0.0064	0.0000	0.0000
DIS GE-term	χ_y	0.5637	0.5637	0.0000
Composite NKPC Slope	κ	0.0797	0.0833	0.0833

Notes: Output Y is evaluated at a common wage markup $\mathcal{M}^w = \bar{H}^\sigma = 1.235$, the smallest wage markup consistent with the participation of rich households at the baseline steady state; no log-linear result depends on it.

to all hand-to-mouth households is an assumption. Fiscal policy enters the steady state and the first-order dynamics only through $R \equiv \delta(1-\tau) = 0.30$, because the redistribution share δ and the transfer tilt τ change households' disposable resources only through this combination (Appendix D.10.1). We select this value externally rather than target it. For any $R < 1$ the sign of the mechanism is unchanged and R scales only its magnitude. Appendix D.11 reports the comparative statics along the fiscal branch from $R = 1$ to $R = 0$.

TABLE C.3: Calibration Ledger for the Distributional Block

Object	Value	Status
Hand-to-mouth share	$\lambda = 0.40$	Externally calibrated (hand-to-mouth share)
Demand elasticities	$(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$	Externally calibrated (Auer et al.: gradient estimated, top-income level normalized; import-margin to variety-CES mapping assumed)
Steady-state markup	$\bar{\mathcal{M}} = 1.34$	Implied outcome, untargeted; equals the production-approach formula $(1-\alpha)/s_L$ in model data
Fiscal parameter	$R = 0.30$ ($\delta = 1$, $\tau = 0.70$)	Externally selected, not targeted; enters only through R (Appendix D.10.1; CEX-implied value -0.042 infeasible)
CEX two-type ratio	2.93	Non-targeted validation; model reaches at most 2.759 (at $R = 0$), baseline value 1.905

Notes: The CEX two-type ratio is the mean consumption of the top 60 percent of households relative to the bottom 40 percent. It is computed each quarter from the cross-sectional standard deviation of log consumption in Coibion et al. (2017), under the assumption that log consumption is normally distributed. Their consumption measure covers nondurables, services and smaller durables but not house or car purchases, and it is not equivalized; the seasonally adjusted standard deviation comes from their replication data. The quarterly ratios are averaged over 1980Q1–2008Q4 (116 quarters; mean standard deviation 0.64).

The CEX moment exposes a limitation of the calibration: the baseline understates consumption dispersion, and the fiscal block cannot close the gap. The moments in Coibion et al. (2017) imply a rich-to-poor ratio of 2.93 at $\lambda = 0.40$. The model reaches at most 2.759, at the feasible boundary $R = 0$, so matching the data would require $R = -0.042$. The raw CEX ratio includes durables and is not equivalized, but it would have to fall below the baseline value 1.905 to reverse this conclusion.

The comparison is a diagnostic of overall dispersion, not a test of the model's types. The CEX groups are ranked by consumption, whereas the model's types differ by asset access. Equal group sizes do not imply equal membership, and dispersion within types also raises the CEX ratio. Within the model, higher steady-state inequality $\bar{H}$ along the fiscal branch flattens the slope κ and raises the markup response to a cost-push shock, as Proposition D.1 states. If the CEX gap reflected too little dispersion between the model's two types, the calibration would therefore understate the demand-composition feedback. The data do not establish that mapping, and even under it the size of the understatement is unknown, because the model cannot reach the CEX ratio.

The model's distributional object is consumption dispersion, so the income and wealth sorters of the empirical sections do not discipline the consumption block. Section 4.2 uses the top-10% income share only as the cross-country sorter, because it is widely available, and the top-10% wealth share as a robustness sorter.

The steady-state markup $\bar{\mathcal{M}} = 1.34$ is an implied outcome rather than a target, because the demand elasticities are calibrated externally and markup estimates measure different objects. The elasticity level, which pins the markup $\varepsilon/(\varepsilon - 1)$, is a free parameter in Auer et al. (2024), who sweep it themselves. The sensitivity exercise therefore sweeps three paths through the plane of the mean elasticity and the elasticity gap: one holds the gap fixed and one the ratio, each varying the level, and one holds the mean fixed and varies the gap alone. The fixed-gap path is primary, because the gap comes from the estimated gradient.

Table C.4 shows that the estimated rich–poor gradient fixes the sign and order of magnitude of the dispersion-pure cost-push gap: the gap stays positive across the published range of markups, and how much it moves depends on which elasticity object is held fixed. Panel A is the fixed-ratio path, Panel B the fixed-gap path, and Panel C reports three bounds. Along the fixed-gap path, which holds $\Delta\varepsilon = 3.6$, the dispersion-pure Taylor-rule gap moves from $+1.7$ to $+6.1$ percent around the baseline $+3.3$ percent. That range is taken over the implied markup span 1.21–1.61, from the 1980 to the 2016 aggregate markup of De Loecker et al. (2020), a span that contains the 2016 cost-weighted aggregate of Edmond et al. (2023) and the French firm-level median and mean of Burstein et al. (2025), along which the elasticity ratio drifts from 1.75 to 3.01, away from the estimated gradient. Holding the ratio at $\varepsilon_P/\varepsilon_R = 2.2$ instead narrows

the range to +2.5 to +4.1 percent over the same markup span. The cost-push result is therefore not tied to one markup level, but broader reparameterizations produce more variation.

TABLE C.4: Sensitivity of the Dispersion-Pure Cost-Push Gap along Three Elasticity Paths

	$\bar{\mathcal{M}}$	ε_R	ε_P	$\varepsilon_P/\varepsilon_R$	$\Delta\varepsilon$	Gap (%)
Panel A: fixed-ratio path ($\varepsilon_P/\varepsilon_R = 2.2$ held, level varies)						
DLEU 1980 aggregate / French firm median	1.21	4.34	9.54	2.20	5.20	+2.48
EMX cost-weighted aggregate	1.25	3.78	8.31	2.20	4.53	+2.78
Baseline (implied, untargeted)*	1.34	3.00	6.60	2.20	3.60	+3.32
French firm-level mean	1.39	2.73	6.01	2.20	3.28	+3.54
DLEU 2016 aggregate	1.61	2.06	4.53	2.20	2.47	+4.12
Panel B: fixed-gap path ($\Delta\varepsilon = 3.6$ held, level varies)						
Low end of markup span	1.21	4.77	8.37	1.75	3.60	+1.70
High end of markup span	1.61	1.79	5.39	3.01	3.60	+6.13
Panel C: bounds						
(a) 10% gap, fixed-mean path	1.44	1.36	9.06	6.65	7.69	+10.00
(b) Ceiling at ratio 2.2	1.91	1.66	3.66	2.20	1.99	+4.31
(c) Supremum over λ, $R^\dagger$	1.34	3.00	6.60	2.20	3.60	+6.72

Notes: Each row reports the dispersion-pure cost-push gap of TE-TANK against its own markup-matched TANK(ε^*), $\Psi_\pi/\Psi_\pi^* - 1$ in percent, at the baseline $\lambda = 0.40$ and $R = 0.30$ unless stated otherwise. Panel A holds the elasticity ratio at $\varepsilon_P/\varepsilon_R = 2.2$ and moves the level, so $\Delta\varepsilon$ adjusts. Panel B holds the gap at $\Delta\varepsilon = 3.6$ and moves the level, so the ratio drifts from 1.75 to 3.01. The markup span 1.21–1.61 of both panels runs from the 1980 to the 2016 aggregate markup of De Loecker et al. (2020) and contains the cost-weighted aggregate of Edmond et al. (2023) and the French firm-level median and mean of Burstein et al. (2025); DLEU and EMX denote De Loecker et al. (2020) and Edmond et al. (2023). The starred row (∗) is the baseline calibration $(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$, whose markup is an untargeted outcome; it lies on all three paths. In Panel C, row (a) holds the mean elasticity at $\bar{\varepsilon} = 4.44$ and varies $\Delta\varepsilon$ alone. Row (b) is the maximum over all elasticity levels at the fixed ratio, attained above the markup span. Row (c), daggered (†), is the supremum over $\lambda \in (0, 1)$ and $R \in [0, 1]$ at the baseline elasticities subject to $\chi_y < 1$: it is approached as $\lambda \to 0.504$ at $R = 0.0$ and attained at no admissible point.

Panel C of Table C.4 bounds how far the gap can move away from the baseline. Row (a) runs along the fixed-mean path: at the baseline λ and R, a 10 percent gap requires $\Delta\varepsilon \approx 7.7$, or $(\varepsilon_R, \varepsilon_P) = (1.36, 9.06)$ and a ratio of 6.6, about three times the calibrated one. The implied markup 1.44 lies inside the published span, so what rules out this parameterization is not the markup but near-unit-elastic demand from rich households. Condition (D.1) fails there, but the numerical check of Appendix D.5.2 finds that the symmetric price is still each firm's global profit maximum, so the maintained pricing assumption that Section 2 states holds. On the same path it fails once the gap

exceeds 7.87, when profit acquires a second, higher peak at a price high enough to sell almost only to rich households (Appendix D.9).

Rows (b) and (c) bound what the elasticity level and the distributional parameters can add. At the fixed ratio 2.2, the gap cannot exceed 4.3 percent at any elasticity level, but that ceiling lies at $\bar{\mathcal{M}} = 1.91$, above the published span; within the span the largest value is the 4.1 percent in the last row of Panel A. At $(\varepsilon_R, \varepsilon_P) = (3.0, 6.6)$ the gap is strictly decreasing in R and, for $R < 1$, strictly increasing in λ. Over all admissible (λ, R) it therefore has a supremum of 6.7 percent, approached at $R = 0.0$ as $\chi_y \to 1$ but attained at no admissible point.

C.3 Homogeneous-Elasticity Benchmarks

Table C.5 shows that, under Rotemberg pricing with a common adjustment cost ζ, the dispersion-pure gap against the markup-matched benchmark does not depend on how the shock is normalized, while gaps against benchmarks with a different steady-state markup do. In the starred markup-matched row of Panel A, which compares TE-TANK with four homogeneous TANK benchmarks, the pass-through and output-contraction ratios are both 1.033. Against the population-mean-preserving benchmark $\bar{\varepsilon} = 4.44$, pass-through rises by 17.9 percent, and Panel B attributes 14.1 percent of it to the higher steady-state markup; the two components compound rather than add. Almost all of this markup-level step comes from the flatter Phillips curve, because the demand locus of TANK($\bar{\varepsilon}$) is only 1.58 percent flatter than that of TANK(ε^*).

Panel C shows that the markup-matched comparison is invariant to the shock normalization, while the comparison with the population-mean-preserving benchmark is not. For an equal normalized NKPC wedge, the impact-inflation ratio is 1.033 against the markup-matched benchmark. An equal underlying real marginal-cost shifter instead enters as $u_t = \delta_2 \hat{u}_t$, while an equal log desired-markup shifter enters as $u_t = \delta_2 q_t$; the two structural-shifter columns are therefore identical. Because markup matching under a common ζ also matches δ_2, all three ratios remain 1.033 in the primary row. Against population-mean-preserving TANK, by contrast, the equal-wedge ratio is 1.179 while either structural-shifter ratio is 1.006. Section 3.3 shows that under Calvo pricing with a common θ, δ_2 differs across the two markup-matched economies, so the primary comparison loses this invariance.

TABLE C.5: Benchmark Dependence of the TE-TANK/TANK Comparison

Panel A: Slopes and TE-TANK/TANK ratios by homogeneous benchmark

	κ	χ_y	TE-TANK / TANK ratio	
TANK benchmark $\bar{\varepsilon}$			Pass-through	Output
3.933 (ε^*, markup-matched)*	0.0833	0.5637	1.033	1.033
4.44 (population-mean-preserving)	0.0977	0.5900	1.179	1.161
4.00 (standard TANK)	0.0852	0.5676	1.052	1.050
4.80 (pair mean)	0.1079	0.6055	1.284	1.252
TE-TANK $(\varepsilon_R, \varepsilon_P)=(3, 6.6)$	0.0797	0.5637	1.000	1.000
RANK (ε^*, ref.)	0.0833	0.0000	–	–

Panel B: Decomposition of the TE-TANK/TANK(4.44) gap: dispersion-pure × markup-level

Object	Total TE / TANK(4.44)	Dispersion-pure TE / interm.	Markup-level interm. / TANK(4.44)
Taylor pass-through Ψ_π	+17.94%	+3.32%	+14.15%
Output contraction $\lvert\Psi_y\rvert$	+16.10%	+3.32%	+12.37%

Panel C: TE-TANK/TANK impact-inflation ratio by shock normalization

TANK benchmark	Equal NKPC wedge	Equal real-MC shifter	Equal desired-markup shifter
3.933 (ε^*, markup-matched)*	1.033	1.033	1.033
4.44 (population-mean-preserving)	1.179	1.006	1.006
4.00 (standard TANK)	1.052	1.029	1.029
4.80 (pair mean)	1.284	0.991	0.991

Notes: Each upper-panel row is a homogeneous-elasticity TANK benchmark. The starred markup-matched row is the main-text benchmark. The next rows preserve the population mean, use the standard TANK elasticity, and use the unweighted pair mean. The κ and χ_y columns report each benchmark's own coefficients; the last two columns report TE-TANK relative to that benchmark. TE-TANK and markup-matched RANK are shown for reference. The second panel decomposes the total against the population-mean-preserving benchmark through the exactly markup-matched ε^* intermediate. The final panel reports impact-inflation ratios for equal normalized NKPC wedges, equal real marginal-cost shifters, and equal log desired-markup shifters. The last two coincide because both structural shifters enter the pricing gap with loading δ_2.

D Model Derivations and Core Robustness

This appendix derives the equilibrium conditions of Section 2 (Appendices D.1–D.8), states how they change under an alternative labor-market closure (Appendix D.8.1), and compares the firm-level superelasticity of demand with other demand systems and estimates (Appendix D.9). Appendix D.10 proves Propositions 1 and 2 of the main text and derives the fiscal sufficient statistic, the IS amplification, and the cost-push pass-through solution. Appendix D.11 states and proves Proposition D.1, on steady-state inequality and cost-push amplification. Appendix D.12 decomposes the technology and monetary policy shocks, and Appendix D.13 shows that consumption-dependent elasticities scale the composition channel by $1-\gamma$.

D.1 Maintained Assumptions

The results of Section 2 hold under the following restrictions; Propositions 1 and 2 state the ones their proofs use, and Proposition D.1 and the derivations of this appendix maintain them. Preferences and technology satisfy $\sigma > 0$, $\varphi \geq 0$, $\alpha \in [0,1)$, and $\zeta > 0$. The population split is interior, $\lambda \in (0,1)$, and the elasticities satisfy $\varepsilon_P \geq \varepsilon_R > 1$; the weak inequality lets the same statements cover the homogeneous-elasticity benchmarks, for which $\varepsilon_P = \varepsilon_R$. The fiscal instruments lie in the unit square, $(\delta, \tau) \in [0,1]^2$, so that $R = \delta(1-\tau) \in [0,1]$. Employment is demand-determined and common across types, $N_t^P = N_t^R \equiv N_t$, at the wage schedule of Debortoli and Galí (2018), $W_t/P_t = \mathcal{M}^w C_t^\sigma N_t^\varphi$ in average consumption C_t, with a constant wage markup $\mathcal{M}^w$ that satisfies the participation condition $\mathcal{M}^w \geq H_t^\sigma$ at the steady state and for shocks that are not too large, so that both types are willing to supply the hours firms demand (Appendix D.2). The cost-push results also maintain $\chi_y = \Gamma G < 1$, which keeps the IS semi-elasticity finite (Appendix D.10.5), and the log-linear results are evaluated at the steady state of Proposition 1. Two of these deserve comment. The restriction $\varepsilon_R > 1$ is what keeps the desired markup finite, and the participation condition is tighter for rich households, whose consumption exceeds the average: at $\mathcal{M}^w = H_t^\sigma$ they are exactly on their labor-supply curve, and at any smaller wage markup they would supply fewer hours than firms demand. The constraint is slack for hand-to-mouth households, whose consumption is below the average. The population split needs $\lambda > 0$ for the demand-composition feedback to have any content: at $\lambda = 0$ the feedback vanishes for every R.

Finally, at every steady state considered, the symmetric price is each firm's global profit maximizer when all other firms charge it, so the root of the pricing first-order condition is an equilibrium. This is not automatic: an elasticity gap makes demand more convex in the price, because a higher price loses mostly the elastic poor customers. A sufficient condition in primitives bounds the elasticity gap relative to the rich elasticity,

$$(\varepsilon_P - \varepsilon_R)^2 < \frac{4\,\varepsilon_R(\varepsilon_R - 1)(1 - \alpha + \alpha\varepsilon_R)}{1-\alpha}. \tag{D.1}$$

Under (D.1) each firm's profit is single-peaked in its own price at any composition of demand. At the baseline calibration the two sides of (D.1) are 12.96 and 60. Because the condition is sufficient but not necessary, we also verify global optimality numerically: it holds at every calibration reported in the paper. On the fixed-mean path of Appendix C.2 it fails only once the elasticity gap exceeds 7.87, above the widest gap the paper reports, 7.69. Appendix D.5.2 derives the condition and describes the check, and Appendix D.9 shows a pair just past that threshold.

D.2 Household Problem

Fiscal policy The government collects a fraction δ of firm profits and redistributes them according to

$$\begin{aligned} T_t^R &= \left(1 + \frac{\tau\lambda}{1-\lambda}\right)\delta D_t, \\ T_t^P &= (1-\tau)\,\delta D_t, \end{aligned} \tag{D.2}$$

where $\tau \in [0,1]$ governs the tilt of transfers toward rich households ($\tau = 0$ is the most progressive scheme, equal per-capita transfers). The government budget balances each period: $(1-\lambda)T_t^R + \lambda T_t^P = \delta D_t$. The rule is the liquid–illiquid profit structure of Debortoli and Galí (2018): a fraction $1-\delta$ of profits is paid as dividends on tradable shares, which only rich households hold, and the remaining fraction δ is distributed by their two-agent transfer rule with weight τ on shareholders, so that T_t^R and T_t^P are their T_t^U and T_t^K. Appendix D.10.1 shows that only $R = \delta(1-\tau)$ matters, the combination that also governs their consumption-gap index.

Rich households Rich households maximize expected lifetime utility (1) subject to the budget constraint

$$C_t^R + \frac{B_t^R}{P_t^R} + Q_t F_t^R = \frac{B_{t-1}^R(1+i_{t-1})}{P_t^R} + \frac{W_t}{P_t^R}N_t^R + \big((1-\delta)D_t + Q_t\big)F_{t-1}^R + T_t^R, \tag{D.3}$$

where B_t^R are nominal bond holdings, F_t^R are equity shares, Q_t is the share price, D_t are real dividends, W_t is the nominal wage, and T_t^R are real transfers. Employment is demand-determined: each household supplies the hours $N_t^R = N_t$ that firms demand at the going wage, as in Debortoli and Galí (2018), so hours are not a choice variable. With consumption expenditure aggregated as in Appendix D.3, attach the multiplier ξ_t to (D.3). The first-order conditions for C_t^R, B_t^R, and F_t^R are $\xi_t = (C_t^R)^{-\sigma}Z_t$, $\xi_t = \beta\mathbb{E}_t[\xi_{t+1}(1+i_t)/\Pi_{t+1}^R]$, and $\xi_t Q_t = \beta\mathbb{E}_t[\xi_{t+1}((1-\delta)D_{t+1} + Q_{t+1})]$; eliminating ξ_t gives the bond and equity Euler equations, which are those of the TANK economy of Debortoli and Galí (2018). The real wage is set by their wage schedule in average consumption,

$$\frac{W_t}{P_t} = \mathcal{M}^w\, C_t^\sigma\, N_t^\varphi, \qquad C_t \equiv (1-\lambda)C_t^R + \lambda C_t^P, \tag{D.4}$$

$$1 = \mathbb{E}_t\left[\Lambda_{t+1,t}\frac{1+i_t}{\Pi_{t+1}^R}\right], \tag{D.5}$$

$$Q_t = \mathbb{E}_t\left[\Lambda_{t+1,t}\big((1-\delta)D_{t+1} + Q_{t+1}\big)\right],$$

with $\Lambda_{t+1,t} \equiv \beta(C^R_{t+1}/C^R_t)^{-\sigma} Z_{t+1}/Z_t$, $\Pi^R_{t+1} \equiv P^R_{t+1}/P^R_t$, and a constant gross wage markup $\mathcal{M}^w > 1$. We read the schedule as standing in for non-competitive wage setting: a union, say, sets one wage for all workers as a markup over the marginal rate of substitution evaluated at average consumption, and firms choose employment at that wage. Average consumption C_t is the population-weighted sum of the two type composites, each in its own units. It equals aggregate real consumption expenditure deflated by P_t whenever the two type-specific price indices coincide, $P^R_t = P^P_t = P_t$, which holds exactly in the symmetric Rotemberg equilibrium and to first order under Calvo pricing, where P_t denotes the average price level of Appendix D.8; these are the only cases in which the schedule is evaluated. The schedule requires that every household be willing to supply N_t at that wage, $\mathcal{M}^w C^\sigma_t N^\varphi_t \geq (C^h_t)^\sigma N^\varphi_t$, that is, $\mathcal{M}^w \geq (C^h_t/C_t)^\sigma$ for $h \in \{R, P\}$. On the admissible branch $C^P_t < C_t < C^R_t$, so the constraint is tighter for the rich and reads $\mathcal{M}^w \geq H^\sigma_t$ with $H_t \equiv C^R_t/C_t$. Debortoli and Galí (2018) require the real wage to be at least every household's marginal rate of substitution, $C_t(s)^\sigma N^\varphi_t$. As Debortoli and Galí (2024) do, we impose the condition at the steady state and for shocks that are not too large: a wage markup strictly above the steady-state bound satisfies it in a neighborhood of the steady state. We do not calibrate $\mathcal{M}^w$: it is a level parameter that scales steady-state hours and output (Appendix D.6) and enters no coefficient of the log-linear system. At the baseline steady state the bound is $\bar{H}^\sigma = 1.235$, and along the fiscal branch it is largest at $R = 0$, where it equals 1.342. The alternative closure in which the wage clears the rich household's intratemporal condition, $W_t/P_t = (C^R_t)^\sigma N^\varphi_t$, with hand-to-mouth households rationed at the common hours, is the schedule (D.4) with the time-varying wage markup $\mathcal{M}^w_t = H^\sigma_t$, the smallest markup at which every household supplies the hours demanded.

Poor households Poor households cannot save or borrow, so they spend all labor income plus transfers each period:

$$C^P_t = \frac{W_t}{P^P_t} N^P_t + T^P_t. \tag{D.6}$$

They supply the common hours $N^P_t = N_t$ at the wage (D.4), and (D.6) determines C^P_t. Their participation constraint, $\mathcal{M}^w \geq (C^P_t/C_t)^\sigma$, is slack whenever $C^P_t < C_t$, because $\mathcal{M}^w > 1$. The household block, the transfer rule (D.2), and the wage schedule (D.4) are thus those of the TANK economy of Debortoli and Galí (2018). Besides the two demand elasticities, the model departs from it only in decreasing returns to labor, $\alpha > 0$, whereas they assume linear production.

D.3 CES Aggregation

Each household type $h \in \{R, P\}$ minimizes expenditure $\int_0^1 P_t(i) C_t^h(i)\, di$ subject to the CES aggregator $C_t^h = \left(\int_0^1 [C_t^h(i)]^{1-1/\varepsilon_h} di\right)^{\varepsilon_h/(\varepsilon_h-1)}$. This is the standard Dixit–Stiglitz expenditure-minimization problem; standard steps (first-order conditions across varieties, substitution into the aggregator, and the price-index definition $P_t^h \equiv (\int_0^1 P_t(j)^{1-\varepsilon_h} dj)^{1/(1-\varepsilon_h)}$) yield the CES demand function

$$C_t^h(i) = \left(\frac{P_t(i)}{P_t^h}\right)^{-\varepsilon_h} C_t^h. \tag{D.7}$$

From variety demand to aggregate output. Price adjustment absorbs goods. We adopt the standard Rotemberg convention that adjustment costs are paid in the same composites of varieties that households consume. Aggregate adjustment resources $X_t \equiv k_t Y_t$, with $Y_t \equiv \sum_h s_h Y_t^h$ and $k_t \equiv (\zeta/2)\int_0^1 \big(P_t(j)/P_{t-1}(j) - 1\big)^2 dj$, are the sum of the firm-level costs $(\zeta/2)\big(P_t(j)/P_{t-1}(j) - 1\big)^2 Y_t$ in the firm's problem of Appendix D.5. They are purchased as type-specific bundles in proportion to each type's output, $X_t^h = k_t Y_t^h$, so that $\sum_h s_h X_t^h = X_t$, and each bundle is spread across varieties with the same CES weights as type h's consumption, $X_t^h(i) = (P_t(i)/P_t^h)^{-\varepsilon_h} X_t^h$. Output of variety i absorbed by type h, $Y_t^h(i) = C_t^h(i) + X_t^h(i)$, is therefore

$$Y_t^h(i) = \left(\frac{P_t(i)}{P_t^h}\right)^{-\varepsilon_h} Y_t^h, \qquad Y_t^h \equiv C_t^h + X_t^h,$$

for any configuration of prices, not only at symmetric prices. Because each firm has measure zero, its own price leaves k_t, X_t^h, Y_t^h, and P_t^h unchanged, so the adjustment cost of firm i enters its demand only through aggregates it takes as given. The same convention gives $C_t^h = Y_t^h - X_t^h = (1 - k_t) Y_t^h$, and in the symmetric equilibrium $k_t = (\zeta/2)(\Pi_t - 1)^2$, so

$$C_t^h = \Delta^p(\Pi_t)\, Y_t^h, \qquad \Delta^p(\Pi_t) \equiv 1 - \frac{\zeta}{2}(\Pi_t - 1)^2, \tag{D.8}$$

where Δ^p is the resource loss from price adjustment; this makes $C_t^R/C_t = Y_t^R/Y_t$ exact. Aggregate output of variety i, summed across household types with population share s_h, is

$$Y_t(i) = \sum_{h\in\{R,P\}} s_h Y_t^h(i) = \sum_{h\in\{R,P\}} s_h \left(\frac{P_t(i)}{P_t^h}\right)^{-\varepsilon_h} Y_t^h. \tag{D.9}$$

D.4 Heterogeneity Index

Equity and transfers Total equity supply is one and only rich households hold shares, so market clearing gives $F_t^R = F_{t-1}^R = 1/(1-\lambda)$, and the purchase and resale-value terms of (D.3) cancel period by period, $Q_t\big(F_{t-1}^R - F_t^R\big) = 0$. Bonds are in zero net supply. The transfer rule (D.2) can be written as $T_t^P = (1-\tau)\delta D_t = RD_t$ and $T_t^R = \delta(1-\lambda+\tau\lambda)D_t/(1-\lambda) = (\delta - \lambda R)D_t/(1-\lambda)$, so the rich household's profit-related income is

$$(1-\delta)D_t F_{t-1}^R + T_t^R = \frac{(1-\delta)+\delta-\lambda R}{1-\lambda} D_t = C\, D_t, \qquad C \equiv \frac{1-\lambda R}{1-\lambda}. \tag{D.10}$$

With $P_t^R = P_t$ and $N_t^R = N_t$, the rich budget constraint therefore reduces to $C_t^R = (W_t/P_t)N_t + CD_t$.

Levels Divide by Y_t and use $C_t^R = H_t C_t = H_t \Delta^p(\Pi_t) Y_t$ from (D.8), labor demand $W_t/P_t = MC_t(1-\alpha)A_t N_t^{-\alpha}$ with the realized markup $\mathcal{M}_t \equiv 1/MC_t$, and $Y_t = A_t N_t^{1-\alpha}$, so that the labor share is $W_t N_t/(P_t Y_t) = (1-\alpha)/\mathcal{M}_t$:

$$H_t = \frac{1}{\Delta^p(\Pi_t)}\left[\frac{1-\alpha}{\mathcal{M}_t} + C\,\frac{D_t}{Y_t}\right].$$

The resource identity $C_t = \Delta^p(\Pi_t)Y_t = W_t N_t/P_t + D_t$ gives $D_t/Y_t = \Delta^p(\Pi_t) - (1-\alpha)/\mathcal{M}_t$, and substituting yields

$$H_t = C - \frac{D}{\Delta^p(\Pi_t)\,\mathcal{M}_t}, \qquad D \equiv (1-\alpha)(C-1) = \frac{(1-\alpha)\lambda(1-R)}{1-\lambda}. \tag{D.11}$$

At the zero-inflation steady state, where $\Delta^p(1) = 1$, this is equation (11).

Log-linearization Because Δ^p is quadratic in $\Pi_t - 1$, $1/\Delta^p(\Pi_t) = 1$ to first order, and H_t moves only through $\mathcal{M}_t$: $\hat{h}_t = [D/(\bar{\mathcal{M}}\bar{H})]\,\hat{\mu}_t \equiv \Gamma\hat{\mu}_t$, with $\Gamma = (1-\alpha)\lambda(1-R)/[(1-\lambda)\bar{\mathcal{M}}\bar{H}]$. This relation uses only the rich budget constraint and labor demand, so it holds under any wage closure at common hours. Substituting $\hat{\mu}_t = -G\tilde{y}_t$ from (D.24) gives $\hat{h}_t = -\chi_y \tilde{y}_t$ with $\chi_y = \Gamma G$, which is (13).

D.5 Firm Problem and Flexible-Price Markup

This subsection derives the firm's pricing first-order condition, from which both the Phillips curve and the flexible-price markup follow. The firm chooses $P_t(i)$ to maximize discounted real profits $\max_{P_t(i)} \mathbb{E}_t \sum_{s=t}^{\infty} (\Lambda_{s,t}/P_s)\big[P_s(i)Y_s(i) - P_s\mathcal{C}(Y_s(i)) - (\zeta/2)\big(P_s(i)/P_{s-1}(i) - 1\big)^2 Y_s P_s\big]$, where $\Lambda_{s,t} \equiv \beta^{s-t}U'_{c,s}/U'_{c,t}$ is the stochastic discount

factor, subject to the demand constraint (D.9), $Y_s(i) = \sum_h s_h (P_s(i)/P_s^h)^{-\varepsilon_h} Y_s^h$, which holds at any price of firm i under the convention of Appendix D.3. The firm takes P_s, P_s^h, Y_s, and Y_s^h, and hence aggregate adjustment resources, as given. It values its adjustment cost at P_s, which equals the cost of the adjustment composite because $P_s^R = P_s^P = P_s$ in the symmetric equilibrium. Differentiate with respect to $P_t(i)$: the revenue and cost terms enter at $s = t$, and the Rotemberg adjustment-cost term enters at both $s = t$ (current-period cost) and $s = t+1$ (next period's cost, since $P_t(i)$ enters $P_{t+1}(i)/P_t(i)$). Evaluating each term under the symmetric equilibrium $P_t(i) = P_t$ and denoting real marginal cost $MC_t \equiv \mathcal{C}'(Y_t(i))$ gives the first-order condition

$$\begin{aligned}&\sum_h s_h \frac{Y_t^h}{P_t}(1-\varepsilon_h)\left(\frac{P_t(i)}{P_t^h}\right)^{-\varepsilon_h} + MC_t \sum_h s_h \frac{\varepsilon_h}{P_t(i)}\left(\frac{P_t(i)}{P_t^h}\right)^{-\varepsilon_h} Y_t^h \\ &- \zeta\left(\Pi_t - 1\right)\Pi_t \frac{Y_t}{P_t} + \mathbb{E}_t\left[\Lambda_{t+1,t}\zeta\left(\Pi_{t+1}-1\right)\Pi_{t+1}\frac{Y_{t+1}}{P_t}\right] = 0. \end{aligned} \tag{D.12}$$

Imposing the symmetric equilibrium $P_t(i) = P_t^h = P_t$, dividing by $\zeta Y_t/P_t$, and defining $H_t^h \equiv Y_t^h/Y_t$ as type h's output relative to the aggregate average gives

$$\sum_h \frac{s_h}{\zeta} H_t^h (1-\varepsilon_h) + MC_t \sum_h \frac{s_h}{\zeta}\varepsilon_h H_t^h + \mathbb{E}_t\left[\Lambda_{t+1,t}\left(\Pi_{t+1}-1\right)\Pi_{t+1}\frac{Y_{t+1}}{Y_t}\right] = \left(\Pi_t - 1\right)\Pi_t. \tag{D.13}$$

At the zero-inflation steady state ($\Pi_{t+1} = \Pi_t = \Pi = 1$) the forward-looking term and the right-hand side vanish, giving steady-state marginal cost

$$\overline{MC} = \frac{\sum_h s_h H^h (\varepsilon_h - 1)}{\sum_h s_h \varepsilon_h H^h} = \frac{\sum_h s_h H^h \varepsilon_h \left(1 - \frac{1}{\varepsilon_h}\right)}{\sum_h s_h \varepsilon_h H^h}, \tag{D.14}$$

in which H^h without a time subscript denotes the steady-state value of H_t^h, so that $H^R = \bar{H}$.

D.5.1 Flexible-Price Markup

With $\zeta = 0$ the adjustment-cost terms of (D.12) drop out, and at the symmetric equilibrium, after multiplying by P_t/Y_t, the condition reads $\sum_h s_h(1-\varepsilon_h)H_t^h + MC_t \sum_h s_h \varepsilon_h H_t^h = 0$. The flexible-price markup is therefore

$$\mathcal{M}_t^n \equiv \frac{1}{MC_t} = \frac{\sum_h s_h \varepsilon_h H_t^h}{\sum_h s_h \left(\varepsilon_h - 1\right) H_t^h}. \tag{D.15}$$

Goods-market clearing, $Y_t = \sum_h s_h Y_t^h$, gives the adding-up condition $\sum_h s_h H_t^h = (1-\lambda)H_t + \lambda H_t^P = 1$, so $H_t^P = [1-(1-\lambda)H_t]/\lambda$ and the denominator equals the

numerator minus one. The numerator is the effective elasticity: differentiating (D.9) gives $-\partial \ln Y_t(i)/\partial \ln P_t(i) = \sum_h s_h \varepsilon_h (P_t(i)/P_t^h)^{-\varepsilon_h} Y_t^h / Y_t(i)$, which at the symmetric allocation equals $\sum_h s_h \varepsilon_h H_t^h = \varepsilon_{\text{eff},t} = H_t(1-\lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P$, each type weighted by its expenditure share $s_h H_t^h$. Hence $\mathcal{M}_t^n = \varepsilon_{\text{eff},t}/(\varepsilon_{\text{eff},t} - 1)$, which is equation (7).

D.5.2 Pricing Second-Order Condition

Condition (D.1) makes each firm's static profit single-peaked in its own price, so the root of the first-order condition (D.12) is the firm's optimal price. Consider the zero-inflation steady state, in which one firm sets the log relative price $x \equiv \ln(P(i)/P)$ while all other firms charge P. Let $\omega_h \equiv s_h \bar{H}^h$ denote the expenditure shares, so that the firm's relative demand is $\mathcal{D}(x) = \sum_h \omega_h \exp(-\varepsilon_h x)$, with $\mathcal{D}(0) = 1$ and $-\mathcal{D}'(0) = \varepsilon_{\text{eff}}$. Since $Y(i) = AN(i)^{1-\alpha}$, the real wage bill per unit of aggregate output is $(1-\alpha)\overline{MC}\,\mathcal{D}(x)^{1/(1-\alpha)}$, and real profit per unit of aggregate output is

$$\varpi(x) = \exp(x)\,\mathcal{D}(x) - (1-\alpha)\,\overline{MC}\,\mathcal{D}(x)^{1/(1-\alpha)}. \tag{D.16}$$

The condition $\varpi'(0) = 0$ is the steady-state condition (D.14), $\overline{MC} = (\varepsilon_{\text{eff}} - 1)/\varepsilon_{\text{eff}}$. Differentiating twice and using $\mathcal{D}''(0) = \text{Var}_\omega(\varepsilon) + \varepsilon_{\text{eff}}^2$ gives

$$\varpi''(0) = \frac{1}{\varepsilon_{\text{eff}}}\left[\text{Var}_\omega(\varepsilon) - \frac{\varepsilon_{\text{eff}}(\varepsilon_{\text{eff}} - 1)(1 - \alpha + \alpha\varepsilon_{\text{eff}})}{1-\alpha}\right], \qquad \text{Var}_\omega(\varepsilon) = \omega_R \omega_P (\varepsilon_P - \varepsilon_R)^2. \tag{D.17}$$

The second term in brackets is the usual curvature of a monopolist with decreasing returns. The first is the expenditure-weighted variance of the elasticity across the firm's customers, and it works against concavity: a price increase loses mostly elastic poor customers, so the elasticity of the remaining demand falls as the price rises. With a common elasticity the variance is zero and $\varpi''(0) < 0$.

The same formula holds at any critical point x^* of ϖ, up to a positive factor, once ω_h is replaced by the expenditure shares at that price, $\omega_h(x^*) \propto \omega_h \exp(-\varepsilon_h x^*)$, and ε_{eff} by their weighted elasticity: $\varpi(x^* + x')$ has the form (D.16) in x', and $\varpi'(x^*) = 0$ plays the role of the steady-state normalization. The variance is at most $(\varepsilon_P - \varepsilon_R)^2/4$, and the second term is increasing in the effective elasticity, which is at least ε_R. Condition (D.1) therefore makes the bracket negative at every critical point, whatever the expenditure shares there. Every critical point is then a strict local maximum, so ϖ has at most one. Since $\varpi(x) \to -\infty$ as $x \to -\infty$ and $\varpi(x) \to 0 < \varpi(0)$ as $x \to \infty$, it has exactly one, $x = 0$, which is the global maximum of static profit. The Rotemberg cost is non-negative and zero on a constant price path, so the constant symmetric price then also maximizes the discounted objective in the zero-inflation steady state, which is the maintained pricing assumption that Section 2 states.

At the baseline calibration, $\mathrm{Var}_\omega(\varepsilon) = 2.49$ against 34.2 for the second term, so $\varpi''(0) = -8.07$. Condition (D.1) does not involve λ or R, so at the baseline elasticities it holds on the whole fiscal branch and at every (λ, R), including the supremum in Panel C of Table C.4. The numerical check locates every critical point of (D.16) and compares profits, so it also covers points at which (D.1) fails; at the high-markup end of the fixed-gap path, for example, the symmetric price remains the global maximizer. The local condition $\varpi''(0) < 0$ alone does not ensure a global maximum. On the fixed-mean path, which holds the population mean at $\bar{\varepsilon} = 4.44$ and widens the gap, profit acquires a second, higher peak at a high relative price once the elasticity gap exceeds 7.87, and beyond that threshold the root of Proposition 1(i) is not an equilibrium under either pricing convention. Every calibration the paper reports lies below it; the widest gap, the 10 percent point of Panel C at 7.69 points, passes the check (Appendix D.9).

D.6 Steady State

Substituting (12) into (11) and clearing the denominator, as in the proof of Proposition 1(i), reduces the fixed point to a single quadratic in $\bar{H}$,

$$A\bar{H}^2 + \big[B - A(C - D)\big]\bar{H} - \big[B(C - D) + D\big] = 0, \tag{D.18}$$

which determines $\bar{H}$ and, through (12), $\bar{\mathcal{M}}$; neither depends on the wage markup $\mathcal{M}^w$, which enters only the level of hours and output. That level follows from the markup definition (D.22) at $A = 1$, with $C = Y$ and $Y = N^{1-\alpha}$, which give $\bar{\mathcal{M}} = (1-\alpha)\bar{N}^{-[\sigma(1-\alpha)+\varphi+\alpha]}/\mathcal{M}^w$ and hence, since $(1-\alpha)/[\sigma(1-\alpha)+\varphi+\alpha] = 1/G$,

$$\bar{N}^{\,\sigma(1-\alpha)+\varphi+\alpha} = \frac{1-\alpha}{\bar{\mathcal{M}}\,\mathcal{M}^w}, \qquad \bar{Y} = \left(\frac{1-\alpha}{\bar{\mathcal{M}}\,\mathcal{M}^w}\right)^{1/G} = \left(\frac{1-\alpha}{\bar{\mathcal{M}}\,\mathcal{M}^w}\right)^{\frac{1-\alpha}{\alpha+\varphi+\sigma(1-\alpha)}}. \tag{D.19}$$

Participation requires $\mathcal{M}^w \geq \bar{H}^\sigma$ at the steady state (Appendix D.2). Along the fiscal branch $\bar{H}$ is largest at $R = 0$ (Proposition 1(ii)), so one constant $\mathcal{M}^w \geq \bar{H}(R{=}0)^\sigma = 1.342$ covers every steady state on the fiscal branch at the baseline λ; the baseline bound is $\bar{H}^\sigma = 1.235$. The hours and output of the closure with $\mathcal{M}^w_t = H^\sigma_t$ are the case $\mathcal{M}^w = \bar{H}^\sigma$.

D.7 Dynamic IS Curve

Substituting $C^R_t = H_t C_t$ and $\Pi^R_{t+1} = \Pi_{t+1}$ into the Euler equation (D.5) gives

$$1 = \mathbb{E}_t\left[\beta\left(\frac{H_{t+1}C_{t+1}}{H_t C_t}\right)^{-\sigma}\frac{Z_{t+1}}{Z_t}\frac{1+i_t}{\Pi_{t+1}}\right].$$

Log-linearizing around the zero-inflation steady state, with $C_t = Y_t$ to first order and $\mathbb{E}_t z_{t+1} = \rho_z z_t$, and subtracting natural output, with $r_t^n = (1-\rho_z)z_t + \sigma \mathbb{E}_t[\Delta \hat{y}_{t+1}^n]$, gives

$$\tilde{y}_t = \mathbb{E}_t[\tilde{y}_{t+1}] + \mathbb{E}_t[\Delta \hat{h}_{t+1}] - \frac{1}{\sigma}\left(\hat{\imath}_t - \mathbb{E}_t[\pi_{t+1}] - r_t^n\right),$$

which contains the expected change in the heterogeneity index, as in TANK (Debortoli and Galí, 2018). Substituting $\hat{h}_t = -\chi_y \tilde{y}_t$ from (13), with $\chi_y = \Gamma G$, turns that term into $-\chi_y(\mathbb{E}_t[\tilde{y}_{t+1}] - \tilde{y}_t)$; collecting terms and dividing by $1-\chi_y$ gives (14), whose IS semi-elasticity $1/[\sigma(1-\chi_y)] = 1/[\sigma(1-\Gamma G)]$ exceeds the RANK value $1/\sigma$ by the factor $1/(1-\chi_y)$.

D.8 New Keynesian Phillips Curve

Linearized pricing condition Around the zero-inflation steady state, $(\Pi_t - 1)\Pi_t \approx \pi_t$, and the forward term of (D.13) becomes $\beta \mathbb{E}_t \pi_{t+1}$, because $\Lambda_{t+1,t}$, Π_{t+1}, and Y_{t+1}/Y_t multiply a term that is zero in steady state. With $H_t^h \approx H^h(1+\hat{h}_t^h)$ and $MC_t \approx \overline{MC}(1+\widehat{mc}_t)$,

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \frac{1}{\zeta}\sum_h s_h H^h \left[\left(1-\varepsilon_h(1-\overline{MC})\right)\hat{h}_t^h + \varepsilon_h \overline{MC}\,\widehat{mc}_t\right]. \tag{D.20}$$

Two identities collapse the bracket. Adding-up, $(1-\lambda)H_t + \lambda H_t^P = 1$, gives $\lambda H^P \hat{h}_t^P = -(1-\lambda)H\hat{h}_t$, with $\hat{h}_t \equiv \hat{h}_t^R$ and $H \equiv H^R = \bar{H}$, so the coefficient on $\hat{h}_t$ is $(1-\lambda)\bar{H}(1-\overline{MC})(\varepsilon_P - \varepsilon_R)/\zeta = \delta_2 \eta_\mu$, using $1-\overline{MC} = 1/\varepsilon_{\text{eff}}$ from (D.14) and (19). The coefficient on $\widehat{mc}_t$ is $\overline{MC}\sum_h s_h H^h \varepsilon_h/\zeta = \overline{MC}\,\varepsilon_{\text{eff}}/\zeta$, which gives

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \delta_2\left(\widehat{mc}_t + \eta_\mu \hat{h}_t\right), \qquad \delta_2 = \frac{\varepsilon_{\text{eff}} - 1}{\zeta} = \frac{1}{\zeta}\left[\bar{H}(1-\lambda)(\varepsilon_R - \varepsilon_P) + \varepsilon_P - 1\right]. \tag{D.21}$$

Since the log desired markup $\ln[\varepsilon_{\text{eff},t}/(\varepsilon_{\text{eff},t}-1)]$ deviates by $\eta_\mu \hat{h}_t$ and $\hat{\mu}_t = -\widehat{mc}_t$, inflation responds to the gap between the realized and the desired markup.

Marginal cost from the wage schedule Real marginal cost is the real wage over the marginal product of labor, $MC_t = (W_t/P_t)/[(1-\alpha)A_t N_t^{-\alpha}]$. Hours are common and demand-determined, and the real wage is the schedule (D.4) in average consumption, $W_t/P_t = \mathcal{M}^w C_t^\sigma N_t^\varphi$. The realized markup $\mathcal{M}_t \equiv 1/MC_t$ is therefore

$$\mathcal{M}_t = \frac{(1-\alpha)\,A_t\,C_t^{-\sigma}N_t^{-\alpha-\varphi}}{\mathcal{M}^w}. \tag{D.22}$$

With $C_t = Y_t$ to first order (the Rotemberg resource cost is second order) and $n_t = (y_t - a_t)/(1-\alpha)$ from (4),

$$\hat{\mu}_t = -\widehat{mc}_t = \Psi_a a_t - G\hat{y}_t, \qquad \Psi_a \equiv \frac{1+\varphi}{1-\alpha}, \tag{D.23}$$

and the constant $\mathcal{M}^w$ drops out. The distribution does not enter marginal cost: the wage responds to average consumption and hours, not to who consumes, so $\hat{h}_t$ reaches the Phillips curve only through the desired markup.

Natural output Under flexible prices $\mathcal{M}_t = \mathcal{M}^n(H_t)$ from (D.15), so H_t^n solves (D.11) with $\Delta^p = 1$, an equation that involves neither A_t nor Z_t. Hence $H_t^n = \bar{H}$ and $\hat{\mu}_t^n = 0$, and setting $\hat{\mu}_t = 0$ in (D.23) gives $\hat{y}_t^n = \Psi_a a_t / G$, the natural output of Debortoli and Galí (2018) with decreasing returns to labor. Subtracting,

$$\hat{\mu}_t = -G\tilde{y}_t, \qquad \text{equivalently} \qquad \widehat{mc}_t = G\tilde{y}_t. \tag{D.24}$$

Shocks that moved the composition of demand directly, such as stochastic redistribution, would move H_t^n and add a cost-push term.

Result Substituting (D.24) into (D.21) gives (15) with $\kappa_y = \delta_2 G$ and $\kappa_h = \delta_2 \eta_\mu = (1-\lambda)\bar{H}(1-\overline{MC})(\varepsilon_P - \varepsilon_R)/\zeta$. The loading on inequality is the composition channel alone. It is zero when $\varepsilon_P = \varepsilon_R$, so with a common elasticity the Phillips curve (15) is that of RANK at the same steady state, whatever the distribution of consumption, as in Debortoli and Galí (2018); the distribution then enters only the IS curve.

Cost-push normalization The disturbance u_t is appended to (D.21) as an additive wedge with unit coefficient. A shifter $\hat{u}_t$ of real marginal cost enters as $u_t = \delta_2 \hat{u}_t$. The markup gap $\hat{\mu}_t = -\widehat{mc}_t$ of (D.23) then remains the labor-cost markup, price over the marginal cost of labor, and the markup over total marginal cost is $\hat{\mu}_t - \hat{u}_t$. Its impact response per unit of $\hat{u}_t$ is $\delta_2 G\, \Theta \Psi_\pi - 1$, by Proposition D.1(iii), and is negative at the baseline calibration, because prices absorb only part of the shifter on impact. A shifter q_t of the log desired markup, which then deviates by $\eta_\mu \hat{h}_t + q_t$, enters the markup gap and hence also as $u_t = \delta_2 q_t$, so the two normalizations coincide, including in homogeneous TANK, where $\delta_2 = (\varepsilon - 1)/\zeta$.

Calvo pricing Under Rotemberg pricing all firms stay at the symmetric price, so the first-order condition never involves the curvature of a firm's own demand curve, and $\delta_2 = (\varepsilon_{\text{eff}} - 1)/\zeta$ carries no curvature term. Suppose instead that a firm can reset its price with probability $1-\theta$ each period, and let $\lambda_\theta \equiv (1-\theta)(1-\beta\theta)/\theta$. There are no

adjustment costs, so $Y_t^h = C_t^h$. A resetting firm's relative price differs from the average, so the shape of its own demand enters. The derivation has five steps.

Reset problem. A firm that resets in period t chooses $P_t^\#$ and still charges it in period $t+k$ with probability θ^k. By (D.9), its output in $t+k$ is

$$Y_{t+k|t} = \sum_h s_h \left(\frac{P_t^\#}{P_{t+k}^h} \right)^{-\varepsilon_h} Y_{t+k}^h. \tag{D.25}$$

It produces with (4) at the common wage, so its real cost of producing Y is $\mathcal{C}_{t+k}(Y) = (W_{t+k}/P_{t+k})(Y/A_{t+k})^{1/(1-\alpha)}$. Its real marginal cost therefore depends on its own output,

$$MC_{t+k|t} \equiv \mathcal{C}'_{t+k}(Y_{t+k|t}) = MC_{t+k} \left(\frac{Y_{t+k|t}}{Y_{t+k}} \right)^{\alpha/(1-\alpha)}, \tag{D.26}$$

where $MC_{t+k} \equiv \mathcal{C}'_{t+k}(Y_{t+k})$ is marginal cost at aggregate output $Y_{t+k} = \sum_h s_h Y_{t+k}^h$. The firm takes all aggregates as given and solves

$$\max_{P_t^\#} \mathbb{E}_t \sum_{k=0}^{\infty} \theta^k \Lambda_{t+k,t} \left[\frac{P_t^\#}{P_{t+k}} Y_{t+k|t} - \mathcal{C}_{t+k}(Y_{t+k|t}) \right] \quad \text{subject to (D.25)},$$

with the stochastic discount factor $\Lambda_{t+k,t}$ of Appendix D.5. The problem involves no firm-specific state, so all resetting firms choose the same $P_t^\#$.

Reset condition in levels. By (D.25), $\partial Y_{t+k|t}/\partial P_t^\# = -\varepsilon_{t+k|t} Y_{t+k|t}/P_t^\#$, where

$$\varepsilon_{t+k|t} \equiv \sum_h \omega_{h,t+k|t}\, \varepsilon_h, \qquad \omega_{h,t+k|t} \equiv \frac{s_h \left(P_t^\#/P_{t+k}^h\right)^{-\varepsilon_h} Y_{t+k}^h}{Y_{t+k|t}},$$

is the firm's own-price elasticity. It is the average of the two elasticities under the firm's own expenditure shares $\omega_{h,t+k|t}$. These shares depend on the firm's relative price and on the composition of demand, $H_{t+k}^h = Y_{t+k}^h/Y_{t+k}$; at the average price they equal $s_h H_{t+k}^h$. Set the derivative of the objective to zero, multiply by $-P_t^\#$, and write $\mathcal{M}_{t+k|t}^d \equiv \varepsilon_{t+k|t}/(\varepsilon_{t+k|t}-1)$ for the firm's desired markup. This gives

$$\mathbb{E}_t \sum_{k=0}^{\infty} \theta^k \Lambda_{t+k,t}\, Y_{t+k|t}\, (\varepsilon_{t+k|t}-1) \left[\frac{P_t^\#}{P_{t+k}} - \mathcal{M}_{t+k|t}^d\, MC_{t+k|t} \right] = 0. \tag{D.27}$$

With a common elasticity, $\varepsilon_{t+k|t} - 1$ and $\mathcal{M}_{t+k|t}^d$ are constants, and (D.27) is the textbook reset condition. Here both depend on the reset price and on H_{t+k}. Let lowercase letters denote logs and $x_{t+k|t} \equiv p_t^\# - p_{t+k}$. With $P_{t+k}^h = P_{t+k}$, which the price-index step below establishes to first order, the term that multiplies $\theta^k \Lambda_{t+k,t}$, per unit of Y_{t+k}, is minus the marginal profit $\varpi'_{t+k}(x_{t+k|t})$, where ϖ_{t+k} is (D.16) at the expenditure shares

and marginal cost of period $t+k$.

Log-linearization. In the zero-inflation steady state, $P_t^{\#} = P_{t+k} = P_{t+k}^h$, $\varepsilon_{t+k|t} = \varepsilon_{\text{eff}}$, and $\bar{\mathcal{M}}\,\overline{MC} = 1$, so the bracket in (D.27) is zero. The factor in front of the bracket, $\theta^k \Lambda_{t+k,t} Y_{t+k|t}(\varepsilon_{t+k|t} - 1)$, can therefore be evaluated at its steady-state value $(\beta\theta)^k \bar{Y}(\varepsilon_{\text{eff}} - 1)$. Deviations of the discount factor, of demand, and of the elasticity multiply a zero and are second order. The bracket deviates by $x_{t+k|t} - \widehat{\mathcal{M}}^d_{t+k|t} - \widehat{mc}_{t+k|t}$, with $\widehat{\mathcal{M}}^d_{t+k|t} \equiv \ln(\mathcal{M}^d_{t+k|t}/\bar{\mathcal{M}})$ and $\widehat{mc}_{t+k|t} \equiv \ln(MC_{t+k|t}/\overline{MC})$. Dividing by the constant $\bar{Y}(\varepsilon_{\text{eff}} - 1)$ gives

$$\sum_{k=0}^{\infty} (\beta\theta)^k \,\mathbb{E}_t\left[x_{t+k|t} - \widehat{\mathcal{M}}^d_{t+k|t} - \widehat{mc}_{t+k|t}\right] = 0. \tag{D.28}$$

The reset price sets the relative price equal to a discounted average of the desired markup times the firm's own marginal cost. The price-index step below shows that $P^h_{t+k} = P_{t+k}$ to first order. The firm's relative demand is then $Y_{t+k|t}/Y_{t+k} = \mathcal{D}_{t+k}(x_{t+k|t})$, with $\mathcal{D}_{t+k}(x) = \sum_h \omega_{h,t+k} \exp(-\varepsilon_h x)$ and $\omega_{h,t+k} \equiv s_h H^h_{t+k}$. This is the demand $\mathcal{D}(x)$ of Appendix D.5.2 at the current composition. Three properties of $\mathcal{D}$ give the two deviations.

(a) Its own-price elasticity is $\varepsilon(x) \equiv -\mathcal{D}'(x)/\mathcal{D}(x) = \sum_h \omega_h \varepsilon_h \exp(-\varepsilon_h x) / \sum_h \omega_h \exp(-\varepsilon_h x)$, the average of the two elasticities under the expenditure shares at price x. Differentiating, $\varepsilon'(x) = -\operatorname{Var}_{\omega(x)}(\varepsilon)$, so at $x = 0$ the superelasticity is $\mathrm{d}\ln\varepsilon/\mathrm{d}x = -\operatorname{Var}_\omega(\varepsilon)/\varepsilon_{\text{eff}} < 0$: a higher own price loses mostly elastic poor customers. At $x = 0$ the elasticity is $\varepsilon_{\text{eff},t+k} = \sum_h \omega_{h,t+k}\varepsilon_h$, which moves with the composition, $\partial\varepsilon_{\text{eff}}/\partial \ln H = (1-\lambda)\bar{H}(\varepsilon_R - \varepsilon_P)$. The cross term in $x_{t+k|t}\hat{h}_{t+k}$ is second order.

(b) The desired markup is $\varepsilon/(\varepsilon-1)$, so $\mathrm{d}\ln\mathcal{M}^d = -\mathrm{d}\ln\varepsilon/(\varepsilon_{\text{eff}} - 1)$ at the steady state. By (a),

$$\widehat{\mathcal{M}}^d_{t+k|t} = \frac{\operatorname{Var}_\omega(\varepsilon)}{\varepsilon_{\text{eff}}(\varepsilon_{\text{eff}} - 1)}\, x_{t+k|t} + \eta_\mu \hat{h}_{t+k},$$

because the coefficient on $\hat{h}_{t+k}$ is $(1-\lambda)\bar{H}(\varepsilon_P - \varepsilon_R)/[\varepsilon_{\text{eff}}(\varepsilon_{\text{eff}} - 1)] = \eta_\mu$ by (19). At the average price the desired markup is the flexible-price markup (D.15), as under Rotemberg pricing. The own-price term is new.

(c) By (D.26), $\widehat{mc}_{t+k|t} = \widehat{mc}_{t+k} + (\alpha/(1-\alpha)) \ln \mathcal{D}_{t+k}(x_{t+k|t})$. Adding-up, $\sum_h s_h H^h_{t+k} = 1$, gives $\mathcal{D}_{t+k}(0) = 1$ at every composition, so H_{t+k} does not enter, and $\widehat{mc}_{t+k|t} = \widehat{mc}_{t+k} - (\alpha\varepsilon_{\text{eff}}/(1-\alpha))\, x_{t+k|t}$.

Substituting (b) and (c), the bracket in (D.28) is

$$x_{t+k|t}\left[1+\frac{\alpha\varepsilon_{\text{eff}}}{1-\alpha}-\frac{\text{Var}_\omega(\varepsilon)}{\varepsilon_{\text{eff}}(\varepsilon_{\text{eff}}-1)}\right]-\left(\widehat{mc}_{t+k}+\eta_\mu\hat{h}_{t+k}\right)=\Xi\left(x_{t+k|t}-x^*_{t+k}\right),$$
$$\Xi\, x^*_t\equiv\widehat{mc}_t+\eta_\mu\hat{h}_t,$$

with Ξ as in (26). Decreasing returns contribute $\alpha\varepsilon_{\text{eff}}/(1-\alpha)$: a higher price lowers the firm's output and hence its marginal cost. The own-price dependence of the elasticity contributes $-\text{Var}_\omega(\varepsilon)/[\varepsilon_{\text{eff}}(\varepsilon_{\text{eff}}-1)]$: a higher price raises the desired markup. Here x^*_t is the static desired relative price, at which the bracket is zero in period t, and $\widehat{mc}_t$ is marginal cost at aggregate output. By (D.17), $\varpi''(0)=-(\varepsilon_{\text{eff}}-1)\Xi$: Ξ is static profit curvature per unit of $\varepsilon_{\text{eff}}-1$, and the maintained $\varpi''(0)<0$ is $\Xi>0$. Equivalently, marginal profit at relative price x is $-(\varepsilon_{\text{eff}}-1)\Xi\,(x-x^*_t)$ to first order, since at $x=0$ its derivatives with respect to $\widehat{mc}_t$ and $\hat{h}_t$ are $\varepsilon_{\text{eff}}-1$ and $(\varepsilon_{\text{eff}}-1)\eta_\mu$, the coefficients of (D.21) times ζ. Since Ξ is a positive constant, (D.28) becomes $\sum_k(\beta\theta)^k\mathbb{E}_t[p^\#_t-p_{t+k}-x^*_{t+k}]=0$. Using $\sum_k(\beta\theta)^k=1/(1-\beta\theta)$ and subtracting p_t,

$$p^\#_t-p_t=(1-\beta\theta)\sum_{k=0}^{\infty}(\beta\theta)^k\,\mathbb{E}_t\left[(p_{t+k}-p_t)+x^*_{t+k}\right].\tag{D.29}$$

Price indices. Let $p_t\equiv\int_0^1\ln P_t(j)\,dj$ be the average log price and V_t the cross-sectional variance of log prices. A second-order expansion of the type-h index $P^h_t=\left(\int_0^1 P_t(j)^{1-\varepsilon_h}dj\right)^{1/(1-\varepsilon_h)}$ around p_t gives $\ln P^h_t=p_t+((1-\varepsilon_h)/2)V_t$ plus higher-order terms. The two types weight varieties with different elasticities, but the elasticity enters only through the dispersion V_t. The firms that keep their price are a random draw, so two laws of motion hold exactly:

$$p_t=(1-\theta)\,p^\#_t+\theta\,p_{t-1},\qquad V_t=\theta\,V_{t-1}+\theta(1-\theta)\left(p^\#_t-p_{t-1}\right)^2.$$

Starting from the symmetric steady state, V_t is a weighted sum of squared reset-price changes and hence second order. Therefore $\ln P^R_t=\ln P^P_t=p_t$ to first order. One price level deflates profits and enters both demands and both Euler equations, and at the average price the firm's relative price is zero for both types. Aggregate hours, $N_t=\int_0^1 N_t(i)\,di$, differ from $(Y_t/A_t)^{1/(1-\alpha)}$ only through the dispersion of relative output, also a second-order term. So $\widehat{mc}_t$ is the marginal cost of (D.24), and (13) also carries over. Inflation follows

$$\pi_t\equiv p_t-p_{t-1}=(1-\theta)\left(p^\#_t-p_{t-1}\right).\tag{D.30}$$

Phillips curve. Split off the $k=0$ term of (D.29), write $p_{t+k}-p_t=(p_{t+k}-p_{t+1})+$

π_{t+1} for $k \geq 1$, and use (D.29) at $t+1$ with the law of iterated expectations. Since $(1-\beta\theta)\sum_{k\geq 1}(\beta\theta)^k = \beta\theta$,

$$p_t^{\#} - p_t = (1-\beta\theta)\, x_t^* + \beta\theta\, \mathbb{E}_t\big[(p_{t+1}^{\#} - p_{t+1}) + \pi_{t+1}\big].$$

By (D.30), $p_t^{\#} - p_t = (p_t^{\#} - p_{t-1}) - \pi_t = \theta\pi_t/(1-\theta)$, at t and at $t+1$. Substituting,

$$\frac{\theta}{1-\theta}\,\pi_t = (1-\beta\theta)\, x_t^* + \frac{\beta\theta}{1-\theta}\,\mathbb{E}_t\pi_{t+1}.$$

Multiplying by $(1-\theta)/\theta$ and using $\Xi x_t^* = \widehat{mc}_t + \eta_\mu \hat{h}_t$ gives

$$\pi_t = \beta\mathbb{E}_t\pi_{t+1} + \lambda_\theta\, x_t^* = \beta\mathbb{E}_t\pi_{t+1} + \delta_2^C\big(\widehat{mc}_t + \eta_\mu \hat{h}_t\big), \qquad \delta_2^C = \frac{\lambda_\theta}{\Xi}.$$

The composition term is the one of (D.21); only the loading changes. Substituting (D.24) and (13), the slope is $\kappa^C = \delta_2^C G(1-\Gamma\eta_\mu)$, which is (21) with δ_2^C in place of δ_2, and since markup-matched TANK(ε^*) shares G and Γ and has $\mathrm{Var}_\omega(\varepsilon) = 0$, the slope ratio is (27). With one elasticity, $\Xi = (1-\alpha+\alpha\varepsilon)/(1-\alpha)$, the inverse of the decreasing-returns factor in the Calvo slope of Galí (2015, ch. 3).

If each type's demand has Kimball curvature (Kimball, 1995), with superelasticity $\varsigma_h \equiv \mathrm{d}\ln\varepsilon_h/\mathrm{d}x$ at the symmetric price (not to be confused with the population share s_h), the elasticity in (a) becomes $\sum_h \omega_h(x)\varepsilon_h(x)$, and the same differentiation gives $\mathrm{d}\ln\varepsilon/\mathrm{d}x|_{x=0} = s_{\mathrm{tot}} \equiv \sum_h w_h\varsigma_h - \mathrm{Var}_\omega(\varepsilon)/\varepsilon_{\mathrm{eff}}$, with weights $w_h \equiv \omega_h\varepsilon_h/\varepsilon_{\mathrm{eff}}$. Step (b) then gives $\mathrm{d}\ln\mathcal{M}^d/\mathrm{d}x = -s_{\mathrm{tot}}/(\varepsilon_{\mathrm{eff}}-1)$, so

$$\Xi = \frac{1-\alpha+\alpha\varepsilon_{\mathrm{eff}}}{1-\alpha} + \frac{s_{\mathrm{tot}}}{\varepsilon_{\mathrm{eff}}-1}.$$

With a common $\varsigma_h = \varsigma$, Ξ gains $\varsigma/(\varepsilon_{\mathrm{eff}}-1)$ in TE-TANK and in a benchmark with the same ς, while the variance term is unchanged. The steady state, η_μ, Γ, and χ_y use only $\varepsilon_{\mathrm{eff}}$ and are unchanged. This is the third column of Table 3.

At $R = 1$ the expenditure shares equal the population shares, so $\mathrm{Var}_\omega(\varepsilon) = \lambda(1-\lambda)(\varepsilon_P - \varepsilon_R)^2 > 0$ and $\Xi_{\mathrm{TE}} < (1-\alpha+\alpha\bar{\varepsilon})/(1-\alpha)$. Under Calvo pricing the TE-TANK slope therefore exceeds the RANK($\bar{\varepsilon}$) slope even without steady-state inequality: the nesting at $R = 1$ proved in Appendix D.10.2 is a Rotemberg property.

D.8.1 An Alternative Closure: The Wage at the Rich Household's Marginal Rate of Substitution

An alternative closure keeps common hours but lets the wage clear the rich household's intratemporal condition, $W_t/P_t = (C_t^R)^\sigma N_t^\varphi$, so that hand-to-mouth households

are rationed below the hours they would choose at that wage. Since $C_t^R = H_t C_t$, this is the schedule (D.4) with the time-varying wage markup $\mathcal{M}_t^w = H_t^\sigma$, the smallest markup at which every household supplies the hours demanded. The steady-state ratios, Γ, η_μ, and δ_2 are unchanged, but marginal cost gains the term $\sigma\hat{h}_t$, so $\widehat{mc}_t = \sigma\hat{h}_t + G\tilde{y}_t$, $\chi_y = \Gamma G/(1+\Gamma\sigma)$, $\kappa_h = \delta_2(\sigma+\eta_\mu)$, and the slope gains the factor $1/(1+\Gamma\sigma)$,

$$\kappa = \delta_2 G\,\frac{1-\Gamma\eta_\mu}{1+\Gamma\sigma},$$

with $1/(1+\Gamma\sigma) = 0.842$ at the baseline. This factor is a second real rigidity, common to any TANK model in which the rich households' marginal rate of substitution sets the wage: a boom compresses inequality, so rich consumption and the wage rise by less.[12] It lowers the slope of common-elasticity TANK(ε^*) below the RANK slope by 15.8 percent, flattens TE-TANK to $\kappa^{\mathrm{TE}} = 0.0671$ against 0.0701 in markup-matched TANK, and cancels from the ratio $\kappa^{\mathrm{TE}}/\kappa^{\mathrm{TANK}^*} = 1-\Gamma\eta_\mu$, so the composition channel and the Calvo slope comparison of Section 3.3 are the same under both closures, while the Calvo pass-through entries differ slightly. The factor also enters Θ through $\tilde{\sigma} = \sigma(1-\Gamma b)/(1+\Gamma\sigma)$ with $b = G-\sigma$, against $\sigma(1-\Gamma G)$ under the schedule. It would cancel from $\kappa\Theta$ only if $\phi_y = 0$, and at the calibrated ϕ_y it does not. Together with the difference in $\tilde{\sigma}$, this makes the markup-matched pass-through gap 3.11 percent under this closure against 3.32 percent under the schedule. Along the fiscal branch the wage factor makes the benchmark slopes fall, by 19.9 percent in markup-matched TANK against 0.0 percent under the schedule, and TE-TANK's by 41.2 percent against 27.0 percent; wedge pass-through then rises in TE-TANK, by 17.2 percent; the markup comparative static of Proposition D.1(iii) holds under the additional sufficient policy condition $\phi_y \le (1-\rho_u)(\varphi+\alpha)/(1-\alpha) = 0.200$, which the calibrated $\phi_y = 0.125$ satisfies. The markup response is now $G\Theta\Psi_\pi/(1+\Gamma\sigma) = G(\phi_\pi-\rho_u)/D$ with $D \equiv (1-\beta\rho_u)\big[\sigma(1-\rho_u)(1-\Gamma b)+\phi_y(1+\Gamma\sigma)\big] + (\phi_\pi-\rho_u)\,\delta_2 G(1-\Gamma\eta_\mu)$. The last term is the slope under the schedule times $\phi_\pi-\rho_u$ and falls with $\bar{H}$ (Step 2 of the proof of Proposition D.1); the first has derivative $(1-\beta\rho_u)\sigma\Gamma'\big[\phi_y-(1-\rho_u)b\big]$, which is non-positive under the condition, so D falls and the response rises. The admissibility condition weakens to $\Gamma b < 1$, so that χ_y reaches one on the branch only from $\lambda = 0.66$. Participation holds by construction, with the rich on their labor-supply curve.

[12] A related object appears in a worker–capitalist economy with flexible wages, where the real wage equals the workers' marginal rate of substitution, so marginal cost tracks workers' consumption rather than aggregate output (Broer et al., 2020, pp. 84–86).

D.9 Superelasticity of Demand

In TE-TANK a firm's demand has a negative superelasticity, $S = -0.63$ at the baseline, the opposite sign of the Kimball curvature that macro models assume. The sign comes from aggregating customers with different elasticities, not from the curvature of any one household's demand, since each household's demand is CES. The price elasticity of demand ε is the percent fall in sales per percent rise in the good's own price: at the model's effective elasticity, a one percent price increase lowers sales by 3.93 percent. The superelasticity $S \equiv \mathrm{d}\ln\varepsilon/\mathrm{d}x$, with x the log own relative price, measures how this elasticity changes along one demand curve. It does not measure how the elasticity changes with a household's income or consumption, which is the non-homotheticity of Appendix D.13. With $S > 0$, the Kimball case, a firm that prices above its rivals faces more elastic demand and hesitates to move away from them, so prices are strategic complements, a real rigidity. In TE-TANK a firm that raises its price loses mostly its price-sensitive poor customers, so its desired markup rises with its own price, which works toward strategic substitutability.

Local and global curvature The static profit (D.16) of a firm that sets x while all other firms charge the symmetric price shows how S enters price setting. Under the firm's own expenditure shares $\omega_h(x) \propto \omega_h \exp(-\varepsilon_h x)$, step (a) of Appendix D.8 gives $\varepsilon'(x) = -\operatorname{Var}_{\omega(x)}(\varepsilon)$, so $S = -\operatorname{Var}_{\omega}(\varepsilon)/\varepsilon_{\text{eff}} < 0$ at $x = 0$; with Kimball demand within each type, $S = s_{\text{tot}}$. By steps (b) and (c), the desired markup rises with x at the rate $-S/(\varepsilon_{\text{eff}} - 1)$ and the firm's marginal cost falls at the rate $\alpha\varepsilon_{\text{eff}}/(1-\alpha)$, so the static desired relative price solves

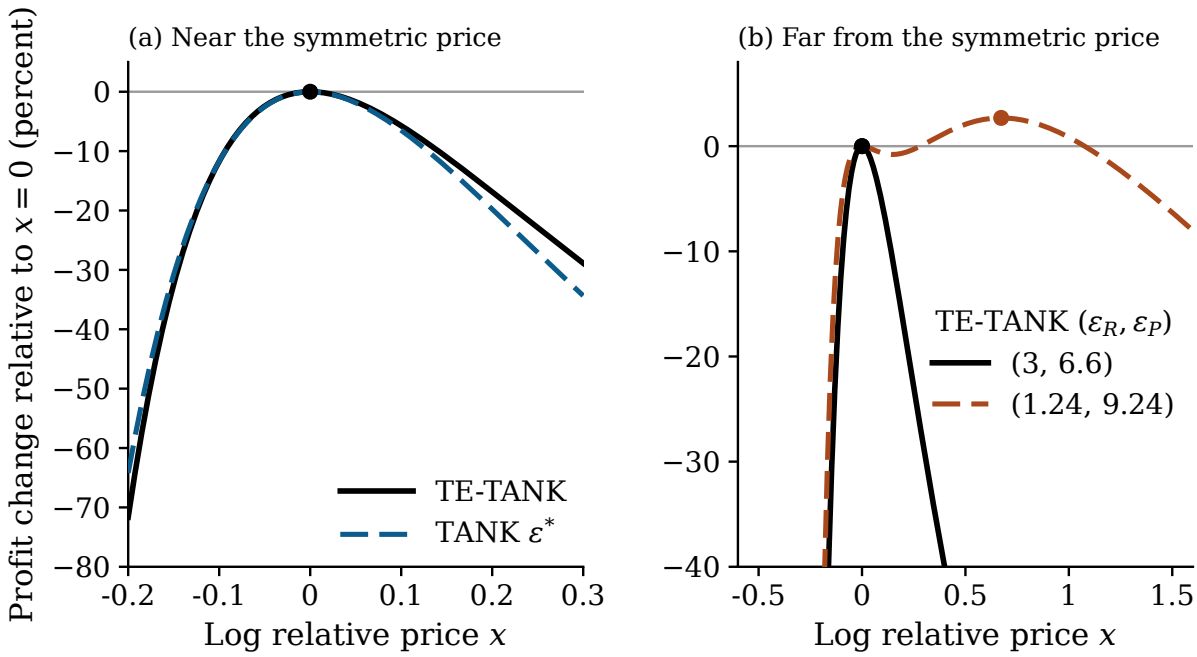
$$x_t^* = -\Big[\frac{S}{\varepsilon_{\text{eff}} - 1} + \frac{\alpha\varepsilon_{\text{eff}}}{1-\alpha}\Big] x_t^* + \widehat{mc}_t + \eta_\mu \hat{h}_t, \qquad \text{hence} \qquad x_t^* = \frac{\widehat{mc}_t + \eta_\mu \hat{h}_t}{\Xi},$$

with the curvature term $\Xi = (1 - \alpha + \alpha\varepsilon_{\text{eff}})/(1-\alpha) + S/(\varepsilon_{\text{eff}} - 1)$, which is (26). The negative S lowers Ξ from $\Xi^* = 2.967$ in markup-matched TANK, whose customers share one elasticity, to $\Xi_{\text{TE}} = 2.751$, so a firm moves its desired price by more per unit change in marginal cost. Figure D.1 shows the same curvature in profit: in panel (a), the static profit of TE-TANK is less concave at the symmetric price than that of markup-matched TANK and falls less steeply when the firm raises its price. Under Calvo pricing a resetting firm's relative price differs from zero, and the lower Ξ raises the loading $\delta_2^C = \lambda_\theta/\Xi$ by 7.8 percent. Under Rotemberg pricing every firm keeps $x = 0$, and S never enters the Phillips curve. The Calvo parameter $\theta = 0.753$ anchors the adjustment cost ζ, as Appendix C.1 describes, and Section 3.3 holds θ fixed across the two economies; we report no separate Calvo impulse responses.

The same variance term sets a second, global condition. By (D.17), $\varpi''(0) = -(\varepsilon_{\text{eff}} - 1)\Xi$, so the local condition $\varpi''(0) < 0$ is $\Xi > 0$, which holds at the baseline. The global condition is that ϖ has no second, higher peak, and panel (b) of Figure D.1 shows how it fails as the elasticity gap widens. At the baseline, profit has a single peak at the symmetric price and falls below the plotted range at high prices. On the fixed-mean path, which holds the population mean at $\bar{\varepsilon} = 4.44$, a second peak appears at a high relative price and overtakes profit at the symmetric price once the gap exceeds 7.87. Every calibration the paper reports lies below this threshold, the widest being the 10 percent point of Table C.4, Panel C, with $\varepsilon_R = 1.36$ and $\varepsilon_P = 9.06$, at which the symmetric price is still the global maximizer. The dashed line shows the pair just above the threshold, with a gap of 8.0, $\varepsilon_R = 1.24$ and $\varepsilon_P = 9.24$: profit has a second peak at $x = 0.67$, 2.7 percent above profit at the symmetric price, at which rich households buy 99.8 percent of the firm's sales. Under Calvo pricing a resetting firm jumps to its new price at no cost, so it would choose such a higher peak directly. Under Rotemberg pricing the adjustment cost is quadratic in each period's price change, so a firm can move toward the higher peak in small steps, keep the total adjustment cost small, and keep the higher profit once it is there. We therefore maintain global optimality under both conventions, as Section 2.2 states; Appendix D.5.2 derives the sufficient condition (D.1) and describes the numerical check, which every reported calibration passes.

Comparison with other demand systems and estimates Table D.1 shows that the two-CES mixture of TE-TANK is the only entry with a negative superelasticity, and that macro calibrations use values several times larger than micro estimates. In Panel A, CES has $S = 0$, the other standard demand systems are subconvex at the model's effective elasticity in the classification of Mrázová and Neary (2017), and the Kimball aggregator leaves S free. In Panel B, Smets and Wouters (2007) and Klenow and Willis (2016) set $S = 10$. In Panel C, the household-level median of Beck and Lein (2020) is 1.93, and Dossche et al. (2010) find curvature "one to two orders of magnitude smaller than the value economists usually impose". Klenow and Willis (2016, p. 469) conclude that the micro evidence points to a superelasticity "closer to 2 than to 10". TE-TANK adds a negative between-type term to the household-level curvature and so moves the firm-level superelasticity further from the macro values. At $\varsigma = 1.93$ the firm's total superelasticity is 1.30, and the net Calvo comparison changes sign only above $\varsigma = 5.86$, which exceeds the 90th percentile of the within-household estimates in Beck and Lein (2020, Table 3, Panel B), 2.23. The between-type term also gives the micro test of Section 3.3: a store's pooled superelasticity lies below the household-level superelasticity of its customers, by 0.63 at the baseline.

FIGURE D.1: Static Profit of a Firm That Deviates from the Symmetric Price



Notes: Static profit (D.16) of one firm that sets the log relative price x while all other firms charge the symmetric price, in the zero-inflation steady state, as the percent change $100\,[\varpi(x)/\varpi(0) - 1]$. Panel (a) shows TE-TANK at the baseline calibration (solid black) and markup-matched TANK($\varepsilon^* = 3.93$) (dashed blue). Panel (b) shows a wider range of x, with the vertical axis cut at -40 percent: TE-TANK at the baseline elasticities (solid black) and the TE-TANK pair just above the elasticity gap of 7.87 at which the symmetric price stops being the global profit maximizer on the fixed-mean path, $(\varepsilon_R, \varepsilon_P) = (1.24, 9.24)$ (dashed rust). Circles mark peaks of profit.

D.10 Proofs and Derivations

This subsection proves Propositions 1 and 2 of the main text and derives three results that the main text states without derivation: the fiscal sufficient statistic, the IS amplification, and the cost-push pass-through solution. Appendix D.11 states and proves Proposition D.1, which builds on all of them. Appendices D.4, D.7, D.8, and D.5.1 supply the intermediate derivations. Throughout, $A = (1-\lambda)(\varepsilon_R - \varepsilon_P)$, $B = \varepsilon_P$, $C = (1+\delta\lambda(\tau-1))/(1-\lambda)$, and $D = (1-\alpha)\lambda(1+\delta(\tau-1))/(1-\lambda)$, as in equation (10).

D.10.1 Fiscal Sufficient Statistic

This subsection shows that, with the non-fiscal parameters and the policy rule fixed, the redistribution share δ and the transfer tilt τ enter every steady-state allocation object and every coefficient of the first-order system only through the rebate parameter $R = \delta(1-\tau)$. The statement refers to the symmetric equilibrium, after transfers and equity values have been eliminated from the household budget constraints. The share price depends on δ separately, but it enters no equilibrium condition other than its own pricing equation. The non-fiscal parameters are $(\beta, \alpha, \varphi, \sigma, \lambda, \zeta, \varepsilon_R, \varepsilon_P, \mathcal{M}^w)$, the policy rule is (ϕ_π, ϕ_y), and the first-order system is that for $(\tilde{y}_t, \pi_t, \hat{h}_t, \hat{\imath}_t)$. The derivation uses the rebate parameter and its complement,

$$R \equiv \delta(1-\tau), \qquad 1+\delta(\tau-1) = 1-R. \tag{D.31}$$

Appendix D.4 eliminates equity and transfers from the rich budget constraint (D.3): equity holdings are constant, so the share-price terms cancel, and the transfer rule (D.2) gives $T_t^P = RD_t$ and rich profit-related income CD_t in (D.10). Changing (δ, τ) at fixed R therefore changes the split of rich income between dividends and transfers but neither type's disposable resources. The equity Euler equation gives $\bar{Q} = \beta(1-\delta)\bar{D}/(1-\beta)$, which depends on δ separately, but the share price enters no equilibrium condition other than its own pricing equation, and in particular neither household's flow resources. The fiscal parameters thus enter the reduced equilibrium only through

$$C = \frac{1+\delta\lambda(\tau-1)}{1-\lambda} = \frac{1-\lambda R}{1-\lambda}, \qquad D = \frac{(1-\alpha)\lambda\big(1+\delta(\tau-1)\big)}{1-\lambda} = \frac{(1-\alpha)\lambda(1-R)}{1-\lambda},$$

both functions of $R = \delta(1-\tau)$ through $1+\delta(\tau-1) = 1-R$. The quadratic primitives A and B contain no fiscal parameters; hence $\bar{H}$, the root of (D.18), depends on (δ, τ) only through (C, D), that is, through R. Then $\bar{\mathcal{M}} = \varepsilon_{\text{eff}}(\bar{H})/(\varepsilon_{\text{eff}}(\bar{H})-1)$; the coefficients δ_2, κ_y, and κ_h (functions of $\bar{H}$ and $\bar{\mathcal{M}}$ only); $\Gamma = D/(\bar{\mathcal{M}}\bar{H})$; $\chi_y = \Gamma G$;

and $\kappa = \kappa_y - \chi_y \kappa_h$ all inherit dependence on R alone. No other occurrence of δ or τ appears in the derivations of Appendix D.4, Appendix D.8, or the dynamic IS block of Appendix D.7. Therefore fiscal pairs with the same R yield the same steady-state allocation and first-order dynamics. Finally, subtracting one from (11) gives $\bar{H} - 1 = (C-1)\big(1 - (1-\alpha)/\bar{\mathcal{M}}\big)$ with $C - 1 = \lambda(1-R)/(1-\lambda)$ and $\bar{\mathcal{M}} > 1$, so $\bar{H} = 1 \iff R = 1$, and $(\delta, \tau) = (1, 0)$ gives $R = 1$; Step 3 of the proof of Proposition 2 establishes $\text{sign}(\chi_y) = \text{sign}(\Gamma) = \text{sign}(1-R)$.

D.10.2 Existence, Uniqueness, and Fiscal Comparative Statics (Proposition 1)

Proof of Proposition 1. (i) The fixed point $\bar{H} = C - D/\bar{\mathcal{M}}$ is the steady-state heterogeneity index of Appendix D.4 evaluated at $\Delta^p(\bar{\Pi}) = 1$, that is, equation (11); the flex-price markup $\bar{\mathcal{M}} = (A\bar{H} + B)/(A\bar{H} + B - 1)$ is equation (12) written with (A, B), and Appendix D.5.1 derives it. Substituting $1/\bar{\mathcal{M}} = (A\bar{H} + B - 1)/(A\bar{H} + B)$ and clearing the denominator:

$$
\begin{aligned}
(\bar{H} - C)(A\bar{H} + B) &= -D(A\bar{H} + B - 1) \\
\Rightarrow \quad A\bar{H}^2 + \big[B - A(C-D)\big]\bar{H} - \big[B(C-D) + D\big] &= 0,
\end{aligned}
$$

which is the quadratic (D.18). *Existence and uniqueness on the fiscal square.* Let $e(H) \equiv AH + B = \varepsilon_{\text{eff}}(H)$ and

$$
H_{\max} \equiv \frac{\varepsilon_P - 1}{(1-\lambda)(\varepsilon_P - \varepsilon_R)},
$$

so that $e(H) > 1$, equivalently $\mathcal{M}(H) > 1$, exactly on $H < H_{\max}$. The steady state is a zero of

$$
q(H; R) \equiv C(R) - D(R) + \frac{D(R)}{e(H)} - H.
$$

Using $C - 1 = \lambda(1-R)/(1-\lambda)$ and $D = (1-\alpha)(C-1)$, for $R < 1$,

$$
q(1) = (C-1)\left[\alpha + \frac{1-\alpha}{\bar{\varepsilon}}\right] > 0,
$$

since $e(1) = \bar{\varepsilon} > 1$. Under the maintained $\varepsilon_R > 1$, $H_{\max} - 1/(1-\lambda) = (\varepsilon_R - 1)/((1-\lambda)(\varepsilon_P - \varepsilon_R)) > 0$, while $C \le 1/(1-\lambda)$, so $\lim_{H \uparrow H_{\max}} q(H; R) = C - H_{\max} < 0$. Finally $q_{HH}(H; R) = 2DA^2/e(H)^3 > 0$: q is convex, positive at $H = 1$, and negative at the upper boundary, so it crosses zero exactly once on $(1, H_{\max})$. The crossing is strict: if $q_H(\bar{H}; R) \ge 0$, convexity would imply $q(H; R) \ge 0$ for every $H > \bar{H}$, contradicting its negative limit at $H_{\max}$. Thus $q_H(\bar{H}; R) < 0$. At the root, $\bar{H} = C - D + D/e(\bar{H}) < C \le 1/(1-\lambda)$, so the poor household's consumption relative to the average, $\bar{H}^P = [1 - (1-\lambda)\bar{H}]/\lambda$, is strictly positive. At $R = 1$, $C = 1$ and $D = 0$, so $q = 1 - H$ and $\bar{H} = 1$. For every $R \in [0, 1]$ the quadratic equals $-e(H)q(H; R)$; it is positive at $H_{\max}$,

where $e = 1$ and $q = C - H_{\max} < 0$, and its leading coefficient A is negative, so its other root exceeds $H_{\max} > 1/(1-\lambda)$, where $\mathcal{M} < 1$ and poor-household consumption is negative. Hence for every $R \in [0,1)$ there is exactly one admissible steady state, with $1 < \bar{H} < C$, and at $R = 1$ it is $\bar{H} = 1$. In the homogeneous TANK case $A = 0$, the system is linear and trivially unique. Existence and uniqueness therefore hold on the entire fiscal square $(\delta, \tau) \in [0,1]^2$. The root solves the pricing first-order condition. By the maintained pricing assumption, the symmetric price it implies is each firm's global profit maximizer, so the root is a steady state and not only a stationary point of profit. Appendix D.5.2 shows that condition (D.1) guarantees this assumption.

(ii) Differentiate the fixed-point condition $q(\bar{H}; R) = 0$. Its partial derivatives are

$$\begin{aligned} q_R(\bar{H}; R) &= -\frac{\lambda}{1-\lambda} + \frac{(1-\alpha)\lambda}{1-\lambda} - \frac{(1-\alpha)\lambda/(1-\lambda)}{e(\bar{H})} \\ &= -\frac{\lambda}{1-\lambda}\left[\alpha + \frac{1-\alpha}{e(\bar{H})}\right] < 0, \\ q_H(\bar{H}; R) &= -\left[1 + \frac{DA}{e(\bar{H})^2}\right] < 0, \end{aligned}$$

where the second inequality follows from the strict downcrossing established in part (i). The implicit-function theorem therefore gives

$$\frac{\mathrm{d}\bar{H}}{\mathrm{d}R} = -\frac{q_R}{q_H} = -\frac{\lambda}{1-\lambda} \cdot \frac{\alpha + (1-\alpha)/e(\bar{H})}{1 + DA/e(\bar{H})^2} < 0.$$

This derivative extends continuously to $R = 1$ from the left. For the markup, $\bar{\mathcal{M}} = e/(e-1)$ implies

$$\frac{\mathrm{d}\bar{\mathcal{M}}}{\mathrm{d}\bar{H}} = -\frac{A}{[e(\bar{H}) - 1]^2} = \frac{(1-\lambda)(\varepsilon_P - \varepsilon_R)}{[e(\bar{H}) - 1]^2} > 0.$$

Combining the two derivatives yields

$$\frac{\mathrm{d}\bar{\mathcal{M}}}{\mathrm{d}R} = -\frac{\lambda(\varepsilon_P - \varepsilon_R)\left[\alpha + (1-\alpha)/e(\bar{H})\right]}{[e(\bar{H}) - 1]^2\ [1 + DA/e(\bar{H})^2]} < 0.$$

Thus redistribution lowers both steady-state inequality and the desired markup. Using $D = \Gamma \bar{\mathcal{M}} \bar{H}$ and the definition of η_μ in (19), the feedback denominator satisfies

$$1 + \frac{DA}{e(\bar{H})^2} = 1 - \Gamma \eta_\mu > 0. \tag{D.32}$$

Hence the endogenous-markup feedback amplifies the direct redistribution effect by $(1 - \Gamma\eta_\mu)^{-1}$. □

Benchmark nesting. Subtracting one from (11) gives

$$\bar{H} - 1 = \frac{\lambda(1-R)}{1-\lambda}\left(1 - \frac{1-\alpha}{\bar{\mathcal{M}}}\right).$$

Because $\bar{\mathcal{M}} > 1 > 1-\alpha$, the second factor is strictly positive, so $\bar{H} = 1 \iff R = 1$, which on the fiscal square requires $(\delta, \tau) = (1, 0)$. There $\varepsilon_{\text{eff}}(1) = \bar{\varepsilon}$ and $\bar{\mathcal{M}} = \bar{\varepsilon}/(\bar{\varepsilon}-1)$, and Γ and χ_y in (13) vanish with $1-R$, so the RANK nesting holds:

$$\bar{H} = 1 \iff R = 1 \iff \Gamma = 0 \iff \chi_y = 0 \iff \hat{h}_t \equiv 0 \quad \text{along every first-order equilibrium path.} \tag{D.33}$$

The last equivalence concerns the loading: $\chi_y = 0$ sets $\hat{h}_t \equiv 0$ under every admissible disturbance, so the elasticity-gap term of the NKPC and the heterogeneity term of the DIS curve drop out, and under Rotemberg pricing every structural equation reduces to its RANK($\bar{\varepsilon}$) counterpart to first order (Appendix D.8 shows that Calvo pricing breaks this nesting). With $\varepsilon_R = \varepsilon_P = \varepsilon$ instead, $A = 0$ and the steady-state system is linear, with $\bar{H} = C - D(\varepsilon-1)/\varepsilon$ and fixed $\bar{\mathcal{M}} = \varepsilon/(\varepsilon-1)$: standard TANK can retain consumption inequality, but its desired markup does not respond to it.

D.10.3 Endogenous Flattening of the Phillips Curve (Proposition 2)

Proof of Proposition 2. Step 1: part (i), the factorization. The reduced-form slope in (20) is $\kappa = \kappa_y - \kappa_h \chi_y$, with $\kappa_y = \delta_2 G$ and $\kappa_h = \delta_2 \eta_\mu$ from (19), and $\chi_y = \Gamma G$ from (13). Hence

$$\kappa = \delta_2 G - \delta_2 \eta_\mu \Gamma G = \delta_2 G\left(1 - \Gamma \eta_\mu\right),$$

which is (21). The entire correction to the feedback-off slope is the desired-markup elasticity η_μ times the feedback strength Γ: under the wage schedule (D.4) real marginal cost does not depend on $\hat{h}_t$, so the demand-composition feedback reaches the slope only through the desired markup. The benchmark slope $\kappa_y = \delta_2 G$ is strictly positive: $G = (\varphi+\alpha)/(1-\alpha) + \sigma > 0$ since $\varphi \geq 0$, $\alpha \in [0,1)$, and $\sigma > 0$; and $\delta_2 = (\varepsilon_{\text{eff}}(\bar{H}) - 1)/\zeta > 0$ by (D.21), because $\zeta > 0$ and $\varepsilon_{\text{eff}}(\bar{H}) = e(\bar{H}) > 1$ at the admissible steady state: for $R < 1$, Proposition 1(i) gives $\bar{\mathcal{M}} = e(\bar{H})/[e(\bar{H}) - 1] > 1$, and at $R = 1$ it gives $\bar{H} = 1$, hence $e(1) = \bar{\varepsilon} > 1$.

Step 2: part (ii), the two bounds on the composition channel. For the strict lower bound, $D = \Gamma \bar{\mathcal{M}} \bar{H}$, $\bar{\mathcal{M}} = e/(e-1)$, and (19) give $\Gamma \eta_\mu = -DA/e(\bar{H})^2$, so $1 - \Gamma\eta_\mu = -q_H(\bar{H}; R) > 0$ by (D.32), whose sign is the strict downcrossing established in the proof of Proposition 1(i). In the degenerate cases the bound is immediate: $R = 1$ gives $D = 0$ and $\varepsilon_P = \varepsilon_R$ gives $A = 0$, so $\Gamma\eta_\mu = 0$ in either case. For the upper bound, both

factors of $\Gamma\eta_\mu$ are non-negative on the fiscal square:

$$\Gamma = \frac{(1-\alpha)\lambda(1-R)}{(1-\lambda)\,\bar{\mathcal{M}}\bar{H}} \;\geq\; 0, \qquad \eta_\mu = \frac{(1-\lambda)\bar{H}(\varepsilon_P - \varepsilon_R)}{e(\bar{H})\big[e(\bar{H})-1\big]} \;\geq\; 0,$$

the first because $R \leq 1$ and its other factors are strictly positive, with equality if and only if $R = 1$, and the second because $e(\bar{H}) > 1$ by Step 1, with equality if and only if $\varepsilon_P = \varepsilon_R$. Hence $0 < 1 - \Gamma\eta_\mu \leq 1$, and Step 1 yields $0 < \kappa = \kappa_y(1 - \Gamma\eta_\mu) \leq \kappa_y$, that is, $\kappa \in (0, \kappa_y]$: the reduced-form Phillips-curve slope is strictly positive at every point of the fiscal square.

Step 3: part (iii), when the flattening is strict. Write the comparison in level form, $\kappa - \kappa_y = -\kappa_h\chi_y$. Step 1 gives $\delta_2 > 0$ and Step 2 gives $\eta_\mu \geq 0$, so $\kappa_h = \delta_2\eta_\mu \geq 0$, with equality if and only if $\varepsilon_P = \varepsilon_R$, because the remaining factors of η_μ, $(1-\lambda)\bar{H}$ and $e(\bar{H})[e(\bar{H})-1]$, are strictly positive. Next, $\chi_y = \Gamma G$ with $G > 0$ by Step 1, so $\operatorname{sign}(\chi_y) = \operatorname{sign}(\Gamma)$; and the factors of Γ in Step 2 other than $1 - R$ are strictly positive, so $\operatorname{sign}(\Gamma) = \operatorname{sign}(1-R)$. Since $R \leq 1$ on the fiscal square, $\chi_y \geq 0$ with $\chi_y > 0 \iff R < 1$. Therefore

$$\kappa < \kappa_y \iff \kappa_h\chi_y > 0 \iff \eta_\mu > 0 \text{ and } \Gamma > 0 \iff \varepsilon_P > \varepsilon_R \text{ and } R < 1.$$

Equality $\kappa = \kappa_y$ holds at $R = 1$, where $\Gamma = \chi_y = 0$ and the composition channel equals one, and under a common elasticity at every R: with $\varepsilon_P = \varepsilon_R = \varepsilon$ the slope is $\delta_2 G = (\varepsilon - 1)G/\zeta$ at every point of the fiscal square, so steady-state inequality by itself does not move the slope. □

D.10.4 Elasticity Dispersion and IS Amplification

This subsection shows that, under strict dispersion $\varepsilon_P > \varepsilon_R$, unequal consumption $R < 1$, and $\chi_y^{\text{TANK}} < 1$, TE-TANK has a higher heterogeneity index and a higher markup than mean-matched TANK($\bar{\varepsilon}$), and therefore a smaller loading χ_y and an IS curve that is more rate-sensitive than RANK's but less rate-sensitive than the benchmark's. Here $\bar{\varepsilon} \equiv (1-\lambda)\varepsilon_R + \lambda\varepsilon_P$ is the population-weighted mean elasticity, mean-matched TANK is common-elasticity TANK with $\varepsilon = \bar{\varepsilon}$, and the fiscal parameters are common to both economies. Formally,

$$\bar{H}^{\text{TE}} > \bar{H}^{\text{TANK}}, \qquad \bar{\mathcal{M}}^{\text{TE}} > \bar{\mathcal{M}}^{\text{TANK}}, \tag{D.34}$$

and consequently

$$\chi_y^{\text{TE}} < \chi_y^{\text{TANK}}, \qquad 1 < \frac{1}{1-\chi_y^{\text{TE}}} < \frac{1}{1-\chi_y^{\text{TANK}}}. \tag{D.35}$$

The ordering follows from the steady-state fixed point. Both economies share (C, D), since (λ, δ, τ) are common, and each steady state solves $H = C - D/\mathcal{M}(H)$. For $R < 1$ the TANK fixed point is $\bar{H}^{\text{TANK}} = C - D/\mathcal{M}(\bar{\varepsilon}) > 1$. For any $H > 1$, the TE effective elasticity is $\varepsilon^{\text{TE}}_{\text{eff}}(H) = \bar{\varepsilon} + A(H-1) < \bar{\varepsilon}$ because $A = (1-\lambda)(\varepsilon_R - \varepsilon_P) < 0$, and $\mathcal{M}(\cdot)$ is decreasing in the elasticity, so $\mathcal{M}^{\text{TE}}(H) > \mathcal{M}(\bar{\varepsilon})$. Hence the TE fixed-point residual $q(H) = C - D/\mathcal{M}^{\text{TE}}(H) - H$ of the proof of Proposition 1(i) (the same q, since $1/\mathcal{M}(H) = 1 - 1/e(H)$) is strictly positive at $\bar{H}^{\text{TANK}}$:

$$q\big(\bar{H}^{\text{TANK}}\big) = D\left[\frac{1}{\mathcal{M}(\bar{\varepsilon})} - \frac{1}{\mathcal{M}^{\text{TE}}(\bar{H}^{\text{TANK}})}\right] > 0.$$

Since q is convex, positive at $H = 1$, and crosses zero exactly once on the admissible band (proof of Proposition 1(i)), and since $\bar{H}^{\text{TANK}} < C \leq 1/(1-\lambda) < H_{\max}$ places $\bar{H}^{\text{TANK}}$ inside that band, $q > 0$ exactly to the left of the TE root, so $\bar{H}^{\text{TE}} > \bar{H}^{\text{TANK}}$. With $\bar{H}^{\text{TE}} > 1$, also $\bar{\mathcal{M}}^{\text{TE}} = \mathcal{M}^{\text{TE}}(\bar{H}^{\text{TE}}) > \mathcal{M}(\bar{\varepsilon}) = \bar{\mathcal{M}}^{\text{TANK}}$. Both factors of $\bar{\mathcal{M}}\bar{H}$ are therefore strictly larger in TE-TANK, so

$$\bar{\mathcal{M}}^{\text{TE}}\bar{H}^{\text{TE}} > \bar{\mathcal{M}}^{\text{TANK}}\bar{H}^{\text{TANK}} \;\Rightarrow\; \Gamma^{\text{TE}} < \Gamma^{\text{TANK}} \;\Rightarrow\; \chi_y^{\text{TE}} < \chi_y^{\text{TANK}}.$$

Finally, $\chi_y^{\text{TE}} > 0$ because $R < 1$ gives $\Gamma^{\text{TE}} > 0$ (Step 3 of the proof of Proposition 2), which gives the left inequality of (D.35).

At the baseline calibration, the amplification factor $1/(1-\chi_y)$ is 1.000 in RANK, 2.292 in TE-TANK, and 2.439 in mean-matched TANK ($\bar{\varepsilon} = 4.44$). Markup-matched TANK(ε^*) shares TE-TANK's $\bar{\mathcal{M}}\bar{H}$ and hence its amplification factor, so the ordering is a markup-level effect and vanishes in the dispersion-pure comparison.

D.10.5 Cost-Push Pass-Through

This subsection solves the model for the cost-push wedge in isolation and shows that, under the conditions (D.36), impact inflation per unit of the wedge is $\Psi_\pi = 1/[(1 - \beta\rho_u) + \kappa\Theta]$, so that two economies with the same non-elasticity primitives, policy rule, and wedge process differ in pass-through if and only if they differ in the product $\kappa\Theta$. Appendix D.11 uses the solution and the conditions. Throughout, the wedge $u_t = \rho_u u_{t-1} + \varepsilon^u_t$ of Section 2.6 is the only disturbance, so $z_t = a_t = v_t = 0$, $r^n_t = 0$, and $\hat{y}_t = \tilde{y}_t$, and we maintain

$$\beta \in (0,1), \quad \sigma > 0, \quad \kappa > 0, \quad \phi_\pi > 1, \quad \phi_y \geq 0, \quad \rho_u \in [0,1), \quad 0 \leq \chi_y < 1, \tag{D.36}$$

in both economies when two are compared, so that $\tilde{\sigma} \equiv \sigma(1-\chi_y) > 0$.

Solution The equilibrium reduces to (20), (14), and (9). Guess $\pi_t = \Psi_\pi u_t$ and $\tilde{y}_t = \Psi_y u_t$, so that $\mathbb{E}_t[\pi_{t+1}] = \rho_u \Psi_\pi u_t$ and $\mathbb{E}_t[\tilde{y}_{t+1}] = \rho_u \Psi_y u_t$. Matching coefficients in (20) gives $(1-\beta\rho_u)\Psi_\pi = \kappa\Psi_y + 1$; substituting the Taylor rule into (14) gives $\Psi_y\left[\tilde{\sigma}(1-\rho_u) + \phi_y\right] = -(\phi_\pi - \rho_u)\Psi_\pi$. Hence

$$\begin{aligned} \pi_t &= \Psi_\pi u_t, \qquad \tilde{y}_t = -\Theta\Psi_\pi u_t, \\ \Psi_\pi &= \frac{1}{(1-\beta\rho_u) + \kappa\Theta} > 0, \qquad \Theta = \frac{\phi_\pi - \rho_u}{\tilde{\sigma}(1-\rho_u) + \phi_y} > 0, \end{aligned} \tag{D.37}$$

where the signs follow from (D.36), and $\Psi_y = -\Theta\Psi_\pi < 0$: a positive wedge is stagflationary. This is the unique bounded solution. The standard determinacy condition is $\kappa(\phi_\pi - 1) + (1-\beta)\phi_y > 0$ (Bullard and Mitra, 2002), which is necessary and sufficient when $\phi_\pi, \phi_y \geq 0$ and $\tilde{\sigma} > 0$; under (D.36) it holds. Given $\tilde{\sigma} > 0$, the condition involves $\tilde{\sigma}$ nowhere else, so within the admissible region $\chi_y < 1$ the demand-composition feedback cannot threaten determinacy through χ_y: it holds at every point of the fiscal branch $R \in [0,1]$ at the baseline calibration, as does $0 \leq \chi_y < 1$, whose maximum along the branch is 0.725.

Comparative statics and the benchmark comparison Differentiating (D.37),

$$\frac{\partial \Psi_\pi}{\partial \kappa} = -\Theta\,\Psi_\pi^2 < 0, \qquad \frac{\partial \Psi_\pi}{\partial \tilde{\sigma}} = \frac{(1-\rho_u)\,\kappa\,(\phi_\pi - \rho_u)}{\left[\tilde{\sigma}(1-\rho_u) + \phi_y\right]^2}\,\Psi_\pi^2 > 0 : \tag{D.38}$$

a flatter Phillips curve and a less rate-sensitive IS curve both raise pass-through. For two economies that share β and ρ_u, $\Psi_\pi^{\text{TE}}/\Psi_\pi^{\text{TANK}} = \left[(1-\beta\rho_u) + \kappa^{\text{TANK}}\Theta^{\text{TANK}}\right]/\left[(1-\beta\rho_u) + \kappa^{\text{TE}}\Theta^{\text{TE}}\right]$, so TE-TANK has greater pass-through if and only if $\kappa^{\text{TE}}\Theta^{\text{TE}} < \kappa^{\text{TANK}}\Theta^{\text{TANK}}$. The sufficient condition that Section 2.6 states, $\kappa^{\text{TE}} \leq \kappa^{\text{TANK}}$ and $\chi_y^{\text{TE}} \leq \chi_y^{\text{TANK}}$ with at least one inequality strict, follows because Θ is increasing in χ_y through $\tilde{\sigma} = \sigma(1-\chi_y)$. Against the mean-matched benchmark both inequalities of that sufficient condition hold analytically whenever $R < 1$ and $\varepsilon_P > \varepsilon_R$. Appendix D.10.4 shows $\chi_y^{\text{TE}} < \chi_y^{\text{TANK}}$. For the slopes, the common-elasticity slope is $\delta_2 G = (\bar{\varepsilon} - 1)G/\zeta$ at every R (Proposition 2(iii)), whereas $e(\bar{H}^{\text{TE}}) = \bar{\varepsilon} - (1-\lambda)(\varepsilon_P - \varepsilon_R)(\bar{H}^{\text{TE}} - 1) < \bar{\varepsilon}$ because $\bar{H}^{\text{TE}} > 1$, and $1 - \Gamma\eta_\mu < 1$, so

$$\kappa^{\text{TE}} = \frac{e(\bar{H}^{\text{TE}}) - 1}{\zeta}\,G\,(1 - \Gamma\eta_\mu) < \frac{\bar{\varepsilon} - 1}{\zeta}\,G = \kappa^{\text{TANK}(\bar{\varepsilon})}.$$

Under Rotemberg pricing with a common ζ, TE-TANK therefore has strictly greater cost-push pass-through than both the markup-matched and the mean-matched benchmark at every $R < 1$.

Admissible region The IS semi-elasticity $1/[\sigma(1-\chi_y)]$ of (14) (Appendix D.7) is finite and positive if and only if $\chi_y < 1$, that is, $\Gamma G < 1$. The restrictions on primitives in Appendix D.1 do not imply $\chi_y < 1$. Because $\bar{\mathcal{M}}\bar{H} > 1$ on the admissible branch and $(1-\alpha)G = \varphi + \alpha + \sigma(1-\alpha)$,

$$\chi_y = \frac{\lambda(1-R)\left[\varphi + \alpha + \sigma(1-\alpha)\right]}{(1-\lambda)\,\bar{\mathcal{M}}\bar{H}} < \frac{\lambda(1-R)\left[\varphi + \alpha + \sigma(1-\alpha)\right]}{1-\lambda},$$

so $\lambda(1-R)[\varphi + \alpha + \sigma(1-\alpha)] \leq 1-\lambda$ is a sufficient condition in primitives, in TE-TANK and in both homogeneous benchmarks alike, since all three have $\bar{\mathcal{M}}\bar{H} > 1$. At the baseline φ, α, and σ and at $R = 0$ it reads $\lambda \leq 1/3$; at $\lambda = 0.40$ it holds for $R \geq 0.25$ and fails below. Below that point $\chi_y < 1$ nevertheless holds: the loading is increasing in $\bar{H}$ along the branch (Proposition D.1(i)) and reaches 0.725 at $R = 0$. The condition is sufficient, not necessary, but it is not slack far from the baseline: at $\lambda = 0.8$ and $R = 0$ the loading is $\chi_y = 2.12$, the IS curve has the wrong sign, and the equilibrium is indeterminate under the calibrated Taylor rule in TE-TANK and in both homogeneous benchmarks. Because χ_y is largest at $R = 0$, the smallest hand-to-mouth share at which the baseline branch leaves the admissible region is the λ at which $\chi_y(R{=}0) = 1$, which is $\lambda = 0.50$ at the baseline elasticities. Outside $\chi_y < 1$ the Taylor principle no longer guarantees determinacy, although it does not fail everywhere there.

D.11 Comparative Statics

Steady-state inequality and cost-push amplification Under Rotemberg pricing and per unit of the cost-push wedge, higher steady-state inequality raises the impact markup response to a cost-push shock: the shock opens a negative output gap, inequality rises, and spending tilts further toward the less price-elastic rich. Proposition D.1 states this comparative static and proves it. Homogeneous-elasticity TANK shares its sign, as the benchmark below shows, and Section 3.5 shows that what the elasticity gap adds depends on the pricing convention.

The comparative static runs along the admissible fiscal branch, the steady states traced out as the rebate parameter $R = \delta(1-\tau)$ moves over $[0,1]$ with all non-fiscal parameters held fixed. We index the branch by steady-state inequality, the heterogeneity index $\bar{H}$ (rich households' consumption relative to average consumption), which equals one at $R = 1$ and is largest at $R = 0$. Subtracting one from (11) gives $\bar{H} - 1 = (\lambda(1-R)/(1-\lambda))\big(1 - (1-\alpha)/\bar{\mathcal{M}}\big)$, whose second factor is strictly positive because $\bar{\mathcal{M}} > 1 > 1-\alpha$ on the branch (Proposition 1(i)). Inverting is therefore legitimate:

$$1 - R = \frac{(1-\lambda)(\bar{H}-1)}{\lambda\left[1 - (1-\alpha)/\bar{\mathcal{M}}(\bar{H})\right]}. \tag{D.39}$$

Since $d\bar{H}/dR < 0$ (Proposition 1(ii)), $R \mapsto \bar{H}$ is a strictly decreasing bijection from $[0,1]$ onto $[1, \bar{H}(R{=}0)]$, with $\bar{H}(R{=}0) = 1.342$ at the baseline calibration. Every object below is therefore a function of $\bar{H}$ alone, and "increasing in $\bar{H}$" means increasing as redistribution weakens.

Sections 2.5 and 2.6 define the composite objects that the proposition tracks. Write $G = (\varphi + \alpha)/(1-\alpha) + \sigma$ for the coefficient linking real marginal cost to the output gap. One condition on primitives organizes the comparative statics:

$$(\varepsilon_R - 1)\big(1 - \alpha + \alpha\varepsilon_R\big) \;\geq\; \frac{\lambda}{1-\lambda}\big(\varepsilon_P - \varepsilon_R\big). \tag{D.40}$$

Condition (D.40) keeps the rich elasticity far enough above one relative to the elasticity gap, weighted by the relative population share of the poor, so the effective elasticity stays away from one on the admissible branch; as it approaches one, the desired markup diverges and the inequality in Step 1 of the proof fails. It holds strictly at the baseline calibration, where it reads $3.333 \geq 2.400$. No condition on the policy rule is needed: under the wage schedule the markup response is the constant G times $|\Psi_y| = \big[(1-\beta\rho_u)/\Theta + \kappa\big]^{-1}$, and along the branch Θ rises for every $\phi_y \geq 0$ while κ falls (Steps 2 and 3 of the proof).

Proposition D.1 (Steady-State Inequality and Cost-Push Amplification)**.** *Under Rotemberg pricing, index the admissible fiscal branch by $\bar{H} \in \big[1, \bar{H}(R{=}0)\big]$ through (D.39), and maintain the cost-push conditions (D.36) together with strict elasticity dispersion, $\varepsilon_P > \varepsilon_R$. Then:*

(i) Under (D.40), Γ, χ_y, η_μ, and Θ are strictly increasing in $\bar{H}$, while δ_2 and κ are strictly decreasing.

(ii) Under (D.40), the impact output contraction $|\Psi_y| = \Theta\Psi_\pi$ is strictly increasing in $\bar{H}$.

(iii) Under (D.40), the impact markup response to the cost-push wedge,

$$\frac{\partial \hat{\mu}_t}{\partial u_t} = G\,\Theta\Psi_\pi,$$

is strictly increasing in $\bar{H}$. Higher steady-state inequality therefore raises the markup response to a cost-push shock.

Proof of Proposition D.1. Throughout, write $e \equiv e(\bar{H}) = A\bar{H} + B = \varepsilon_{\text{eff}}(\bar{H})$ for the effective demand elasticity firms face, the expenditure-weighted average of the two types' elasticities. Since $A = -|A| < 0$ with $|A| = (1-\lambda)(\varepsilon_P - \varepsilon_R)$, e is strictly decreasing in $\bar{H}$: as inequality rises, spending shifts toward the low-elasticity rich and the average

elasticity falls. Proposition 1(i) gives $1 \le \bar{H} < C \le 1/(1-\lambda)$ and $e > 1$, equivalently $\bar{\mathcal{M}} = e/(e-1) > 1$, everywhere on the branch.

Step 1: the feedback strength Γ is strictly increasing in $\bar{H}$; this is where (D.40) does its work. Substituting (D.39) into $\Gamma = (1-\alpha)\lambda(1-R)/((1-\lambda)\bar{\mathcal{M}}\bar{H})$ and using $\bar{\mathcal{M}} - (1-\alpha) = (\alpha e + 1 - \alpha)/(e-1)$ expresses Γ as a function of $\bar{H}$ alone,

$$\Gamma(\bar{H}) = \frac{(1-\alpha)(\bar{H}-1)}{\bar{H}\big[\bar{\mathcal{M}} - (1-\alpha)\big]} = \frac{(1-\alpha)(\bar{H}-1)(e-1)}{\bar{H}\big[\alpha e + 1 - \alpha\big]}.$$

Two forces compete: the factor $\bar{H} - 1$ rises with inequality, while the factor $e - 1$ falls with it. Log-differentiating and collecting the two e terms separates them,

$$\frac{\mathrm{d}\log\Gamma}{\mathrm{d}\bar{H}} = \frac{1}{\bar{H}(\bar{H}-1)} - \frac{|A|}{(e-1)(\alpha e + 1 - \alpha)},$$

$$\text{so} \qquad \Gamma' > 0 \iff (e-1)(\alpha e + 1 - \alpha) > |A|\,\bar{H}(\bar{H}-1).$$

Because e is strictly decreasing in $\bar{H}$, the left side of this inequality is strictly decreasing and the right side strictly increasing in $\bar{H}$; it therefore suffices to check it weakly at the right endpoint of the branch's closure. That endpoint is $\bar{H} = 1/(1-\lambda)$, where $e\big(1/(1-\lambda)\big) = A/(1-\lambda) + B = \varepsilon_R$ exactly and $|A|(1/(1-\lambda))\big(1/(1-\lambda) - 1\big) = (\lambda/(1-\lambda))(\varepsilon_P - \varepsilon_R)$, so the inequality evaluated there is precisely (D.40). Since $\bar{H} < 1/(1-\lambda)$ strictly on the branch, the weak condition (D.40) delivers the strict inequality at every admissible $\bar{H}$, hence $\Gamma' > 0$. At the left endpoint $\Gamma(1) = 0$ and $\Gamma'(1) = (1-\alpha)(e(1)-1)/(\alpha e(1) + 1 - \alpha) > 0$.

Step 2: the remaining objects of part (i) follow by inspection once e falls and Γ rises with $\bar{H}$. By Appendix D.8, $\delta_2 = (e-1)/\zeta$, so $\delta_2 > 0$ and $\delta_2' < 0$. By (19), $\eta_\mu = |A|\bar{H}/\big[e(e-1)\big]$ has a strictly increasing positive numerator and a strictly decreasing positive denominator, so $\eta_\mu > 0$ is strictly increasing: desired markups become more sensitive to demand composition. Since $\chi_y = \Gamma G$ with $G > 0$, Step 1 makes χ_y strictly increasing; hence $\tilde{\sigma} = \sigma(1-\chi_y)$ is strictly decreasing and $\Theta = (\phi_\pi - \rho_u)/\big[\tilde{\sigma}(1-\rho_u) + \phi_y\big]$ strictly increasing: the demand side contracts more per unit of inflation. Finally, by (21), $\kappa = \delta_2 G(1 - \Gamma\eta_\mu)$. Proposition 2(ii) already establishes $0 < 1 - \Gamma\eta_\mu \le 1$ analytically, in particular the strict positivity $1 - \Gamma\eta_\mu = 1 + DA/e^2 > 0$ of (D.32), which follows from the strict downcrossing of q. So κ is G times a product of two strictly positive factors, δ_2 and $1 - \Gamma\eta_\mu$, of which the first is strictly decreasing and the second decreasing, because $\Gamma \ge 0$ and $\eta_\mu > 0$ are both increasing; therefore $\kappa' < 0$: higher steady-state inequality flattens the Phillips curve.

Step 3: part (ii), the output contraction deepens. By (D.37), $|\Psi_y| = \Theta\Psi_\pi = \Theta/\big[(1 - \beta\rho_u) + \kappa\Theta\big] = \big[(1-\beta\rho_u)/\Theta + \kappa\big]^{-1}$. By part (i), Θ is strictly increasing and κ strictly

decreasing, so both terms in the bracket fall, the bracket is strictly decreasing, and $|\Psi_y|$ is strictly increasing. Only (D.40) is used.

Step 4: part (iii), the markup response rises. By (D.24), $\hat{\mu}_t = -G\tilde{y}_t$, and (D.37) gives $\tilde{y}_t = -\Theta\Psi_\pi u_t$, which yields the response displayed in part (iii); the prefactor is G itself because, under the wage schedule, the distribution does not enter marginal cost. So $\partial\hat{\mu}_t/\partial u_t = G|\Psi_y|$ with $G > 0$ fixed along the branch, and Step 3 makes it strictly increasing in $\bar{H}$ for every $\phi_y \geq 0$. □

Numerical complement: pass-through along the branch Parts (i)–(iii) of Proposition D.1 do not sign inflation pass-through along the branch, so the following results hold at the baseline calibration only. Per unit of the Phillips-curve wedge u_t, inflation pass-through Ψ_π is not monotone along the branch: it falls from 2.739 at $R = 1$ over most of the branch, reaches its minimum at $R = 0.24$, and rises again at low R, ending at 2.597 at $R = 0$, 5.2 percent below its value at $R = 1$, while κ falls from 0.0977 to 0.0713 and the markup response of (iii) rises by 39.8 percent. Per unit of a structural shifter $\hat{u}_t$ of real marginal cost or of the log desired markup, which enters as $u_t = \delta_2\hat{u}_t$ by Appendix D.8, pass-through $\delta_2\Psi_\pi$ is strictly decreasing in $\bar{H}$ along the branch, falling by 25.6 percent between $R = 1$ and $R = 0$, and the markup response per unit of $\hat{u}_t$, $\delta_2 G\Theta\Psi_\pi$, is strictly increasing, rising by 9.8 percent.

By (D.37), $\Psi_\pi = \left[(1-\beta\rho_u) + \kappa\Theta\right]^{-1}$ depends on $\bar{H}$ only through the product $\kappa\Theta$, and by part (i) of Proposition D.1 its two factors move in opposite directions: the flatter Phillips curve raises Ψ_π, and the larger output contraction per unit of inflation lowers it. Condition (D.40) signs the factors but not the product, so we verify the shape numerically: at the baseline calibration, on a 2001-point grid in $R \in [0,1]$, $\kappa\Theta$ first rises and then falls in $\bar{H}$, so Ψ_π falls over most of the branch and rises again near $R = 0$, with its minimum at $R = 0.24$ and the endpoint values reported above; $\delta_2\Psi_\pi$ is strictly decreasing at every grid step, because $\delta_2 = (e-1)/\zeta$ falls by more than Ψ_π moves; and $\delta_2 G\Theta\Psi_\pi$ is strictly increasing at every grid step. Parts (i)–(iii) of the proposition do not use the shape of Ψ_π; they compare steady states on the fiscal branch under (D.40) and are not a global ordering over all parameterizations.

Homogeneous-elasticity benchmark Proposition D.1 is stated for TE-TANK, but parts of it survive the removal of the elasticity gap, and the parts that do not identify what the gap adds. Setting $\varepsilon_R = \varepsilon_P = \varepsilon$ gives $A = 0$, hence $e \equiv \varepsilon$, $\eta_\mu = 0$, and constant $\bar{\mathcal{M}} = \varepsilon/(\varepsilon-1)$ and $\delta_2 = (\varepsilon-1)/\zeta$ along the entire branch; only Γ still moves with $\bar{H}$. Three claims follow, none of which needs (D.40).

(a) The slope is constant, and the demand objects keep their signs. With $\bar{\mathcal{M}}$ constant, Step 1's expression reduces to $\Gamma(\bar{H}) = (1-\alpha)(\bar{H}-1)/\left[\bar{H}(\bar{\mathcal{M}} - (1-\alpha))\right]$, and since

$\mathrm{d}\big[(\bar{H}-1)/\bar{H}\big]/\mathrm{d}\bar{H} = 1/\bar{H}^2 > 0$ with $\bar{\mathcal{M}} > 1 > 1-\alpha$, we get $\Gamma' > 0$ unconditionally. Condition (D.40) is used only to sign Γ' against the countervailing movement of e, which is now absent. Step 2 then gives $\chi'_y > 0$ and $\Theta' > 0$ as before, but $\kappa = \delta_2 G$ with $\eta_\mu = 0$ and δ_2 constant gives $\kappa' = 0$: with a common elasticity the slope is the RANK slope at every point of the branch, as in Debortoli and Galí (2018), so the whole change in the slope along the branch in TE-TANK, -27.0 percent (Table 4), is the elasticity gap. Step 3 applies with $\kappa' = 0$, so $|\Psi_y|$ is strictly increasing.

(b) The markup comparative static is not specific to the elasticity gap. The response $\partial\hat{\mu}_t/\partial u_t = G\Theta\Psi_\pi$ of part (iii) has a prefactor free of η_μ; the elasticity gap enters only inside Ψ_π, through δ_2 and $1-\Gamma\eta_\mu$ in κ. The slope route is the reason the markup response rises by 39.8 percent along the branch in TE-TANK against 11.8 percent in *markup-matched* TANK (Table 4). The sign of the response, however, does not depend on the gap. By (a), $|\Psi_y|$ is strictly increasing for every $\phi_y \geq 0$, and G is constant along the branch, so $G\Theta\Psi_\pi = G|\Psi_y|$ is again increasing: higher steady-state inequality raises the markup response to a cost-push shock in homogeneous-elasticity TANK too, under any output coefficient in the Taylor rule. The elasticity gap changes the magnitude, mainly through κ, not the sign.

(c) Pass-through falls with inequality under either normalization. Because δ_2 is constant along the branch, the comparative static of Ψ_π is also that of the structural pass-through $\delta_2\Psi_\pi$. With $\kappa = \delta_2 G$ constant,

$$\kappa\Theta = \frac{\delta_2 G(\phi_\pi - \rho_u)}{\sigma(1-\rho_u)(1-\Gamma G)+\phi_y},$$

whose denominator has derivative $-\sigma(1-\rho_u)G\,\Gamma' < 0$ in $\bar{H}$. Since $\Psi_\pi = \big[(1-\beta\rho_u)+\kappa\Theta\big]^{-1}$,

$$\frac{\mathrm{d}\Psi_\pi}{\mathrm{d}\bar{H}} < 0 \qquad \text{for every } \phi_y \geq 0, \tag{D.41}$$

so homogeneous-elasticity pass-through is strictly decreasing in steady-state inequality under either normalization and under any output coefficient: the demand block is the only thing that moves along the branch, and it moves against inflation. Along the baseline branch Ψ_π falls by 25.2 percent in markup-matched and by 27.5 percent in mean-matched TANK.

The pass-through sign in TE-TANK In TE-TANK the elasticity gap adds a supply-side factor to $\kappa\Theta$ that works against the demand side. With $\tilde{\sigma} = \sigma(1-\Gamma G)$ and $\kappa = \delta_2 G(1-\Gamma\eta_\mu)$,

$$\kappa\Theta = \frac{\delta_2 G(1-\Gamma\eta_\mu)(\phi_\pi - \rho_u)}{\sigma(1-\rho_u)(1-\Gamma G)+\phi_y}.$$

The denominator falls with $\bar{H}$, as in part (c), which raises $\kappa\Theta$ and lowers Ψ_π. The numerator is new. As the rebate falls, spending tilts toward the price-insensitive rich and the effective elasticity falls, and with it both the marginal-cost loading $\delta_2 = (\varepsilon_{\text{eff}}(\bar{H}) - 1)/\zeta$ and the desired-markup factor $1 - \Gamma\eta_\mu$, which lowers $\kappa\Theta$ and raises Ψ_π. Both come from one primitive, the endogenous effective elasticity, which moves if and only if $\varepsilon_P \neq \varepsilon_R$. Which side dominates is a quantitative question. Along the baseline branch the log change in $\kappa\Theta$ splits into $\Delta \log \delta_2 = -0.242$ and $\Delta \log(1 - \Gamma\eta_\mu) = -0.073$ on the supply side against 0.389 from the denominator, so the supply side is about 0.8 times the demand side. Per unit of the Phillips-curve wedge, Ψ_π therefore falls between the endpoints, by 5.2 percent against 25.2 percent in the markup-matched benchmark, and not monotonically (Appendix D.11). Per unit of a structural shifter, which enters as $u_t = \delta_2 \hat{u}_t$, pass-through is $\delta_2 \Psi_\pi$, so the fall in δ_2 also scales inflation down directly, and structural pass-through falls by 25.6 percent, about as much as in the benchmarks. The fall in δ_2 arises because the Rotemberg cost ζ is held fixed while the effective elasticity falls. Under Calvo pricing with a common reset probability, $\delta_2 = \lambda_\theta/\Xi$ instead rises along the branch, by 14.0 percent, so the TE-TANK slope rises by 6.0 percent rather than falling, while the benchmark slopes are constant under both conventions; pass-through falls under both normalizations and both conventions. Section 3.5 reports the endpoint changes in Table 4.

D.12 Technology and Monetary Policy Shock Transmission

Figure D.2 applies the band decomposition of Section 3.4, in the Rotemberg baseline, to a positive technology shock and to an expansionary monetary policy shock; the preference shock, not shown, adds nothing new. After the technology shock (Figure D.2a), the gray and red bands reproduce the familiar New Keynesian and TANK shapes, with hand-to-mouth aggregation muting the output gain. The blue band adds inflation: the output gap turns negative, inequality rises, firms raise desired markups, and because the shock is deflationary the elasticity gap shrinks the fall in inflation.

After the monetary policy shock (Figure D.2b), TE-TANK nearly coincides with the markup-matched TANK($\varepsilon^* = 3.93$), so the blue band is thin, and both deviate substantially from the RANK corner. The rate cut raises Ricardian consumption and, with sticky prices, labor demand and the real wage. Marginal cost rises, so markups fall by (D.24), and dividends, which follow from $D_t = Y_t[\Delta^p(\Pi_t) - (1 - \alpha)/\mathcal{M}_t]$ of Appendix D.4 as $\hat{d}_t = \hat{y}_t + (1 - \alpha)\hat{\mu}_t/[\bar{\mathcal{M}} - (1 - \alpha)]$, fall because the markup term dominates the rise in output; transfers to both types fall with them. Poor households, whose income is mostly labor income, nevertheless gain and spend it, which raises output further, a channel absent in RANK. Inequality therefore falls. This leaves de-

sired markups unchanged in TANK but lowers them in TE-TANK, as demand tilts toward the price-elastic poor, so firms raise prices by less: the blue band is negative in the inflation panel, while the output gap rises by essentially the same amount in both economies.

D.13 Fixed versus Consumption-Dependent Elasticities

If a household's elasticity also falls with its own consumption, the desired-markup channel scales by $1-\gamma$, with the curvature index γ defined below. Against a markup-matched benchmark with the same average gradient, the channel therefore keeps its sign when γ is less than one and reverses when it exceeds one. Across the assumed calibrations below, which the cross-sectional evidence does not pin down, the pass-through gap ranges from 3.18 percent in the linear case to -0.65 percent when γ is fitted to the model's own consumption spread, against 3.32 percent with fixed elasticities. The model fixes ε_R and ε_P within type, whereas Auer et al. (2024) obtain the gap from non-homothetic preferences and the closest structural model, Mongey and Waugh (2025), from the marginal value of wealth under preferences common to all households; in both, a household's elasticity falls as its own consumption or wealth rises and so would move over the business cycle. This within-household margin, together with the sorting of poor and middle-income households across firms, is why a one-time transfer raises the aggregate markup in Mongey and Waugh (2025). In Olivi et al. (2024), each household's contribution to the endogenous markup wedge combines the composition term kept here with the response of its elasticity to its own spending, the term this appendix adds.

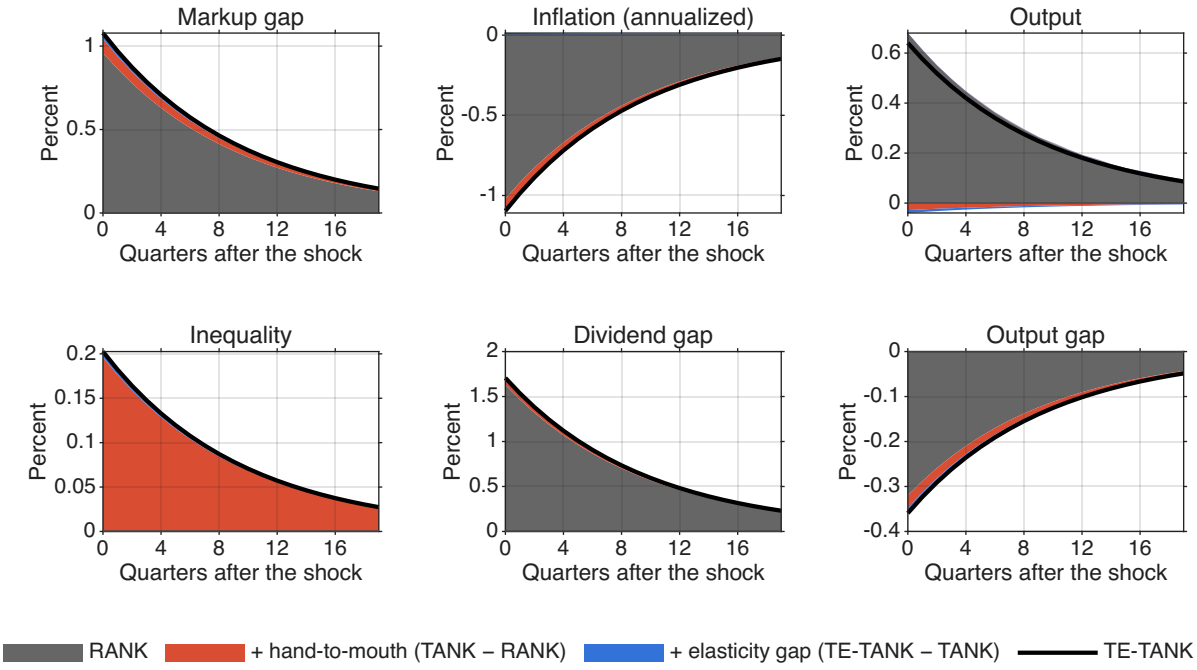
Two forces Let $\eta_h \equiv \mathrm{d}\varepsilon_h/\mathrm{d}\ln C_h < 0$ be the within-type gradient, in elasticity points per log point of own consumption, so that $\varepsilon_{h,t} = \bar{\varepsilon}_h + \eta_h \hat{c}_t^h$ to first order. With the expenditure shares $\omega_R = (1-\lambda)\bar{H}$ and $\omega_P = \lambda \bar{H}^P$, the effective elasticity $\varepsilon_{\text{eff},t} = \sum_h s_h H_t^h \varepsilon_{h,t}$ moves through a composition term, $(\varepsilon_R - \varepsilon_P)(1-\lambda)\bar{H}\hat{h}_t$, the paper's mechanism, and a within-type term, $\omega_R \eta_R \hat{c}_t^R + \omega_P \eta_P \hat{c}_t^P$, which fixed elasticities switch off. Substituting $\hat{c}_t^R = \hat{y}_t + \hat{h}_t$ and $\hat{c}_t^P = \hat{y}_t - (\omega_R/\omega_P)\hat{h}_t$, which follow from $C_t^R = H_t C_t$, the adding-up condition, and $C_t = Y_t$, gives

$$\mathrm{d}\varepsilon_{\text{eff},t} = (1-\lambda)\bar{H}\big[(\varepsilon_R - \varepsilon_P) + (\eta_R - \eta_P)\big]\hat{h}_t + \bar{\eta}\,\hat{y}_t, \qquad \bar{\eta} \equiv \omega_R \eta_R + \omega_P \eta_P.$$

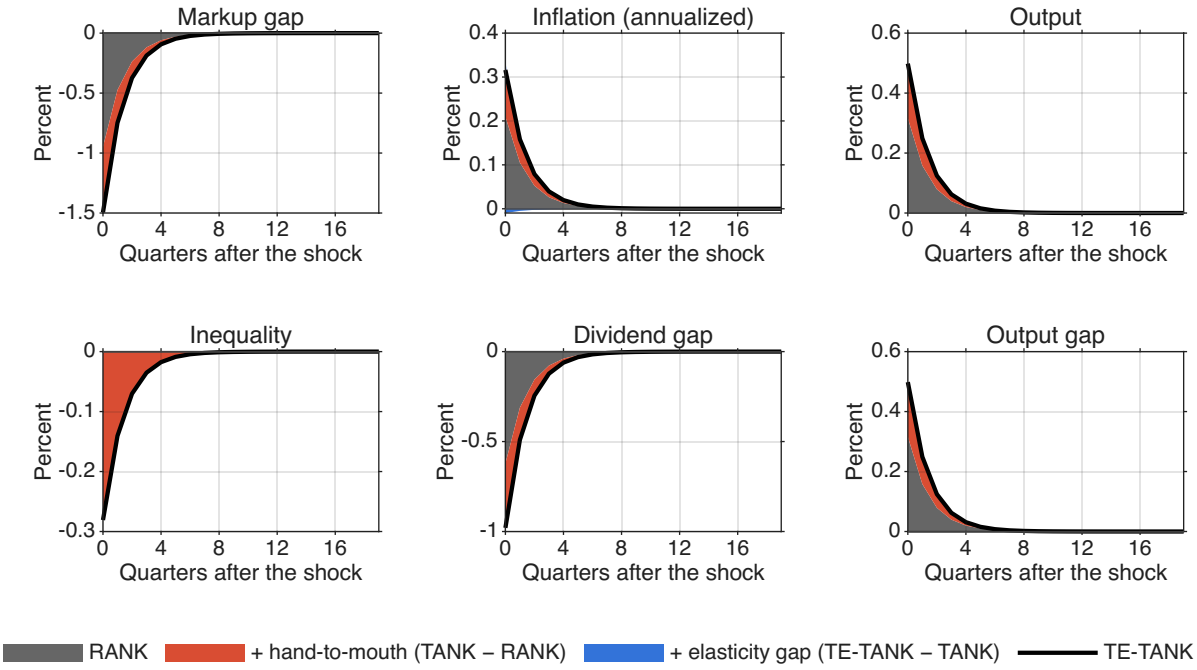
The average gradient $\bar{\eta}$ multiplies aggregate output; only the difference $\eta_R - \eta_P$ reaches the inequality term.

FIGURE D.2: Technology and Monetary Policy Shocks: Channel Decomposition

(A) Positive technology shock



(B) Expansionary monetary policy shock



Notes: Impulse responses to a one-standard-deviation positive technology shock (panel a) and a one-standard-deviation reduction in the nominal interest rate (panel b). Section 3.4 defines the bands, homogeneous corners, and units. Panels as in Figure 2 (without the strips), plus output (top right) and the dividend gap (bottom middle). Inequality is identically zero in RANK, so the gray band is flat in the inequality panel and the inequality response is carried by the red and blue bands. In panel b the red band amplifies output and inflation relative to RANK through the high marginal propensity to consume of hand-to-mouth households, and the negative blue band in the inflation panel is the elasticity-gap damping discussed in the text.

Slopes In the pricing condition (D.13) the elasticities enter through $\sum_h s_h H_t^h[1 - \varepsilon_{h,t}(1 - MC_t)]/\zeta$. Linearizing in $\varepsilon_{h,t}$ as well therefore adds $-[(1 - \overline{MC})/\zeta] \sum_h \omega_h \, \mathrm{d}\varepsilon_{h,t}$ to (D.20), whose $\hat{h}_t$ terms equal $-[(1 - \overline{MC})/\zeta]$ times the composition term because $\sum_h s_h H^h \hat{h}_t^h = 0$; together the Phillips curve loads on $-[(1 - \overline{MC})/\zeta] \, \mathrm{d}\varepsilon_{\mathrm{eff},t}$. The $\bar{\eta}\hat{y}_t$ part adds $\Delta\kappa_y \equiv -(1 - \overline{MC})\bar{\eta}/\zeta > 0$ to κ_y and leaves κ_h unchanged: when output is high, every household is less price-sensitive, and the Phillips curve is steeper. Define the curvature index $\gamma \equiv (\eta_R - \eta_P)/(\varepsilon_P - \varepsilon_R)$, the within-type gradient difference in units of the between-type elasticity gap, positive when the poor's elasticity falls more steeply with consumption than the rich's. The bracket in the last display equals $(\varepsilon_R - \varepsilon_P)(1 - \gamma)$, so $\kappa_h = \delta_2 \eta_\mu$ scales by $1 - \gamma$, while Γ and $\chi_y = \Gamma G$, which depend only on the steady state, are unchanged. With the subscript η marking slopes under within-type gradients and $\kappa^{\mathrm{TANK}^*} = \kappa_y = \delta_2 G$,

$$\kappa_\eta^{\mathrm{TANK}^*} = \kappa^{\mathrm{TANK}^*} + \Delta\kappa_y, \qquad \kappa_\eta^{\mathrm{TE}} = \kappa^{\mathrm{TANK}^*}\left[1 - \Gamma\eta_\mu(1 - \gamma)\right] + \Delta\kappa_y,$$
$$\kappa_\eta^{\mathrm{TE}} - \kappa_\eta^{\mathrm{TANK}^*} = -\chi_y \delta_2 \eta_\mu (1 - \gamma), \qquad \frac{\kappa_\eta^{\mathrm{TANK}^*} - \kappa_\eta^{\mathrm{TE}}}{\kappa_\eta^{\mathrm{TANK}^*}} = \frac{\kappa^{\mathrm{TANK}^*}}{\kappa^{\mathrm{TANK}^*} + \Delta\kappa_y} \Gamma\eta_\mu(1 - \gamma).$$

We assume that the comparison economy TANK(ε^*) shares the gradient $\bar{\eta}$ at $\varepsilon^* = \varepsilon_{\mathrm{eff}}(\bar{H})$, so that only dispersion differs; a power law with a common γ and a linear form with a common slope both satisfy this. The common increment $\Delta\kappa_y$ then cancels from the level difference but not from the ratio, and a benchmark that instead keeps ε^* fixed changes the demand system itself; against it, the common increment outweighs the composition effect even at the income-based curvature below, so TE-TANK would be slightly steeper. Since the ratio factor is positive, the channel keeps its sign for $\gamma < 1$, vanishes at $\gamma = 1$, and reverses for $\gamma > 1$, whatever the size of $\Delta\kappa_y$; pass-through falls with κ and the gradients leave χ_y unchanged, so the markup-matched pass-through gap has the sign of the slope gap.

Calibrating γ The cross-sectional evidence of Auer et al. (2024), which shows how price sensitivity differs across households at a point in time, does not identify the within-household gradient that γ measures. We therefore assume functional forms, calibrate them to the same evidence, and report the implied γ; these values are assumptions, not bounds. In all cases the steady state is unchanged, since $\bar{\varepsilon}_R$ and $\bar{\varepsilon}_P$ stay at their baseline values.

(i) Linear, $\gamma = 0$. If ε is linear in log consumption with the cross-sectional slope $-(\varepsilon_P - \varepsilon_R)/\ln 6 \approx -2.0$ points per log point, the two types share a gradient and the level difference is unchanged at $-\chi_y \delta_2 \eta_\mu$, but the common increment lowers the pass-through gap from 3.32 to 3.18 percent.

(ii)–(iv) Power law. A constant proportional gradient, $\varepsilon = AC^{-\gamma}$, gives $\gamma = \ln(\varepsilon_P/\varepsilon_R)/\ln(C^R/C^P)$, where $\varepsilon_P/\varepsilon_R = 2.2$ and C^R/C^P is the rich-to-poor spread over which that ratio is taken, so any spread above 2.2 gives $\gamma < 1$. (ii) Across the 6-fold range of permanent income over which the elasticity ratio is estimated, $\gamma = 0.44$ and the pass-through gap is 1.76 percent. (iii) The held-out CEX rich-to-poor consumption ratio of 2.93, a spread in the units in which γ is defined, gives $\gamma = 0.73$ and a pass-through gap of 0.81 percent. (iv) The model's own steady-state ratio $\bar{C}^R/\bar{C}^P = \bar{H}^R/\bar{H}^P = 1.905$ gives $\gamma = 1.22 > 1$, and the channel reverses, with a pass-through gap of -0.65 percent; that ratio is neither a calibration target nor a data moment but follows once λ and $R \equiv \delta(1-\tau)$ are set externally (Appendix C.2), and both the CEX ratio and the widest spread the model can reach, 2.759 at $R = 0$, exceed 2.2. Within the power-law family a single $\varepsilon(\cdot)$ with $\gamma < 1$ implies $\varepsilon_P/\varepsilon_R < 1.905$ at the model's types and cannot reproduce the calibrated ratio of 2.2, so cases (ii) and (iii) imply type-specific intercepts, $\varepsilon_h = A_h C_h^{-\gamma}$ with A_P/A_R between 1.37 and 1.66; the curvature index, and hence the scaling, is unchanged.

Natural allocation With gradients the flexible-price markup moves with output. Under flexible prices, with $\hat{h}_t^n = \Gamma\hat{\mu}_t^n$ and Ψ_a of (D.23),

$$\hat{\mu}_t^n = b'\hat{y}_t^n, \quad b' \equiv \frac{-\bar{\eta}}{\varepsilon^*(\varepsilon^*-1)\left[1-\Gamma\eta_\mu(1-\gamma)\right]}, \quad \hat{y}_t^n = \frac{\Psi_a a_t}{G+b'},$$
$$\pi_t - \beta\mathbb{E}_t\pi_{t+1} = \kappa_\eta^{\mathrm{TE}}(\hat{y}_t - \hat{y}_t^n) = \kappa_\eta^{\mathrm{TE}}(\hat{y}_t - \hat{y}_t^{\mathcal{M}}) + \Delta\kappa_y\,\hat{y}_t^{\mathcal{M}}, \qquad \hat{y}_t^{\mathcal{M}} \equiv \Psi_a a_t/G.$$

Over cases (ii)–(iii), a one percent technology shock raises $\hat{\mu}_t^n$ by 0.15–0.23 percent and leaves $\hat{y}_t^n$ 4.9–7.8 percent below the constant-markup output $\hat{y}_t^{\mathcal{M}}$ that fixed elasticities imply. The Phillips curve has no residual against natural output, so divine coincidence is unaffected; the wedge $\Delta\kappa_y\hat{y}_t^{\mathcal{M}}$ appears only against constant-markup output, and only technology shocks move it.

TABLE D.1: Superelasticity of Demand: Model and Literature

Demand system or source	S	Source
Panel A: demand systems at the effective elasticity ε^*		
CES	0	Mrázová and Neary (2017)
Linear demand, $\varepsilon + 1$	4.93	Derived from Mrázová and Neary (2017, pp. 3848, 3868)
CARA demand, ε	3.93	Derived from Mrázová and Neary (2017)
Translog demand, $(\varepsilon - 1)^2/\varepsilon$	2.19	Derived from Mrázová and Neary (2017, pp. 3849, 3868)
Kimball aggregator	free parameter	Kimball (1995)
TE-TANK two-CES mixture, $-\mathrm{Var}_\omega(\varepsilon)/\varepsilon_{\mathrm{eff}}$	-0.63	This paper
Mixture with within-type Kimball, $\varsigma - \mathrm{Var}_\omega(\varepsilon)/\varepsilon_{\mathrm{eff}}$, $\varsigma = 1.93$	1.30	This paper
Within-type ς at which the net Calvo comparison flips	5.86	This paper
Panel B: macro calibrations		
Smets and Wouters (2007), Kimball curvature of goods and labor aggregators	10	Smets and Wouters (2007, p. 592)
Klenow and Willis (2016), calibration	10	Klenow and Willis (2016, p. 458)
Macro estimates reported by Klenow and Willis (2016)	10 and 33	Klenow and Willis (2016, p. 458)
Panel C: micro estimates		
Beck and Lein (2020), household-level median (10th–90th percentile)	1.93 (0.05–2.23)	Beck and Lein (2020, Table 3)
Dossche et al. (2010), sensible curvature value	"around 4"	Dossche et al. (2010, p. 726)
Klenow and Willis (2016), summary of micro evidence	"closer to 2 than to 10"	Klenow and Willis (2016, p. 469)

Notes: The superelasticity is $S \equiv \mathrm{d}\ln\varepsilon/\mathrm{d}x$ at the symmetric price, with x the log own relative price and ε the own-price elasticity of demand. $S > 0$ is Kimball curvature: the elasticity rises with the own price, which makes prices strategic complements. Demand is subconvex if and only if $S > 0$, CES has $S = 0$, and demand with $S < 0$ is superconvex (Mrázová and Neary, 2017, p. 3868). Panel A evaluates each demand system at the model's effective elasticity $\varepsilon^* = 3.93$; linear, CARA and translog demand have $S = \varepsilon + 1$, ε and $(\varepsilon - 1)^2/\varepsilon$. The two-CES mixture has $S = -\mathrm{Var}_\omega(\varepsilon)/\varepsilon_{\mathrm{eff}}$. The next row adds Kimball demand within each type with superelasticity $\varsigma = 1.93$, and the last row of Panel A gives the within-type ς above which the net Calvo comparison with markup-matched TANK changes sign. Panels B and C report the values stated in the sources of the last column; the entries for Dossche et al. and for the summary of Klenow and Willis are quotations.